

2014 AHA/ACC/HRS Guideline for the Management of Patients With Atrial Fibrillation

A Report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society

Developed in Collaboration With the Society of Thoracic Surgeons

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Preamble

The medical profession should play a central role in evaluating the evidence related to drugs, devices, and procedures for the detection, management, and prevention of disease. When properly applied, expert analysis of available data on the benefits and risks of these therapies and procedures can improve the quality of care, optimize patient outcomes, and favorably affect costs by focusing resources on the most effective strategies. An organized and directed approach to a thorough review of evidence has resulted in the production of clinical practice guidelines that assist clinicians in selecting the best management strategy for an individual patient. Moreover, clinical practice guidelines can provide a foundation for other applications, such as performance measures, appropriate use criteria, and both quality improvement and clinical decision support tools.

The American College of Cardiology (ACC) and the American Heart Association (AHA) have jointly engaged in the production of guidelines in the area of cardiovascular disease since 1980. The ACC/AHA Task Force on Practice Guidelines (Task Force), whose charge is to develop, update, or revise practice guidelines for cardiovascular diseases and procedures, directs this effort. Writing committees are charged with the task of performing an assessment of the evidence and acting as an independent group of authors to develop, update, or revise written recommendations for clinical practice.

Experts in the subject under consideration are selected from both organizations to examine subject-specific data and write guidelines. Writing committees are specifically charged to perform a literature review; weigh the strength of evidence for or against particular tests, treatments, or procedures; and include estimates of expected health outcomes where such data exist. Patient-specific modifiers, comorbidities, and issues of patient preference that may influence the choice of tests or therapies are considered, as well as frequency of follow-up and cost-effectiveness. When available, information from studies on cost is considered; however, review of data on efficacy and outcomes constitutes the primary basis for preparing recommendations in this guideline.

In analyzing the data, and developing recommendations and supporting text, the writing committee uses evidence-based methodologies developed by the Task Force.¹ The Classification

of Recommendation (COR) is an estimate of the size of the treatment effect, with consideration given to risks versus benefits, as well as evidence and/or agreement that a given treatment or procedure is or is not useful/effective or in some situations may cause harm; this is defined in Table 1. The Level of Evidence (LOE) is an estimate of the certainty or precision of the treatment effect. The writing committee reviews and ranks evidence supporting each recommendation, with the weight of evidence ranked as LOE A, B, or C, according to specific definitions that are included in Table 1. Studies are identified as observational, retrospective, prospective, or randomized, as appropriate. For certain conditions for which inadequate data are available, recommendations are based on expert consensus and clinical experience and are ranked as LOE C. When recommendations at LOE C are supported by historical clinical data, appropriate references (including clinical reviews) are cited if available.

For issues with sparse available data, a survey of current practice among the clinician members of the writing committee is the basis for LOE C recommendations and no references are cited.

The schema for COR and LOE is summarized in Table 1, which also provides suggested phrases for writing recommendations within each COR.

A new addition to this methodology is the separation of the Class III recommendations to delineate whether the recommendation is determined to be of “no benefit” or is associated with “harm” to the patient. In addition, in view of the increasing number of comparative effectiveness studies, comparator verbs and suggested phrases for writing recommendations for the comparative effectiveness of one treatment or strategy versus another are included for COR I and IIa, LOE A or B only.

In view of the advances in medical therapy across the spectrum of cardiovascular diseases, the Task Force has designated the term *guideline-directed medical therapy* to represent optimal medical therapy as defined by ACC/AHA guideline (primarily Class I)—recommended therapies. This new term, *guideline-directed medical therapy*, is used herein and throughout subsequent guidelines.

Therapies not available in the United States are discussed in the text without a specific COR. For studies performed in large numbers of subjects outside North America, each writing committee reviews the potential impact of different practice patterns and patient populations on the treatment effect and relevance to the ACC/AHA target population to determine whether the findings should inform a specific recommendation.

The ACC/AHA practice guidelines are intended to assist clinicians in clinical decision making by describing a range of generally acceptable approaches to the diagnosis, management, and prevention of specific diseases or conditions. The guidelines attempt to define practices that meet the needs of most patients in most circumstances. The ultimate judgment about care of a particular patient must be made by the clinician and patient in light of all the circumstances presented by that patient. As a result, situations may arise in which deviations from these guidelines may be appropriate. Clinical decision making should involve consideration of the quality and availability of expertise in the area where care is provided. When these guidelines are used as the basis for regulatory or payer decisions, the goal should be improvement in quality of

Table 1. Applying Classification of Recommendations and Level of Evidence

		SIZE OF TREATMENT EFFECT				
		CLASS I <i>Benefit >>> Risk</i> Procedure/Treatment SHOULD be performed/ administered	CLASS IIa <i>Benefit >> Risk</i> <i>Additional studies with focused objectives needed</i> IT IS REASONABLE to per- form procedure/administer treatment	CLASS IIb <i>Benefit ≥ Risk</i> <i>Additional studies with broad objectives needed; additional registry data would be helpful</i> Procedure/Treatment MAY BE CONSIDERED	CLASS III <i>No Benefit</i> or CLASS III <i>Harm</i>	
					Procedure/ Test	Treatment
ESTIMATE OF CERTAINTY (PRECISION) OF TREATMENT EFFECT	LEVEL A Multiple populations evaluated* Data derived from multiple randomized clinical trials or meta-analyses	■ Recommendation that procedure or treatment is useful/effective ■ Sufficient evidence from multiple randomized trials or meta-analyses	■ Recommendation in favor of treatment or procedure being useful/effective ■ Some conflicting evidence from multiple randomized trials or meta-analyses	■ Recommendation's usefulness/efficacy less well established ■ Greater conflicting evidence from multiple randomized trials or meta-analyses	■ Recommendation that procedure or treatment is not useful/effective and may be harmful ■ Sufficient evidence from multiple randomized trials or meta-analyses	
	LEVEL B Limited populations evaluated* Data derived from a single randomized trial or nonrandomized studies	■ Recommendation that procedure or treatment is useful/effective ■ Evidence from single randomized trial or nonrandomized studies	■ Recommendation in favor of treatment or procedure being useful/effective ■ Some conflicting evidence from single randomized trial or nonrandomized studies	■ Recommendation's usefulness/efficacy less well established ■ Greater conflicting evidence from single randomized trial or nonrandomized studies	■ Recommendation that procedure or treatment is not useful/effective and may be harmful ■ Evidence from single randomized trial or nonrandomized studies	
	LEVEL C Very limited populations evaluated* Only consensus opinion of experts, case studies, or standard of care	■ Recommendation that procedure or treatment is useful/effective ■ Only expert opinion, case studies, or standard of care	■ Recommendation in favor of treatment or procedure being useful/effective ■ Only diverging expert opinion, case studies, or standard of care	■ Recommendation's usefulness/efficacy less well established ■ Only diverging expert opinion, case studies, or standard of care	■ Recommendation that procedure or treatment is not useful/effective and may be harmful ■ Only expert opinion, case studies, or standard of care	
	Suggested phrases for writing recommendations		should is recommended is indicated is useful/effective/beneficial	is reasonable can be useful/effective/beneficial is probably recommended or indicated	may/might be considered may/might be reasonable usefulness/effectiveness is unknown/unclear/uncertain or not well established	COR III: No Benefit is not recommended is not indicated should not be performed/ administered/ other is not useful/ beneficial/ effective
Comparative effectiveness phrases†		treatment/strategy A is recommended/indicated in preference to treatment B treatment A should be chosen over treatment B	treatment/strategy A is probably recommended/indicated in preference to treatment B it is reasonable to choose treatment A over treatment B		COR III: Harm potentially harmful causes harm associated with excess morbidity/mortality should not be performed/ administered/ other	

A recommendation with Level of Evidence B or C does not imply that the recommendation is weak. Many important clinical questions addressed in the guidelines do not lend themselves to clinical trials. Although randomized trials are unavailable, there may be a very clear clinical consensus that a particular test or therapy is useful or effective.

*Data available from clinical trials or registries about the usefulness/efficacy in different subpopulations, such as sex, age, history of diabetes mellitus, history of prior myocardial infarction, history of heart failure, and prior aspirin use.

†For comparative-effectiveness recommendations (Class I and IIa; Level of Evidence A and B only), studies that support the use of comparator verbs should involve direct comparisons of the treatments or strategies being evaluated.

care. The Task Force recognizes that situations arise in which additional data are needed to inform patient care more effectively; these areas are identified within each respective guideline when appropriate.

Prescribed courses of treatment in accordance with these recommendations are effective only if followed. Because lack of patient understanding and adherence may adversely affect outcomes, clinicians should make every effort to engage the patient's active participation in prescribed medical regimens and lifestyles. In addition, patients should be informed of the risks, benefits, and alternatives to a particular treatment and should be involved in shared decision making whenever feasible, particularly for COR IIa and IIb, for which the benefit-to-risk ratio may be lower.

The Task Force makes every effort to avoid actual, potential, or perceived conflicts of interest that may arise as a result of relationships with industry and other entities (RWI) among the members of the writing committee. All writing committee members and peer reviewers of the guideline are required to disclose all current healthcare-related relationships, including those existing 12 months before initiation of the writing effort.

In December 2009, the ACC and AHA implemented a new RWI policy that requires the writing committee chair plus a minimum of 50% of the writing committee to have no relevant RWI (Appendix 1 includes the ACC/AHA definition of *relevance*). The Task Force and all writing committee members review their respective RWI disclosures during each conference call

and/or meeting of the writing committee, and members provide updates to their RWI as changes occur. All guideline recommendations require a confidential vote by the writing committee and require approval by a consensus of the voting members. Members may not draft or vote on any recommendations pertaining to their RWI. Members who recused themselves from voting are indicated in the list of writing committee members, and specific section recusals are noted in Appendix 1. Authors' and peer reviewers' RWI pertinent to this guideline are disclosed in Appendixes 1 and 2. In addition, to ensure complete transparency, writing committee members' comprehensive disclosure information—including RWI not pertinent to this document—is available as an [online supplement](#). Comprehensive disclosure information for the Task Force is also available online at <http://www.cardiosource.org/en/ACC/About-ACC/Who-We-Are/Leadership/Guidelines-and-Documents-Task-Forces.aspx>. The ACC and AHA exclusively sponsor the work of the writing committee, without commercial support. Writing committee members volunteered their time for this activity. Guidelines are official policy of both the ACC and AHA.

In an effort to maintain relevance at the point of care for clinicians, the Task Force continues to oversee an ongoing process improvement initiative. As a result, in response to pilot projects, several changes to this guideline will be apparent, including limited narrative text, a focus on summary and evidence tables (with references linked to abstracts in PubMed), and more liberal use of summary recommendation tables (with references that support the LOE) to serve as a quick reference.

In April 2011, the Institute of Medicine released 2 reports: *Finding What Works in Health Care: Standards for Systematic Reviews* and *Clinical Practice Guidelines We Can Trust*.^{2,3} It is noteworthy that the Institute of Medicine cited ACC/AHA practice guidelines as being compliant with many of the proposed standards. A thorough review of these reports and of our current methodology is under way, with further enhancements anticipated.

The recommendations in this guideline are considered current until they are superseded by a focused update, the full-text guideline is revised, or until a published addendum declares it out of date and no longer official ACC/AHA policy.

Jeffrey L. Anderson, MD, FACC, FAHA
Chair, ACC/AHA Task Force on Practice Guidelines

1. Introduction

1.1. Methodology and Evidence Review

The recommendations listed in this document are, whenever possible, evidence based. An extensive evidence review was conducted, focusing on 2006 through October 2012 and selected other references through March 2014. The relevant data are included in evidence tables in the [Online Data Supplement](#). Searches were extended to studies, reviews, and other evidence conducted in human subjects, published in English, and accessible through PubMed, EMBASE, Cochrane, Agency for Healthcare Research and Quality Reports, and other selected databases relevant to this guideline. Key search words included but were not limited to the following: *age, antiarrhythmic, atrial fibrillation, atrial remodeling, atrioventricular conduction, atrioventricular node, cardioversion, classification, clinical trial, complications, concealed*

conduction, cost-effectiveness, defibrillator, demographics, epidemiology, experimental, heart failure, hemodynamics, human, hyperthyroidism, hypothyroidism, meta-analysis, myocardial infarction, pharmacology, postoperative, pregnancy, pulmonary disease, quality of life, rate control, rhythm control, risks, sinus rhythm, symptoms, and tachycardia-mediated cardiomyopathy. Additionally, the writing committee reviewed documents related to atrial fibrillation (AF) previously published by the ACC and AHA. References selected and published in this document are representative and not all-inclusive.

To provide clinicians with a comprehensive set of data, whenever deemed appropriate or when published, the absolute risk difference and number needed to treat or harm are provided in the guideline or data supplement, along with confidence intervals (CI) and data related to the relative treatment effects such as the odds ratio (OR), relative risk (RR), hazard ratio, or incidence rate ratio.

1.2. Organization of the Writing Committee

The 2014 AF writing committee was composed of clinicians with broad expertise related to AF and its treatment, including adult cardiology, electrophysiology, cardiothoracic surgery, and heart failure (HF). The writing committee was assisted by staff from the ACC and AHA. Under the guidance of the Task Force, the Heart Rhythm Society was invited to be a partner organization and provided representation. The writing committee also included a representative from the Society of Thoracic Surgeons. The rigorous methodological policies and procedures noted in the Preamble differentiate ACC/AHA guidelines from other published guidelines and statements.

1.3. Document Review and Approval

This document was reviewed by 2 official reviewers each nominated by the ACC, AHA, and Heart Rhythm Society, as well as 1 reviewer from the Society of Thoracic Surgeons and 43 individual content reviewers (from the ACC Electrophysiology Section Leadership Council, ACC Adult Congenital and Pediatric Cardiology Section Leadership Council, ACC Association of International Governors, ACC Heart Failure and Transplant Section Leadership Council, ACC Imaging Section Leadership Council, ACC Interventional Section Leadership Council, ACC Surgeons' Council, and the Heart Rhythm Society Scientific Documents Committee). All information on reviewers' RWI was distributed to the writing committee and is published in this document (Appendix 2).

This document was approved for publication by the governing bodies of the ACC, AHA, and Heart Rhythm Society and endorsed by the Society of Thoracic Surgeons.

1.4. Scope of the Guideline

The task of the 2014 writing committee was to establish revised guidelines for optimum management of AF. The new guideline incorporates new and existing knowledge derived from published clinical trials, basic science, and comprehensive review articles, along with evolving treatment strategies and new drugs. This guideline supersedes the "ACC/AHA/ESC 2006 Guidelines for the Management of Patients With Atrial Fibrillation"⁴ and the 2 subsequent focused updates from 2011.^{5,6} In addition, the ACC, AHA, American College of Physicians, and American Academy

of Family Physicians submitted a proposal to the Agency for Healthcare Research and Quality to perform a systematic review on specific questions related to the treatment of AF. The data from that report were reviewed by the writing committee and incorporated where appropriate.^{7a,7b}

The 2014 AF guideline is organized thematically, with recommendations, where appropriate, provided with each section. Some recommendations from earlier guidelines have been eliminated or updated as warranted by new evidence or a better understanding of earlier evidence. In developing the 2014 AF guideline, the writing committee reviewed prior published guidelines and related statements. Table 2 lists these publications and statements deemed pertinent to this effort and is intended for use as a resource.

2. Background and Pathophysiology

AF is a common cardiac rhythm disturbance that increases in prevalence with advancing age. Approximately 1% of patients with AF are <60 years of age, whereas up to 12% of patients with AF are 75 to 84 years of age.²⁹ More than one third of patients with AF are ≥80 years of age.^{30,31} In the United States, the percentage of Medicare fee-for-service beneficiaries with AF in 2010 was reported as 2% for those <65 years of age and 9% for those ≥65 years of age.³² For individuals of European descent, the lifetime risk of developing AF after 40 years of age is 26% for men and 23% for women.³³ In African Americans, although risk factors for AF are more prevalent, incidence of AF appears to be lower.³⁴ AF is often associated with structural heart disease and other co-occurring chronic conditions (Table 3; see also <http://www.cms>.

Table 2. Associated Guidelines and Statements

Title	Organization	Publication Year/ Reference
Guidelines		
Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC 7)	NHLBI	2003 ⁸
Assessment of Cardiovascular Risk in Asymptomatic Adults	ACC/AHA	2010 ⁹
Coronary Artery Bypass Graft Surgery	ACC/AHA	2011 ¹⁰
Hypertrophic Cardiomyopathy	ACC/AHA	2011 ¹¹
Percutaneous Coronary Intervention	ACC/AHA/SCAI	2011 ¹²
Secondary Prevention and Risk Reduction Therapy for Patients With Coronary and Other Atherosclerotic Vascular Disease	AHA/ACC	2011 ¹³
Atrial Fibrillation*	CCS	2012 ¹⁴
Atrial Fibrillation	ESC	2012 ¹⁵
Stable Ischemic Heart Disease	ACC/AHA/ACP/AATS/ PCNA/SCAI/STS	2012 ¹⁶
Antithrombotic Therapy	ACCP	2012 ¹⁷
Device-Based Therapy	ACC/AHA/HRS	2012 ¹⁸
Heart Failure	ACC/AHA	2013 ¹⁹
ST-Elevation Myocardial Infarction	ACC/AHA	2013 ²⁰
Unstable Angina/Non-ST-Elevation Myocardial Infarction	ACC/AHA	2014 ²¹
Valvular Heart Disease	AHA/ACC	2014 ²²
Assessment of Cardiovascular Risk	ACC/AHA	2013 ²³
Lifestyle Management to Reduce Cardiovascular Risk	AHA/ACC	2013 ²⁴
Management of Overweight and Obesity in Adults	AHA/ACC/TOS	2013 ²⁵
Treatment of Blood Cholesterol to Reduce Atherosclerotic Cardiovascular Risk in Adults	ACC/AHA	2013 ²⁶
Statements		
Treatment of Atrial Fibrillation	AHRQ	2013 ^{7a,7b}
Oral Antithrombotic Agents for the Prevention of Stroke in Nonvalvular Atrial Fibrillation: A Science Advisory for Healthcare Professionals	AHA/ASA	2012 ²⁷
Expert Consensus Statement on Catheter and Surgical Ablation of Atrial Fibrillation: Recommendations for Patient Selection, Procedural Techniques, Patient Management and Follow-Up, Definitions, Endpoints, and Research Trial Design	HRS/EHRA/ECAS	2012 ²⁸

*Includes the following sections: Catheter Ablation for AF/Atrial Flutter; Prevention and Treatment of AF Following Cardiac Surgery; Rate and Rhythm Management; Prevention of Stroke and Systemic Thromboembolism in AF and Flutter; Management of Recent-Onset AF and Flutter in the Emergency Department; Surgical Therapy; The Use of Antiplatelet Therapy in the Outpatient Setting; and Focused 2012 Update of the CCS AF Guidelines: Recommendations for Stroke Prevention and Rate/Rhythm Control.

AATS indicates American Association for Thoracic Surgery; ACC, American College of Cardiology; ACCP, American College of Chest Physicians; ACP, American College of Physicians; AF, atrial fibrillation; AHA, American Heart Association; AHRQ, Agency for Healthcare Research and Quality; ASA, American Stroke Association; CCS, Canadian Cardiology Society; ECAS, European Cardiac Arrhythmia Society; EHRA, European Heart Rhythm Association; ESC, European Society of Cardiology; HRS, Heart Rhythm Society; JNC, Joint National Committee; NHLBI, National Heart, Lung, and Blood Institute; PCNA, Preventive Cardiovascular Nurses Association; SCAI, Society for Cardiovascular Angiography and Interventions; STS, Society of Thoracic Surgeons; and TOS, The Obesity Society.

Table 3. 10 Most Common Comorbid Chronic Conditions Among Medicare Beneficiaries With AF

Beneficiaries ≥65 y of Age (N=2 426 865) (Mean Number of Conditions=5.8; Median=6)			Beneficiaries <65 y of Age (N=105 878) (Mean Number of Conditions=5.8; Median=6)		
	N	%		N	%
Hypertension	2 015 235	83.0	Hypertension	85 908	81.1
Ischemic heart disease	1 549 125	63.8	Ischemic heart disease	68 289	64.5
Hyperlipidemia	1 507 395	62.1	Hyperlipidemia	64 153	60.6
HF	1 247 748	51.4	HF	62 764	59.3
Anemia	1 027 135	42.3	Diabetes mellitus	56 246	53.1
Arthritis	965 472	39.8	Anemia	48 252	45.6
Diabetes mellitus	885 443	36.5	CKD	42 637	40.3
CKD	784 631	32.3	Arthritis	34 949	33.0
COPD	561 826	23.2	Depression	34 900	33.0
Cataracts	546 421	22.5	COPD	33 218	31.4

AF indicates atrial fibrillation; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; and HF, heart failure.

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gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Chronic-Conditions/Downloads/2012Chartbook.pdf). The mechanisms causing and sustaining AF are multifactorial; AF can be complex and difficult for clinicians to manage. Symptoms of AF range from nonexistent to severe. Frequent hospitalizations, hemodynamic abnormalities, and thromboembolic events related to AF result in significant morbidity and mortality. AF is associated with a 5-fold increased risk of stroke,³⁵ and stroke risk increases with age.³⁶ AF-related stroke is likely to be more severe than non-AF-related stroke.³⁷ AF is also associated with a 3-fold risk of HF^{38–40} and a 2-fold increased risk of both dementia⁴¹ and mortality.³⁵ In the United States, hospitalizations with AF as the primary diagnosis total >467 000 annually, and AF is estimated to contribute to >99 000 deaths per year. Patients with AF are hospitalized twice as often as patients without AF and are 3 times more likely to have multiple admissions; 2.1% of patients with AF died in the hospital, compared with 0.1% without it.^{42,43} AF is also expensive, adding approximately \$8700 per year (estimate from 2004 to 2006) for a patient with AF compared with a patient without AF. It is estimated that treating patients with AF adds \$26 billion to the US healthcare bill annually. AF affects between 2.7 million and 6.1 million American adults, and that number is expected to double over the next 25 years, adding further to the cost burden.^{42,43}

AF Web-based tools are available, including several risk calculators and clinical decision aids (<http://www.cardiosource.org/Science-And-Quality/Clinical-Tools/Atrial-Fibrillation-Toolkit.aspx>); however, these tools must be used with caution because validation across the broad range of patients with AF encountered in clinical practice is incomplete.

2.1. Definitions and Pathophysiology of AF

AF is a supraventricular tachyarrhythmia with uncoordinated atrial activation and consequently ineffective atrial contraction.^{4,28,30} Characteristics on an electrocardiogram (ECG) include 1) irregular R-R intervals (when atrioventricular [AV] conduction is present), 2) absence of distinct repeating P waves, and 3) irregular atrial activity.

Hemodynamic consequences of AF can result from a variable combination of suboptimal ventricular rate control (either

too rapid or too slow), loss of coordinated atrial contraction, beat-to-beat variability in ventricular filling, and sympathetic activation.^{45–47} Consequences for individual patients vary, ranging from no symptoms to fatigue, palpitations, dyspnea, hypotension, syncope, or HF.⁴⁸ The most common symptom of AF is fatigue. The appearance of AF is often associated with exacerbation of underlying heart disease, either because AF is a cause or consequence of deterioration or because it contributes directly to deterioration.^{49,50} For example, initially asymptomatic patients may develop tachycardia-induced ventricular dysfunction and HF (tachycardia-induced cardiomyopathy) when the ventricular rate is not adequately controlled.^{51,52} AF also confers an increased risk of stroke and/or peripheral thromboembolism owing to the formation of atrial thrombi, usually in the left atrial appendage (LAA).

In the absence of an accessory AV pathway, the ventricular rate is determined by the conduction and refractory properties of the AV node and the sequence of wave fronts entering the AV node.^{53–55} L-type calcium channels are responsible for the major depolarizing current in AV nodal cells. Beta-adrenergic receptor stimulation enhances AV nodal conduction, whereas vagal stimulation (muscarinic receptor activation by acetylcholine) impedes AV nodal conduction.⁵⁵ Sympathetic activation and vagal withdrawal such as that with exertion or illness accelerates the ventricular rate. Each atrial excitation wave front that depolarizes AV nodal tissue renders those cells refractory for a period of time, preventing successive impulses from propagating in the node, an effect called concealed conduction.⁵⁵ This effect of concealed conduction into the AV node explains why the ventricular rate can be faster and more difficult to slow when fewer atrial wave fronts are entering the AV node, as in atrial flutter, compared with AF.⁵³

Loss of atrial contraction may markedly decrease cardiac output, particularly when diastolic ventricular filling is impaired by mitral stenosis, hypertension, hypertrophic cardiomyopathy (HCM), or restrictive cardiomyopathy.^{50,56,57} After restoration of sinus rhythm, atrial mechanical function fails to recover in some patients, likely as a consequence of remodeling or underlying atrial disease and duration of AF.⁵⁸ Ventricular contractility is not constant during AF because of variable diastolic filling time and changes in the force-interval relationship.^{59,60} Overall, cardiac output may decrease and filling pressures may increase

Table 4. Definitions of AF: A Simplified Scheme

Term	Definition
Paroxysmal AF	• AF that terminates spontaneously or with intervention within 7 d of onset. • Episodes may recur with variable frequency.
Persistent AF	• Continuous AF that is sustained >7 d.
Long-standing persistent AF	• Continuous AF >12 mo in duration.
Permanent AF	• The term “permanent AF” is used when the patient and clinician make a joint decision to stop further attempts to restore and/or maintain sinus rhythm. • Acceptance of AF represents a therapeutic attitude on the part of the patient and clinician rather than an inherent pathophysiological attribute of AF. • Acceptance of AF may change as symptoms, efficacy of therapeutic interventions, and patient and clinician preferences evolve.
Nonvalvular AF	• AF in the absence of rheumatic mitral stenosis, a mechanical or bioprosthetic heart valve, or mitral valve repair.

AF indicates atrial fibrillation.

compared with a regular rhythm at the same mean rate. In patients undergoing AV nodal ablation, irregular right ventricular (RV) pacing at the same rate as regular ventricular pacing resulted in a 15% reduction in cardiac output.⁶⁰ Irregular R-R intervals also promote sympathetic activation.^{45,46}

2.1.1. AF Classification

AF may be described in terms of the duration of episodes using a simplified scheme shown in Table 4.^{4,28,30} Implanted loop recorders, pacemakers, and defibrillators offer the possibility of reporting frequency, rate, and duration of abnormal atrial rhythms, including AF.^{61,62} Episodes often increase in frequency and duration over time.

Characterization of patients with AF by duration of their AF episodes (Table 4) has clinical relevance in that outcomes of therapy such as catheter ablation are better for paroxysmal AF than for persistent AF.²⁸ When sinus rhythm is restored by cardioversion, however, the ultimate duration of the AF episode(s) is not known. Furthermore, both paroxysmal and persistent AF may occur in a single individual.

“Lone AF” is a historical descriptor that has been variably applied to younger persons without clinical or echocardiographic evidence of cardiopulmonary disease, hypertension, or diabetes mellitus.⁶³ Because the definitions are variable, the term lone AF is potentially confusing and should not be used to guide therapeutic decisions.

2.1.1.1. Associated Arrhythmias

Other atrial arrhythmias are often encountered in patients with AF. Atrial tachycardias are characterized by an atrial rate of ≥100 beats per minute (bpm) with discrete P waves and atrial activation sequences. Atrial activation is most commonly the same from beat to beat.

Focal atrial tachycardia is characterized by regular, organized atrial activity with discrete P waves, typically with an isoelectric segment between P waves (Figure 1).^{64,65} Electrophysiological mapping reveals a focal point of origin.

The mechanism can be automaticity or a microreentry circuit.^{66,67} In multifocal atrial tachycardia, the atrial activation sequence and P-wave morphology vary.⁶⁴

2.1.1.2. Atrial Flutter and Macroreentrant Atrial Tachycardia

Early studies designated atrial flutter with a rate of 240 bpm to 340 bpm as “type I flutter,” and this term has commonly been applied to typical atrial flutter.^{65,68} An electrocardiographic appearance of atrial flutter with a rate faster than 340 bpm was designated as “type II flutter”; the mechanism for type II flutter remains undefined.⁶⁹ It is now recognized that tachycardias satisfying either of these descriptions can be due to reentrant circuits or rapid focal atrial tachycardia.

Typical atrial flutter is a macroreentrant atrial tachycardia that usually proceeds up the atrial septum, down the lateral atrial wall, and through the cavotricuspid (subeustachian) isthmus between the tricuspid valve annulus and inferior vena cava, where it is commonly targeted for ablation. It is also known as “common atrial flutter” or “cavotricuspid isthmus-dependent atrial flutter.”⁶⁴ This sequence of activation (also referred to as “counterclockwise atrial flutter”) produces predominantly negative “sawtooth” flutter waves in ECG leads II, III, and aVF, and a positive deflection in V₁ (Figure 1). The atrial rate is typically 240 bpm to 300 bpm, but conduction delays in the atrial circuit due to scars from prior ablation, surgery, or antiarrhythmic drugs can slow the rate to <150 bpm in some patients.⁶⁵ When the circuit revolves in the opposite direction, flutter waves typically appear positive in the inferior ECG leads and negative in V₁ (reverse typical atrial flutter, also referred to as clockwise typical atrial flutter).⁶⁵ Unusual flutter wave morphologies occur in the presence of substantial atrial disease, prior surgery, or radiofrequency catheter ablation; the P-wave morphology is not a reliable indicator of the type of macroreentrant atrial tachycardia in these situations.^{70–72} Atrial flutter is often a persistent rhythm that requires electrical cardioversion or radiofrequency catheter ablation for termination. It is often initiated by a brief episode of atrial tachycardia or by AF.^{69,73} This relationship between AF and atrial flutter may explain why ≥80% of patients who undergo radiofrequency catheter ablation of typical atrial flutter will have AF within the following 5 years.⁷⁴

AF may be misdiagnosed as atrial flutter when AF activity is prominent on the ECG.^{75,76} Atrial flutter may also arise during treatment with antiarrhythmic agents prescribed to prevent recurrent AF,⁷⁷ particularly sodium channel-blocking antiarrhythmic drugs such as flecainide or propafenone. Catheter ablation of the cavotricuspid isthmus is effective for prevention of recurrent atrial flutter in these patients while allowing continued antiarrhythmic treatment to prevent recurrent AF.⁷⁸

Atypical flutter, or “non-cavotricuspid isthmus-dependent macroreentrant atrial tachycardia,” describes macroreentrant atrial tachycardias that are not one of the typical forms of atrial flutter that use the cavotricuspid isthmus.⁶⁴ A variety of reentrant circuits have been described, including “perimitral flutter” reentry involving the roof of the left atrium (LA), and reentry around scars in the left or right atrium, often from prior surgery or ablation.^{65,67,79} Complex reentry circuits with >1 reentry loop or circuit can occur and often coexist with common atrial flutter. These arrhythmias are not abolished by ablation of the cavotricuspid isthmus, but recognition and distinction of these arrhythmias from common atrial flutter usually requires electrophysiological

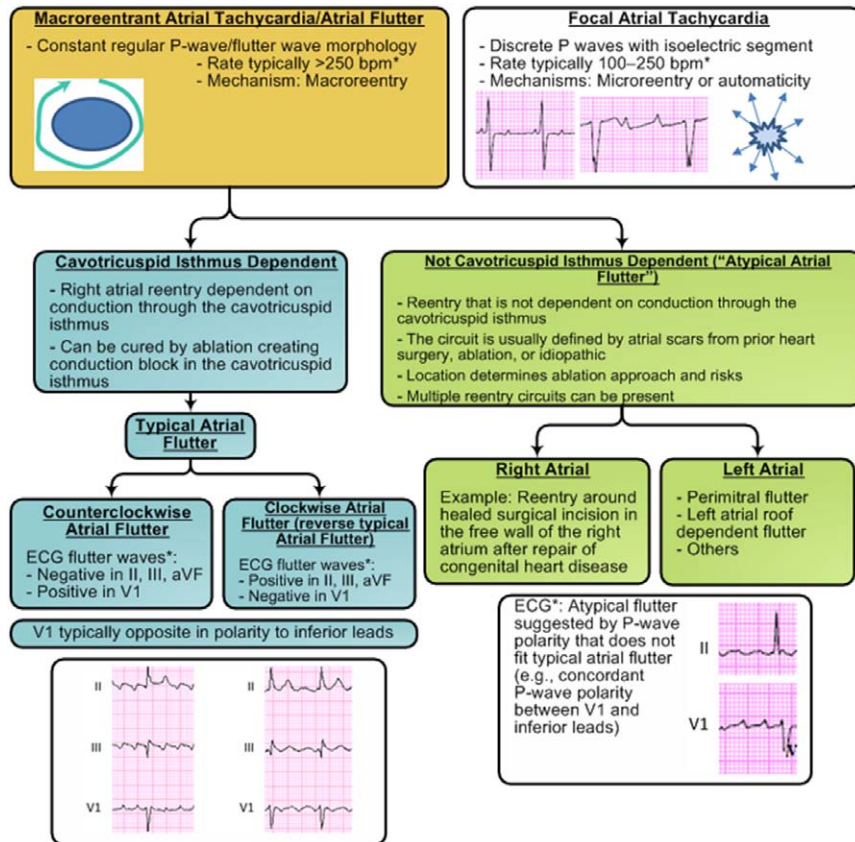


Figure 1. Atrial tachycardias. Diagram summarizing types of atrial tachycardias often encountered in patients with a history of AF, including those seen after catheter or surgical ablation procedures. P-wave morphologies are shown for common types of atrial flutter; however, the P-wave morphology is not always a reliable guide to the reentry circuit location or the distinction between common atrial flutter and other macroreentrant atrial tachycardias. *Exceptions to P-wave morphology and rate are common in scarred atria. AF indicates atrial fibrillation; bpm, beats per minute; and ECG, electrocardiogram.^{72,80}

study with atrial mapping.⁶⁵ A variety of terms have been applied to these arrhythmias according to the reentry circuit location, including LA flutter and LA macroreentrant tachycardia.^{65,67,79,80}

2.2. Mechanisms of AF and Pathophysiology

AF occurs when structural and/or electrophysiological abnormalities alter atrial tissue to promote abnormal impulse formation and/or propagation (Figure 2). These abnormalities are caused by diverse pathophysiological mechanisms,^{28,81,82} such that AF represents a final common phenotype for multiple disease pathways and mechanisms that are incompletely understood.

2.2.1. Atrial Structural Abnormalities

Any disturbance of atrial architecture potentially increases susceptibility to AF.⁸³ Such changes (eg, inflammation, fibrosis, hypertrophy) occur most commonly in the setting of underlying heart disease associated with hypertension, coronary artery disease (CAD), valvular heart disease, cardiomyopathies, and HF, which tend to increase LA pressure, cause atrial dilation, and alter wall stress. Similarly, atrial ischemia from CAD and infiltrative diseases such as amyloidosis, hemochromatosis, and sarcoidosis can also promote AF. Extracardiac factors that promote AF include hypertension, sleep apnea, obesity, use of alcohol/drugs, and hyperthyroidism, all of which have pathophysiological

effects on atrial cellular structure and/or function. Even in patients with paroxysmal AF without recognized structural heart disease, atrial biopsies have revealed inflammatory infiltrates consistent with myocarditis and fibrosis.⁸⁴ In addition, prolonged rapid atrial pacing increases arrhythmia susceptibility and forms the basis for a well-studied model of AF. In the atria of patients with established AF and of animals subjected to rapid atrial pacing, there is evidence of myocyte loss from glycogen deposits and of mitochondrial disturbances and gap-junction abnormalities that cause cell necrosis and apoptosis.^{85–87} These structural abnormalities can heterogeneously alter impulse conduction and/or refractoriness, generating an arrhythmogenic substrate.

Myocardial fibrosis is a common feature of both experimental and human AF.⁸⁸ The atria are more sensitive to profibrotic signaling and harbor a greater number of fibroblasts than the ventricles. Atrial stretch activates the renin-angiotensin-aldosterone system, which generates multiple downstream profibrotic factors, including transforming growth factor- β_1 . Additional mechanisms, including inflammation and genetic factors, can also promote atrial fibrosis. The canine rapid ventricular pacing model of HF causes extensive atrial fibrosis and increases AF susceptibility.⁸⁹ Fibrosis also occurs in the rapid atrial pacing model of AF. Late gadolinium-enhancement magnetic resonance imaging is used to image and quantitate atrial fibrosis noninvasively.^{90–95} Human

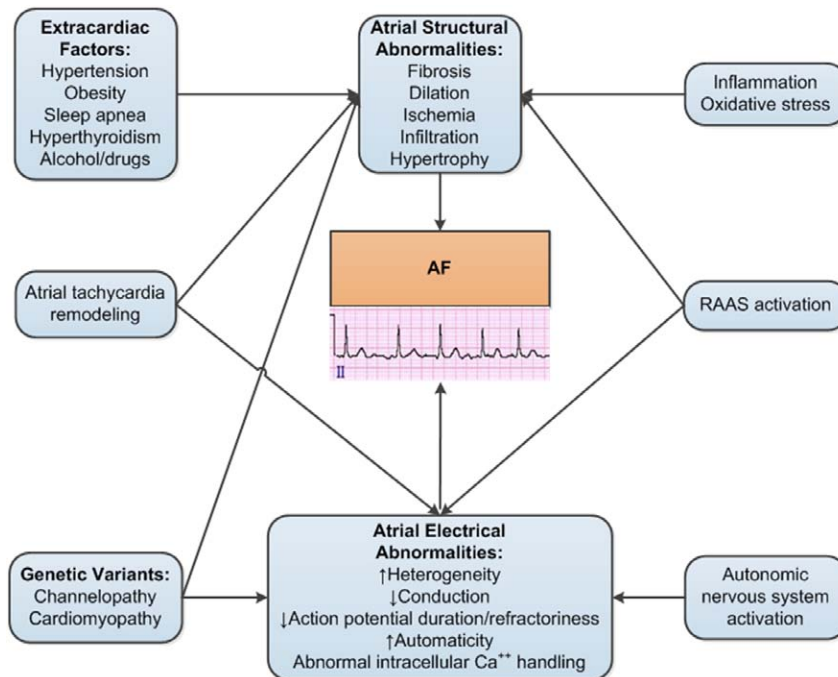


Figure 2. Mechanisms of AF. AF indicates atrial fibrillation; Ca⁺⁺, ionized calcium; and RAAS, renin-angiotensin-aldosterone system.

studies show a strong correlation between regions of low voltage on electroanatomic mapping and areas of late enhancement on magnetic resonance imaging. Preliminary results suggest that the severity of atrial fibrosis correlates with the risk of stroke⁹¹ and decreased response to catheter ablation.⁹⁰

2.2.2. Electrophysiological Mechanisms

AF requires both a trigger for initiation and an appropriate anatomic substrate for maintenance, both of which are potential targets for therapy. Several hypotheses have been proposed to explain the electrophysiological mechanisms that initiate and maintain AF.²⁸ In humans, the situation is complex, and it is likely that multiple mechanisms coexist in an individual patient.

2.2.2.1. Triggers of AF

Ectopic focal discharges often initiate AF.^{96–98} Rapidly firing foci initiating paroxysmal AF arise most commonly from LA myocardial sleeves that extend into the pulmonary veins. These observations led to the development of pulmonary vein isolation as the cornerstone for radiofrequency catheter ablation strategies.²⁸ Unique anatomic and electrophysiological features of the pulmonary veins and atriopulmonary vein junctions may account for their arrhythmogenic nature. Atrial myocardial fibers are oriented in disparate directions around the pulmonary veins and the posterior LA, with considerable anatomic variability among individuals. Conduction abnormalities that promote reentry are likely due to relatively depolarized resting potentials in pulmonary vein myocytes that promote sodium channel inactivation and to the abrupt changes in fiber orientation. Reentry is further favored by abbreviated action potentials and refractoriness in pulmonary vein myocytes.⁹⁹ Isolated pulmonary vein myocytes also demonstrate abnormal automaticity and triggered activity that could promote rapid focal firing. Additional potential sources for abnormal activity include interstitial cells

(similar to pacemaker cells in the gastrointestinal tract)¹⁰⁰ and melanocytes,¹⁰¹ both of which have been identified in pulmonary veins. Although the pulmonary veins are the most common sites for ectopic focal triggers, triggers can also arise elsewhere, including the posterior LA, ligament of Marshall, coronary sinus, venae cavae, septum, and appendages.

Abnormal intracellular calcium handling may also play a role in AF owing to diastolic calcium leak from the sarcoplasmic reticulum, which can trigger delayed afterdepolarizations.^{102–106}

2.2.2.2. Maintenance of AF

Theories proposed to explain the perpetuation and maintenance of AF include 1) multiple independent reentrant wavelets associated with heterogeneous conduction and refractoriness; 2) ≥ 1 rapidly firing foci, which may be responsive to activity from cardiac ganglion plexi; and 3) ≥ 1 rotors, or spiral wave reentrant circuits.^{28,82,88,107–113} With a single rapid focus or rotor excitation, wave fronts may encounter refractory tissue and break up during propagation, resulting in irregular or fibrillatory conduction.^{28,107,110} Both rapid focal firing and reentry may be operative during AF.

These presumed mechanisms have driven the development of therapies. The atrial maze procedure and ablation lines may interrupt paths for multiple wavelets and spiral reentry. When a biatrial phase mapping approach was used, a limited number of localized, rapid drivers (mean of approximately 2 per patient) were identified in a small group of patients with various types of AF.¹¹² In most cases, these localized sources appeared to be reentrant, whereas in others they were consistent with focal triggers, and radiofrequency catheter ablation targeting of these sites often terminated or slowed AF. Other investigators, using a noninvasive continuous biatrial mapping system, report contrasting results, observing mostly evidence for multiple wavelets and focal sites rather than rotor activity.¹¹⁴

Some investigators targeted regions in which electrogram recordings show rapid complex atrial fractionated electrograms, which are thought to be indicative of the substrate for AF or markers for ganglion plexi (see Section 2.2.2.3 for ablation of AF).¹⁰⁹ The relation of complex atrial fractionated electrograms to AF remains controversial.

2.2.2.3. Role of the Autonomic Nervous System

Autonomic stimulation can provoke AF.^{28,98,115} Activation of the parasympathetic and/or sympathetic limbs can provoke atrial arrhythmias.^{108,116} Acetylcholine activates a specific potassium current, I_{K^+ACh} , which heterogeneously shortens atrial action potential duration and refractoriness, increasing susceptibility to reentry. Sympathetic stimulation increases intracellular calcium, which promotes automaticity and triggered activity. Increased parasympathetic and/or sympathetic activity before onset of AF has been observed in some animal models and humans.^{117,118}

Plexi of autonomic ganglia that constitute the intrinsic cardiac autonomic nervous system are located in epicardial fat near the pulmonary vein-LA junctions and the ligament of Marshall. Stimulation of the ganglia in animals elicits repetitive bursts of rapid atrial activity. These plexi are often located proximal to atrial sites where complex atrial fractionated electrograms are recorded. Ablation targeting these regions improved outcomes over pulmonary vein isolation alone in some but not all studies.^{119–121}

In some patients with structurally normal hearts, AF is precipitated during conditions of high-parasympathetic tone, such as during sleep and following meals, and is referred to as “vagal mediated AF.”¹²² Avoidance of drugs, such as digoxin, that enhance parasympathetic tone has been suggested in these patients, but this remains an unproven hypothesis. Catheter ablation targeting ganglion plexi involved in vagal responses abolished AF in only 2 of 7 patients in 1 small series.¹²⁰ Adrenergic stimulation, such as that during exercise, can also provoke AF in some patients.¹²³

2.2.3. Pathophysiological Mechanisms

2.2.3.1. Atrial Tachycardia Remodeling

AF often progresses from paroxysmal to persistent over a variable period of time. Cardioversion of AF and subsequent maintenance of sinus rhythm are more likely to be successful when AF duration is <6 months.¹²⁴ The progressive nature of AF is consistent with studies demonstrating that AF causes electrical and structural remodeling such that “AF begets AF.”^{125,126}

2.2.3.2. Inflammation and Oxidative Stress

Inflammation (eg, associated with pericarditis and cardiac surgery) may be linked to AF and can be correlated with a rise in plasma concentrations of C-reactive protein.⁸¹ Inflammatory infiltrates consistent with myocarditis are often present in the atria of patients with AF and in animals with atrial dilation. Plasma concentrations of C-reactive protein and interleukin-6 are elevated in AF; increased C-reactive protein predicts the development of AF and relapse after cardioversion; and genetic variants in the interleukin-6 promoter region may influence the development of postoperative AF. In the canine pericarditis and atrial tachypacing models, prednisone suppresses AF susceptibility and reduces plasma concentrations of C-reactive protein.¹²⁷

Aging, environmental stress, inflammation, and activation of the renin-angiotensin-aldosterone system can cause oxidative damage in the atrium. Oxidative changes are present in the atrial

tissue of patients with AF and are associated with upregulation of genes involved in the production of reactive oxygen species. In human AF and a porcine model of atrial tachypacing, atrial superoxide production increased, with an apparent contribution of NAD(P)H oxidase.¹²⁸ The antioxidant ascorbate attenuated electrical remodeling in the canine atrial tachypacing model and reduced postoperative AF in a small study in humans.¹²⁹

2.2.3.3. The Renin-Angiotensin-Aldosterone System

Stimulation of the renin-angiotensin-aldosterone system promotes structural and likely electrophysiological effects in the atrium and ventricle that increase arrhythmia susceptibility.^{130–133} In addition to adverse hemodynamic effects, activation of multiple cell signaling cascades promotes increased intracellular calcium, hypertrophy, apoptosis, cytokine release and inflammation, oxidative stress, and production of growth-related factors that also stimulate fibrosis, as well as possible modulation of ion channel and gap-junction dynamics. Components of the renin-angiotensin-aldosterone system (including angiotensin II, angiotensin-converting enzyme [ACE], and aldosterone) are synthesized locally in the atrial myocardium and are increased during atrial tachypacing and AF. Variants in the ACE gene that increase angiotensin II plasma concentrations can elevate risk of AF, whereas selective cardiac overexpression of ACE causes atrial dilation, fibrosis, and increased susceptibility of AF. Therapy with these agents can reduce the occurrence of AF in patients with hypertension or left ventricular (LV) dysfunction but does not help prevent recurrence of AF in the absence of these other indications for these drugs (Section 6.2.1).

Aldosterone plays an important role in angiotensin II-mediated inflammation and fibrosis; in patients with primary hyperaldosteronism, the incidence of AF is increased. In experimental models of HF, spironolactone and eplerenone decreased atrial fibrosis and/or susceptibility of AF. Eplerenone therapy is associated with decreased AF in patients with HF.¹³⁴

2.2.3.4. Risk Factors and Associated Heart Disease

Multiple clinical risk factors, electrocardiographic and echocardiographic features, and biochemical markers are associated with an increased risk of AF (Table 5). One epidemiological analysis found that 56% of the population-attributable risk of AF could be explained by ≥ 1 common risk factor.¹³⁵ Thus, it may be possible to prevent some cases of AF through risk factor modification such as blood pressure control or weight loss.

Many potentially “reversible” causes of AF have been reported, including binge drinking, cardiothoracic and non-cardiac surgery, myocardial infarction (MI), pericarditis, myocarditis, hyperthyroidism, electrocution, pneumonia, and pulmonary embolism.^{10,49,136–138} AF that occurs in the setting of Wolff-Parkinson-White (WPW) syndrome, AV nodal reentrant tachycardia, or atrial ectopic tachycardia may resolve after catheter ablation for these arrhythmias.⁶⁹ It is important to recognize that sparse data support the notion that patients with AF that occurs in the setting of 1 of these potentially “reversible” conditions are, in fact, cured of AF after effective treatment or elimination of the condition. Because long-term follow-up data are not available in these clinical scenarios and AF may recur, these patients should receive careful follow-up.

See [Online Data Supplements 1 and 2](#) for additional data on electrophysiological and pathophysiological mechanisms.

Table 5. Selected Risk Factors and Biomarkers for AF

Clinical Risk Factors	References
Increasing age	139
Hypertension	139
Diabetes mellitus	139
MI	139
VHD	139
HF	38,139
Obesity	140–142
Obstructive sleep apnea	142
Cardiothoracic surgery	137
Smoking	143
Exercise	144–146
Alcohol use	147–149
Hyperthyroidism	150–152
Increased pulse pressure	153
European ancestry	154
Family history	155
Genetic variants	156–159
ECG	
LVH	35
Echocardiographic	
LA enlargement	35,160
Decreased LV fractional shortening	35
Increased LV wall thickness	35
Biomarkers	
Increased CRP	86,161
Increased BNP	162,163

AF indicates atrial fibrillation; BNP, B-type natriuretic peptide; CRP, C-reactive protein; ECG, electrocardiographic; HF, heart failure; LA, left atrial; LV, left ventricular; LVH, left ventricular hypertrophy; MI, myocardial infarction; and VHD, valvular heart disease.

3. Clinical Evaluation: Recommendation

Class I

1. Electrocardiographic documentation is recommended to establish the diagnosis of AF. (*Level of Evidence: C*)

The diagnosis of AF in a patient is based on the patient's clinical history and physical examination and is confirmed by ECG, ambulatory rhythm monitoring (eg, telemetry, Holter monitor, event recorders), implanted loop recorders, pacemakers or defibrillators, or, in rare cases, by electrophysiological study. The clinical evaluations, including additional studies that may be required, are summarized in Appendix 4.

3.1. Basic Evaluation of the Patient With AF

3.1.1. Clinical History and Physical Examination

The initial evaluation of a patient with suspected or proven AF involves characterizing the pattern of the arrhythmia (paroxysmal, persistent, long-standing persistent, or permanent), determining its cause, defining associated cardiac and extracardiac disease, and assessing thromboembolic risk.

Symptoms, prior treatment, family history, and a review of associated conditions and potentially reversible risk factors as outlined in Table 5 should be recorded.

The physical examination suggests AF by the presence of an irregular pulse, irregular jugular venous pulsations, and variation in the intensity of the first heart sound or absence of a fourth sound previously heard during sinus rhythm. Physical examination may also disclose associated valvular heart disease or myocardial abnormalities. The pulse in atrial flutter is often regular and rapid, and venous oscillations may be visible in the jugular pulse.

3.1.2. Investigations

An ECG, or other electrocardiographic recording, is the essential tool for confirming AF. A chest radiograph should be done if pulmonary disease or HF is suspected and may also detect enlargement of the cardiac chambers. As part of the initial evaluation, all patients with AF should have a 2-dimensional transthoracic echocardiogram to detect underlying structural heart disease, assess cardiac function, and evaluate atrial size. Additional laboratory evaluation should include assessment of serum electrolytes and of thyroid, renal, and hepatic function, and a blood count.

Transesophageal Echocardiography (TEE): TEE is the most sensitive and specific technique to detect LA thrombi as a potential source of systemic embolism in AF and can be used to guide the timing of cardioversion or catheter ablation procedures (Section 6.1.1). TEE can also identify features associated with an increased risk of LA thrombus formation, including reduced LAA flow velocity, spontaneous LA contrast, and aortic atheroma. In 5% to 15% of patients with AF, TEE before planned cardioversion revealed an LA or LAA thrombus.^{164,165}

Electrophysiological Study: An electrophysiological study can be helpful when initiation of AF is due to a supraventricular tachycardia, such as AV node reentrant tachycardia, AV reentry involving an accessory pathway, or ectopic atrial tachycardia. Ablation of the supraventricular tachycardia may prevent or reduce recurrences of AF. Electrophysiological study is often warranted in patients with a delta wave on the surface ECG indicating pre-excitation. Some patients with AF also have atrial flutter that may benefit from treatment with radiofrequency catheter ablation. AF associated with rapid ventricular rates and a wide-complex QRS (aberrant conduction) may sometimes be mislabeled as ventricular tachycardia, and an electrophysiological study can help establish the correct diagnosis.

Additional Investigation of Selected Patients With AF: Plasma levels of B-type natriuretic peptide or N-terminal pro-B-type natriuretic peptide may be elevated in patients with paroxysmal and persistent AF in the absence of clinical HF, and levels decrease rapidly after restoration of sinus rhythm. A sleep study may be useful if sleep apnea is suspected.¹⁶⁶

3.1.3. Rhythm Monitoring and Stress Testing

Prolonged or frequent monitoring may be necessary to reveal episodes of asymptomatic AF. Electrocardiography, ambulatory rhythm monitoring (eg, telemetry, Holter monitor, and event recorders), and exercise testing can be useful to judge the adequacy of rate control. Patient-activated electrocardiographic event recorders can help assess the relation to symptoms, whereas auto-triggered event recorders may detect asymptomatic episodes.

Table 6. Summary of Recommendations for Risk-Based Antithrombotic Therapy

Recommendations	COR	LOE	References
Antithrombotic therapy based on shared decision making, discussion of risks of stroke and bleeding, and patient's preferences	I	C	N/A
Selection of antithrombotic therapy based on risk of thromboembolism	I	B	167–170
CHA ₂ DS ₂ -VASc score recommended to assess stroke risk	I	B	171–173
Warfarin recommended for mechanical heart valves and target INR intensity based on type and location of prosthesis	I	B	174–176
With prior stroke, TIA, or CHA ₂ DS ₂ -VASc score ≥2, oral anticoagulants recommended. Options include:			
Warfarin	I	A	171–173
Dabigatran, rivaroxaban, or apixaban	I	B	177–179
With warfarin, determine INR at least weekly during initiation of therapy and monthly when stable	I	A	180–182
Direct thrombin or factor Xa inhibitor recommended if unable to maintain therapeutic INR	I	C	N/A
Reevaluate the need for anticoagulation at periodic intervals	I	C	N/A
Bridging therapy with UFH or LMWH recommended with a mechanical heart valve if warfarin is interrupted. Bridging therapy should balance risks of stroke and bleeding	I	C	N/A
For patients without mechanical heart valves, bridging therapy decisions should balance stroke and bleeding risks against duration of time patient will not be anticoagulated	I	C	N/A
Evaluate renal function before initiation of direct thrombin or factor Xa inhibitors, and reevaluate when clinically indicated and at least annually	I	B	183–185
For atrial flutter, antithrombotic therapy is recommended as for AF	I	C	N/A
With nonvalvular AF and CHA ₂ DS ₂ -VASc score of 0, it is reasonable to omit antithrombotic therapy	IIa	B	183, 184
With CHA ₂ DS ₂ -VASc score ≥2 and end-stage CKD (CrCl <15 mL/min) or on hemodialysis, it is reasonable to prescribe warfarin for oral anticoagulation	IIa	B	185
With nonvalvular AF and a CHA ₂ DS ₂ -VASc score of 1, no antithrombotic therapy or treatment with oral anticoagulant or aspirin may be considered	IIb	C	N/A
With moderate-to-severe CKD and CHA ₂ DS ₂ -VASc scores ≥2, reduced doses of direct thrombin or factor Xa inhibitors may be considered	IIb	C	N/A
For PCI,* BMS may be considered to minimize duration of DAPT	IIb	C	N/A
After coronary revascularization in patients with CHA ₂ DS ₂ -VASc score ≥2, it may be reasonable to use clopidogrel concurrently with oral anticoagulants but without aspirin	IIb	B	186
Direct thrombin dabigatran and factor Xa inhibitor rivaroxaban are not recommended in patients with AF and end-stage CKD or on dialysis because of a lack of evidence from clinical trials regarding the balance of risks and benefits	III: No Benefit	C	177–179, 187–189
Direct thrombin inhibitor dabigatran should not be used with a mechanical heart valve	III: Harm	B	190

*See the 2011 PCI guideline for type of stent and duration of DAPT recommendations.¹²

AF indicates atrial fibrillation; BMS, bare-metal stent; CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age ≥75 years (doubled), Diabetes mellitus, Prior Stroke or TIA or thromboembolism (doubled), Vascular disease, Age 65 to 74 years, Sex category; CKD, chronic kidney disease; COR, Class of Recommendation; CrCl, creatinine clearance; DAPT, dual antiplatelet therapy; INR, international normalized ratio; LMWH, low-molecular-weight heparin; LOE, Level of Evidence; N/A, not applicable; PCI, percutaneous coronary intervention; TIA, transient ischemic attack; and UFH, unfractionated heparin.

These technologies may also provide valuable information to guide drug dosage for rate control or rhythm management.

4. Prevention of Thromboembolism

4.1. Risk-Based Antithrombotic Therapy: Recommendations

See Table 6 for a summary of recommendations from this section.

Class I

1. In patients with AF, antithrombotic therapy should be individualized based on shared decision making after discussion of the absolute risks and RRs of stroke and bleeding and the patient's values and preferences. (*Level of Evidence: C*)

2. Selection of antithrombotic therapy should be based on the risk of thromboembolism irrespective of whether the AF pattern is paroxysmal, persistent, or permanent.^{167–170} (*Level of Evidence: B*)
3. In patients with nonvalvular AF, the CHA₂DS₂-VASc* score is recommended for assessment of stroke risk.^{171–173} (*Level of Evidence: B*)
4. For patients with AF who have mechanical heart valves, warfarin is recommended, and the target international normalized ratio (INR) intensity (2.0 to 3.0 or 2.5 to 3.5) should be based on the

*CHA₂DS₂-VASc indicates Congestive heart failure, Hypertension, Age ≥75 years (doubled), Diabetes mellitus, Prior Stroke or TIA or thromboembolism (doubled), Vascular disease, Age 65 to 74 years, Sex category.

type and location of the prosthesis.^{174–176} (*Level of Evidence: B*)

5. For patients with nonvalvular AF with prior stroke, transient ischemic attack (TIA), or a CHA₂DS₂-VASc score of 2 or greater, oral anticoagulants are recommended. Options include warfarin (INR 2.0 to 3.0)^{171–173} (*Level of Evidence: A*), dabigatran¹⁷⁷ (*Level of Evidence: B*), rivaroxaban¹⁷⁸ (*Level of Evidence: B*), or apixaban.¹⁷⁹ (*Level of Evidence: B*)
6. Among patients treated with warfarin, the INR should be determined at least weekly during initiation of antithrombotic therapy and at least monthly when anticoagulation (INR in range) is stable.^{180–182} (*Level of Evidence: A*)
7. For patients with nonvalvular AF unable to maintain a therapeutic INR level with warfarin, use of a direct thrombin or factor Xa inhibitor (dabigatran, rivaroxaban, or apixaban) is recommended. (*Level of Evidence: C*)
8. Reevaluation of the need for and choice of antithrombotic therapy at periodic intervals is recommended to reassess stroke and bleeding risks. (*Level of Evidence: C*)
9. Bridging therapy with unfractionated heparin (UFH) or low-molecular-weight heparin (LMWH) is recommended for patients with AF and a mechanical heart valve undergoing procedures that require interruption of warfarin. Decisions on bridging therapy should balance the risks of stroke and bleeding. (*Level of Evidence: C*)
10. For patients with AF without mechanical heart valves who require interruption of warfarin or new anticoagulants for procedures, decisions about bridging therapy (LMWH or UFH) should balance the risks of stroke and bleeding and the duration of time a patient will not be anticoagulated. (*Level of Evidence: C*)
11. Renal function should be evaluated before initiation of direct thrombin or factor Xa inhibitors and should be reevaluated when clinically indicated and at least annually.^{183–185} (*Level of Evidence: B*)
12. For patients with atrial flutter, antithrombotic therapy is recommended according to the same risk profile used for AF. (*Level of Evidence: C*)

Class IIa

1. For patients with nonvalvular AF and a CHA₂DS₂-VASc score of 0, it is reasonable to omit antithrombotic therapy.^{183,184} (*Level of Evidence: B*)
2. For patients with nonvalvular AF with a CHA₂DS₂-VASc score of 2 or greater and who have end-stage chronic kidney disease (CKD) (creatinine clearance [CrCl] <15 mL/min) or are on hemodialysis, it is reasonable to prescribe warfarin (INR 2.0 to 3.0) for oral anticoagulation.¹⁸⁵ (*Level of Evidence: B*)

Class IIb

1. For patients with nonvalvular AF and a CHA₂DS₂-VASc score of 1, no antithrombotic therapy or treatment with an oral anticoagulant or aspirin may be considered. (*Level of Evidence: C*)

2. For patients with nonvalvular AF and moderate-to-severe CKD with CHA₂DS₂-VASc scores of 2 or greater, treatment with reduced doses of direct thrombin or factor Xa inhibitors may be considered (eg, dabigatran, rivaroxaban, or apixaban), but safety and efficacy have not been established. (*Level of Evidence: C*)
3. In patients with AF undergoing percutaneous coronary intervention,[†] bare-metal stents may be considered to minimize the required duration of dual antiplatelet therapy. Anticoagulation may be interrupted at the time of the procedure to reduce the risk of bleeding at the site of peripheral arterial puncture. (*Level of Evidence: C*)
4. Following coronary revascularization (percutaneous or surgical) in patients with AF and a CHA₂DS₂-VASc score of 2 or greater, it may be reasonable to use clopidogrel (75 mg once daily) concurrently with oral anticoagulants but without aspirin.¹⁸⁶ (*Level of Evidence: B*)

Class III: No Benefit

1. The direct thrombin inhibitor dabigatran and the factor Xa inhibitor rivaroxaban are not recommended in patients with AF and end-stage CKD or on dialysis because of the lack of evidence from clinical trials regarding the balance of risks and benefits.^{177–179,187–189} (*Level of Evidence: C*)

Class III: Harm

1. The direct thrombin inhibitor dabigatran should not be used in patients with AF and a mechanical heart valve.¹⁹⁰ (*Level of Evidence: B*)

4.1.1. Selecting an Antithrombotic Regimen—Balancing Risks and Benefits

AF, whether paroxysmal, persistent, or permanent and whether symptomatic or silent, significantly increases the risk of thromboembolic ischemic stroke.^{191–194} Nonvalvular AF increases the risk of stroke 5 times, and AF in the setting of mitral stenosis increases the risk of stroke 20 times¹⁹⁵ over that of patients in sinus rhythm. Thromboembolism occurring with AF is associated with a greater risk of recurrent stroke, more severe disability, and mortality.¹⁹⁶ Silent AF is also associated with ischemic stroke.^{191–194} The appropriate use of antithrombotic therapy and the control of other risk factors, including hypertension and hypercholesterolemia, substantially reduce stroke risk.

Antithrombotic agents routinely used for the prevention of thromboembolism in patients with nonvalvular AF include anticoagulant drugs (UFH and LMWH, warfarin, and direct thrombin and factor Xa inhibitors) and antiplatelet drugs (aspirin and clopidogrel). Although anticoagulants have been effective in reducing ischemic stroke in multiple randomized controlled trials (RCTs), their use is associated with an increased risk of bleeding, ranging from minor bleeding to fatal intracranial or extracranial hemorrhage. Platelet inhibitors (alone or in combination) are less effective than warfarin, are better tolerated by some patients, and are associated with a lower risk of intracerebral hemorrhage. However, they have similar overall rates of major bleeding in

[†]See the 2011 percutaneous coronary intervention guideline for type of stent and duration of dual antiplatelet therapy recommendations.¹²

Table 7. Comparison of the CHADS₂ and CHA₂DS₂-VASc Risk Stratification Scores for Subjects With Nonvalvular AF

Definition and Scores for CHADS ₂ and CHA ₂ DS ₂ -VASc		Stroke Risk Stratification With the CHADS ₂ and CHA ₂ DS ₂ -VASc Scores	
	Score		Adjusted Stroke Rate (% per y)
CHADS ₂		CHADS ₂ *	
Congestive HF	1	0	1.9
Hypertension	1	1	2.8
Age ≥75 y	1	2	4.0
Diabetes mellitus	1	3	5.9
Stroke/TIA/TE	2	4	8.5
Maximum score	6	5	12.5
		6	18.2
CHA ₂ DS ₂ -VASc		CHA ₂ DS ₂ -VASc†	
Congestive HF	1	0	0
Hypertension	1	1	1.3
Age ≥75 y	2	2	2.2
Diabetes mellitus	1	3	3.2
Stroke/TIA/TE	2	4	4.0
Vascular disease (prior MI, PAD, or aortic plaque)	1	5	6.7
Age 65–74 y	1	6	9.8
Sex category (ie, female sex)	1	7	9.6
Maximum score	9	8	6.7
		9	15.20

*These adjusted stroke rates are based on data for hospitalized patients with AF and were published in 2001.²⁰¹ Because stroke rates are decreasing, actual stroke rates in contemporary nonhospitalized cohorts might vary from these estimates.

†Adjusted stroke rate scores are based on data from Lip and colleagues.^{15,30,171,202,203} Actual rates of stroke in contemporary cohorts might vary from these estimates.

AF indicates atrial fibrillation; CHADS₂, Congestive heart failure, Hypertension, Age ≥75 years, Diabetes mellitus, Prior Stroke or TIA or Thromboembolism (doubled); CHA₂DS₂-VASc, Congestive heart failure, Hypertension, Age ≥75 years (doubled), Diabetes mellitus, Prior Stroke or TIA or thromboembolism (doubled), Vascular disease, Age 65 to 74 years, Sex category; HF, heart failure; MI, myocardial infarction; PAD, peripheral artery disease; TE, thromboembolism; and TIA, transient ischemic attack.^{202,203}

some studies.^{184,189,197–199} Careful consideration is required to balance the benefits and risks of bleeding in each individual patient.

4.1.1.1. Risk Stratification Schemes (CHADS₂, CHA₂DS₂-VASc, and HAS-BLED)

One meta-analysis has stratified ischemic stroke risk among patients with nonvalvular AF using the following point scoring systems: AF Investigators²⁰⁰; CHADS₂ (Congestive heart failure, Hypertension, Age ≥75 years, Diabetes mellitus, Prior Stroke or TIA or Thromboembolism [doubled]),²⁰¹ or CHA₂DS₂-VASc (Congestive heart failure, Hypertension, Age ≥75 years [doubled], Diabetes mellitus, Prior Stroke or TIA or thromboembolism [doubled], Vascular disease, Age 65 to 74 years, Sex category) (Table 7).

The CHADS₂ score has been validated in multiple nonvalvular AF cohorts, with findings indicating an increase in

stroke rate of approximately 2.0% for each 1-point increase in CHADS₂ score (from 1.9% with a score of 0 to 18.2% with a score of 6).^{201,204} A limitation of the CHADS₂ score is that a score of 1 is considered an “intermediate” risk and those at lowest risk may not be well identified. Furthermore, patients whose only risk factor is a CHADS₂ score of 2 due to prior stroke may have a greater risk than a score of 2 would indicate.

When compared with the CHADS₂ score, the CHA₂DS₂-VASc score¹⁵ for nonvalvular AF has a broader score range (0 to 9) and includes a larger number of risk factors (female sex, 65 to 74 years of age, and vascular disease).^{202,203} In this scheme, women cannot achieve a CHA₂DS₂-VASc score of 0. In a nationwide Danish registry from 1997 to 2008, the CHA₂DS₂-VASc index better discriminated stroke risk among subjects with a baseline CHADS₂ score of 0 to 1 with an improved predictive ability.¹⁷² In another study among patients with AF, the CHA₂DS₂-VASc score more clearly defined anticoagulation recommendations than the CHADS₂ score did.¹⁷³ More patients, particularly older women, were redistributed from the low- to high-risk categories. In a study of Swedish patients with nonvalvular AF, women again had a moderately increased stroke risk compared with men; however, women younger than 65 years of age and without other AF risk factors had a low risk for stroke, and it was concluded that anticoagulant treatment was not required.²⁰⁵ However, the continued evolution of AF-related thromboembolic risk evaluation is needed.

Bleeding risk scores to quantify hemorrhage risk include HAS-BLED (Hypertension, Abnormal renal/liver function, Stroke, Bleeding history or predisposition, Labile INR, Elderly, Drugs/alcohol concomitantly), RIETE (Computerized Registry of Patients With Venous Thromboembolism), and ATRIA (Anticoagulation and Risk Factors in Atrial Fibrillation).^{206–208} Although these scores may be helpful in defining patients at elevated bleeding risk, their clinical utility is insufficient for use as evidence for the recommendations in this guideline. The RIETE score was developed from a large venous thromboembolism cohort and includes 2 points for recent bleeding, 1.5 points for abnormal creatinine levels or anemia, and 1 point for each of the following: >75 years of age, cancer, or pulmonary embolism at baseline. The ATRIA score assigns points to the following variables: anemia, 3; severe renal disease, 3; >75 years of age, 2; prior hemorrhage, 1; and hypertension, 1.

HAS-BLED^{14,30} is a score based on the presence of hypertension (systolic blood pressure >160 mm Hg), abnormal liver or renal function, history of stroke or bleeding, labile INRs, elderly age (>65 years), use of drugs that promote bleeding, or alcohol excess.²⁰⁹ A score ≥3 indicates potentially “high risk” for bleeding and may require closer observation of a patient for adverse risks, closer monitoring of INRs, or differential dose selections of oral anticoagulants or aspirin. HAS-BLED is better at discriminating risk than the HEMORR2HAGES or ATRIA scoring systems are, but all 3 scores had C indexes <0.70 in the receiver operating curves, indicating only modest performance and poor predictive accuracy.²¹⁰

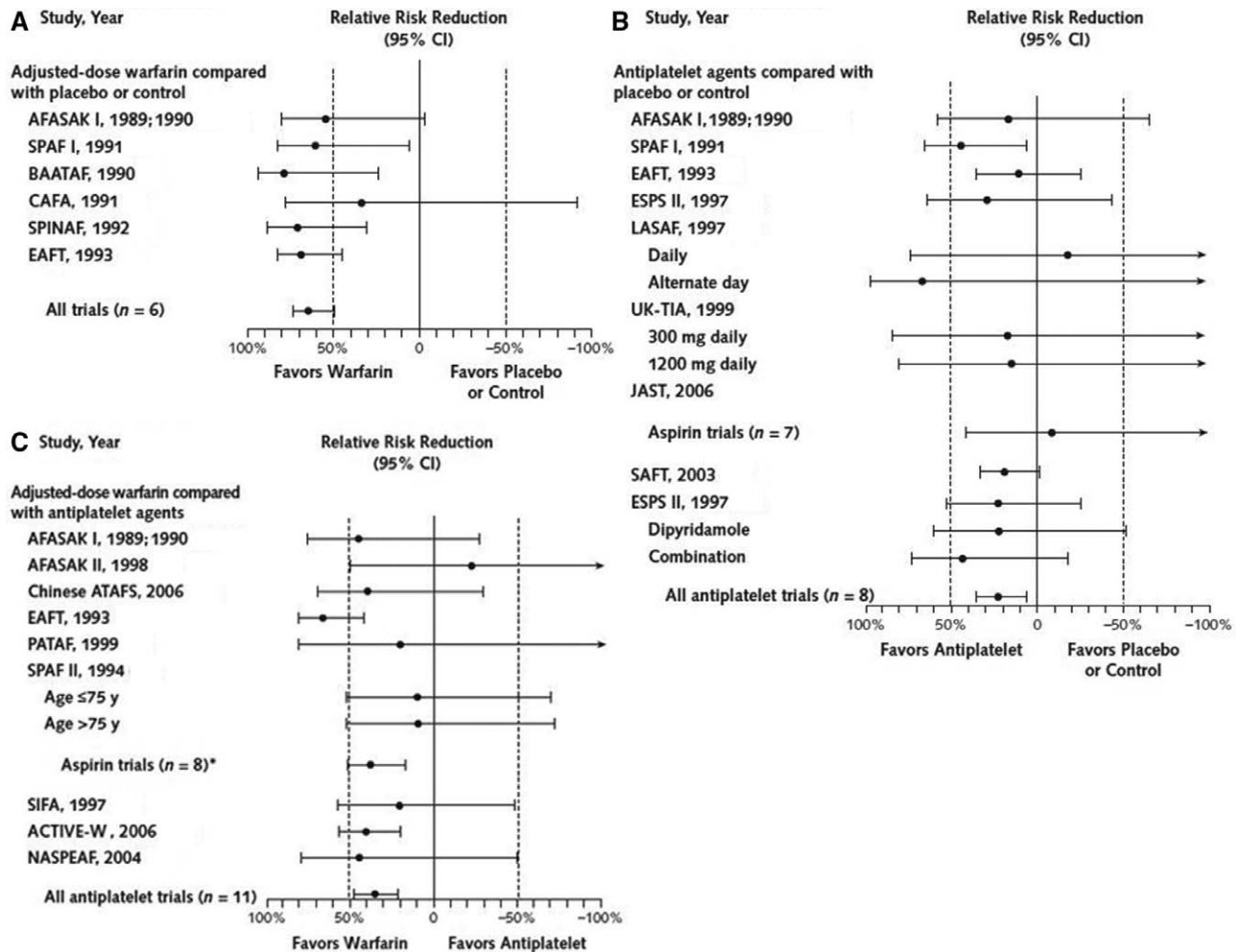


Figure 3. Antithrombotic therapy to prevent stroke in patients who have nonvalvular AF (meta-analysis). ACTIVE-W indicates Atrial Fibrillation Clopidogrel Trial With Irbesartan for Prevention of Vascular Events-W; AF, atrial fibrillation; AFASAK, Atrial Fibrillation, Aspirin and Anticoagulant Therapy Study; ATAFS, Antithrombotic Therapy in Atrial Fibrillation Study; BAATAF, Boston Area Anticoagulation Trial for Atrial Fibrillation; CAFA, Canadian Atrial Fibrillation Anticoagulation; CI, confidence interval; EAF, European Atrial Fibrillation Trial; ESPS, European Stroke Prevention Study; JAST, Japan AF Stroke Prevention Trial; LASAF, Low-Dose Aspirin, Stroke, Atrial Fibrillation; NASPEAF, National Study for Prevention of Embolism in Atrial Fibrillation; PATAF, Primary Prevention of Arterial Thromboembolism in Nonrheumatic Atrial Fibrillation; SAFT, Swedish Atrial Fibrillation Trial; SIFA, Studio Italiano Fibrillazione Atriale; SPAF, Stroke Prevention in Atrial Fibrillation Study; SPINAF, Stroke Prevention in Atrial Fibrillation; and UK-TIA, United Kingdom-Transient Ischemic Attack. Adapted with permission from Hart et al.¹⁸⁴

4.2. Antithrombotic Options

Antithrombotic medications prevent strokes and systemic emboli among patients with AF in part by reducing the formation of platelet-rich or thrombotic clots in the LA or LAA, from which the clots can embolize through the systemic circulation to the brain or other sites. Stroke prevention trials (Figure 3) compared warfarin or aspirin with placebo and compared aspirin with warfarin or clopidogrel and aspirin. Warfarin was also compared with dual antiplatelet agents (clopidogrel and aspirin). Trials have also compared direct thrombin inhibitors and factor Xa inhibitors with warfarin and, in 1 case, with aspirin. Both primary and secondary stroke prevention have been evaluated. The selection of an antithrombotic agent should be based on shared decision making that takes into account risk factors, cost, tolerability, patient preference, potential for drug interactions, and other clinical characteristics, including time in the INR therapeutic range if the patient has been on warfarin, irrespective

of whether the AF pattern is paroxysmal, persistent, or permanent.

Meta-analyses have summarized the effect of antithrombotic therapies for stroke prevention in nonvalvular AF. The largest meta-analysis identified 29 RCTs from 1996 to 2007 that tested antithrombotic therapies of >12 weeks' duration among 28 044 patients.¹⁸⁴ Nine trials were double-blind designs with a mean follow-up of 1.5 years per patient. The average age of the subjects was 71 years, and 35% of the subjects were women. Among 12 of the trials, 3003 subjects were randomized to placebo or control, with an average stroke rate of 4.1% per year among the primary prevention studies and 13% per year among those with prior stroke or TIA.

4.2.1. Antiplatelet Agents

No studies, with the exception of the SPAF (Stroke Prevention in Atrial Fibrillation)-1 trial, show benefit for aspirin alone in preventing stroke among patients with AF.^{183,184,211} Antiplatelet

therapy was compared with placebo or no treatment in 8 trials with a total of 4876 subjects¹⁸⁴ (Figure 3). Seven of these 8 trials compared different doses of aspirin ranging from 25 mg twice a day to 1300 mg once a day.¹⁸⁴ For primary prevention, aspirin was associated with a 19% reduction (95% CI: -1% to 35%) in stroke incidence with an absolute risk reduction of 0.8% per year (number needed to treat: 125). The 95% CI encompassed 0, which includes the possibility that aspirin has no real effect on stroke reduction. For secondary prevention among those with TIA or strokes, aspirin was associated with an absolute risk reduction of 2.5% per year and a corresponding number needed to treat of 40. It is important to recognize that the 19% reduction in stroke incidence observed in this meta-analysis was driven by positive results from only 1 of these RCTs—the SPAF-1 trial. In this trial, aspirin was prescribed at 325 mg once daily, and the impact of aspirin was very heterogeneous between groups. Aspirin was ineffective in preventing strokes in those >75 years of age and did not prevent severe strokes. Moreover, aspirin has not been studied in a population at low risk of AF.

Clopidogrel plus aspirin was evaluated for stroke prevention in the ACTIVE (Atrial Fibrillation Clopidogrel Trial With Irbesartan for Prevention of Vascular Events)-W trial.¹⁹⁸ This trial was terminated early (before planned follow-up was completed) on the recommendation of the Data Safety and Monitoring Board because the combination of antiplatelet agents, clopidogrel (75 mg once daily) plus aspirin (75 mg to 100 mg once daily), proved inferior to warfarin (target INR 2.0 to 3.0) in patients with a mean CHADS₂ score of 2. ACTIVE-W found a 40% RR reduction (95% CI: 18% to 56%; $P<0.001$) for stroke with warfarin compared with the dual antiplatelet regimen. ACTIVE-A compared clopidogrel combined with aspirin versus aspirin alone in patients with AF who were unsuitable candidates for oral anticoagulation and who had ≥ 1 additional stroke risk factor.¹⁹⁹ The combination of clopidogrel and aspirin resulted in a 28% RR reduction (95% CI: 17% to 38%; $P<0.0002$) in all strokes compared with aspirin alone. Major bleeding was significantly greater with the combination and increased by 57% (95% CI: 29% to 92%; $P<0.001$). The absolute differences between the treatment arms were small, with major vascular events decreased by 0.8% per year and major hemorrhages increased by 0.7% per year. The results of ACTIVE-W and ACTIVE-A demonstrate that adjusted-dose warfarin for stroke prevention is significantly better than clopidogrel plus aspirin, and clopidogrel plus aspirin is superior to aspirin alone. The latter benefits are dampened by the significant increase in major bleeding events. No direct comparisons have been made between clopidogrel and aspirin and the new oral anticoagulants that have lower bleeding risks than warfarin. However, there is a direct comparison between aspirin and the factor Xa inhibitor apixaban in the AVERROES (Apixaban Versus Acetylsalicylic Acid to Prevent Strokes) study, a double-blind study of 5599 patients deemed unsuitable for warfarin therapy.¹⁸⁹ Subjects were randomized to apixaban 5 mg twice daily (2.5 mg twice daily for those who had 2 of the following 3 factors: age ≥ 80 years, weight ≤ 60 kg, serum creatinine ≥ 1.5 mg/dL) or to aspirin 81 mg or 325 mg once daily. The primary outcome of the study was the occurrence of a stroke or systemic embolism. After a

mean follow-up of 1.1 years, the study was prematurely terminated owing to the superiority of apixaban over aspirin for preventing the primary outcome. Major bleeding risk between the 2 treatments was similar.

4.2.2. Oral Anticoagulants

See [Online Data Supplement 3](#) for additional data and evidence tables on warfarin versus aspirin and the new oral anticoagulants.

4.2.2.1. Warfarin

Warfarin is a vitamin K antagonist in use since the 1950s as an oral anticoagulant for stroke prevention in patients with AF. Its multiple sites of action in the coagulation cascade are shown in Figure 4. Among 6 RCTs of 2900 subjects in which adjusted-dose warfarin was compared with placebo or no treatment, the mean INR ranged from 2.0 to 2.9.^{184,212} Adjusted-dose warfarin resulted in a 64% RR reduction (95% CI: 49% to 74%) for ischemic and hemorrhagic stroke compared with placebo. The absolute risk reduction was 2.7% per year, which yielded a number needed to treat of 37 for 1 year to prevent 1 stroke and 12 for patients with prior stroke or TIA.¹⁸⁴

A Cochrane Collaboration review of warfarin versus placebo among subjects without prior cerebral events found that warfarin was associated with a significant risk reduction in all strokes, ischemic stroke, and the combined endpoint of stroke, MI, or vascular death.²¹⁴ With an ischemic stroke rate of 4% per year in the control group, the absolute risk reduction was about 2.6% per year for those with no prior stroke or TIA, or about 25 ischemic strokes prevented in 1 year per 1000 subjects treated with warfarin. The RR reductions were consistent across the trials. Intracranial hemorrhage was not significantly increased among the subjects randomized to warfarin, but the patient numbers were small and the CI wide.

For nonvalvular AF, 2 separate Cochrane Collaboration reviews evaluated the efficacy and safety of oral anticoagulants compared with antiplatelet agents.^{215,216} One review included those with no history of stroke or TIA and the other those with a history of stroke or TIA. Among 9598 subjects with AF, the majority (90%) of whom had no prior stroke or TIA, oral anticoagulants were associated with a significant reduction in all strokes and ischemic strokes compared with antiplatelet agents. Assuming an absolute stroke risk of 4% per year with antiplatelet agents, approximately 19 strokes could be prevented per year for every 1000 patients with AF treated with oral anticoagulants. The risk of intracranial hemorrhage was significantly increased among those treated with oral anticoagulants, but major extracranial hemorrhages were not significantly different. After excluding the ACTIVE-W trial, which used clopidogrel and aspirin as the antiplatelet agent comparison, oral anticoagulants were significantly associated with an increased risk of bleeding (OR: 1.90; 95% CI: 1.07 to 3.39).²¹⁵ Similarly, among patients with a prior history of stroke or TIA, oral anticoagulants compared with antiplatelet agents were associated with significant reductions in all major vascular events and recurrent stroke. Bleeding risks—including for any intracranial bleeds and major extracranial bleeds—were increased with oral anticoagulants.

The BAFTA (Birmingham Atrial Fibrillation Treatment of the Aged) study also evaluated the efficacy of warfarin among

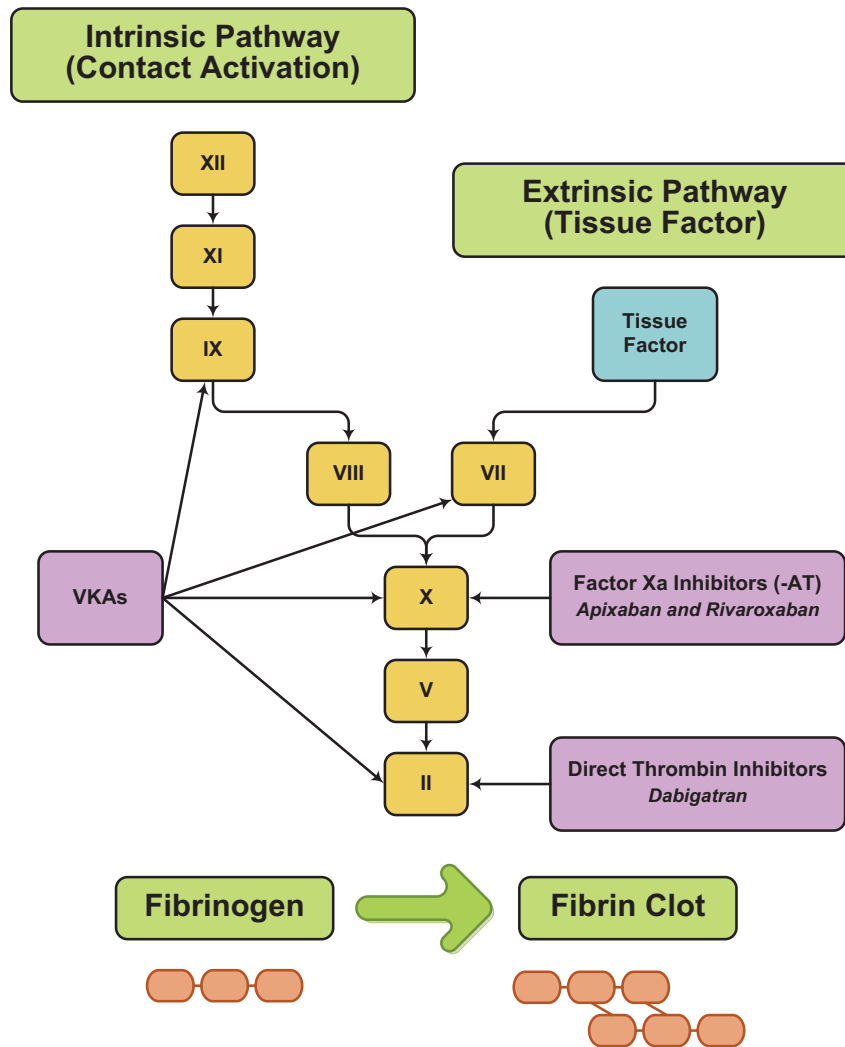


Figure 4. Coagulation cascade. AT indicates antithrombin and VKAs, vitamin K antagonists. Adapted with permission from Nutescu et al.²¹³

higher-risk elderly subjects >75 years of age.¹⁹⁷ BAFTA was designed to compare warfarin with aspirin for the prevention of fatal and nonfatal stroke, intracranial hemorrhage, and other clinically significant arterial embolism in a primary care population of patients ≥ 75 years of age who had AF. Warfarin was superior in preventing stroke or systemic embolism without a significant increase in bleeding risk. The annual risk of extracranial hemorrhage was 1.4% in the warfarin group and 1.6% in the aspirin group.

Despite strong evidence for the efficacy of warfarin, several limitations have led to it being underused.^{217–221} The narrow therapeutic window and increased risk of bleeding, including in the brain, have hindered broader use, especially among the elderly. Interactions with other drugs, effects of alterations in diet, and the requirement for close monitoring with frequent blood tests have also made the dosing of warfarin challenging for clinicians and patients. Even in well-conducted clinical trials, the time in therapeutic range (TTR) of those taking warfarin was reported as 55% to 66%,^{177–179} whereas in some community settings, TTR has been reported as approximately 50%.^{222,223} Despite underutilization of warfarin among eligible patients due to a variety of factors,^{217–221} a meta-analysis of contemporary studies found

risk of stroke or systemic embolism estimated to be at 1.66% per year for warfarin in patients with AF²²⁴ (Figure 5).

See *Online Data Supplements 4 and 5* for additional data on warfarin and antiplatelet therapy.

4.2.2.2. New Target-specific Oral Anticoagulants

Dabigatran is the first new oral anticoagulant approved by the US Food and Drug Administration (FDA) to reduce the risk of stroke and systemic embolism in patients with nonvalvular AF and is a direct thrombin inhibitor. Its site of action in the coagulation cascade is shown in Figure 4. Dabigatran was compared with warfarin in the RE-LY (Randomized Evaluation of Long-Term Anticoagulation Therapy) trial, which was an open-label randomized comparison of dabigatran (110 mg or 150 mg twice daily in a blinded fashion) with adjusted-dose warfarin in 18 113 patients over a median follow-up period of 2 years.¹⁷⁷ The mean CHADS₂ score was 2.1 and the primary outcome was stroke (of any type) and systemic embolism, with any major hemorrhage being the primary safety outcome. Half of the patients were naïve to oral anticoagulants. The mean TTR for those randomized to warfarin was 64%. The primary outcome was assessed first for noninferiority followed by superiority.

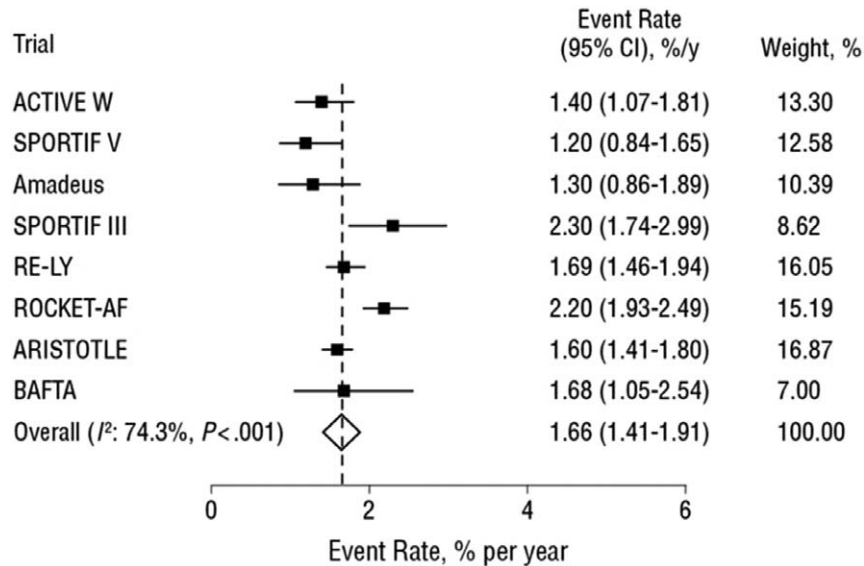


Figure 5. Pooled estimates of stroke or systemic embolism in patients with AF treated with warfarin. ACTIVE W indicates Atrial Fibrillation Clopidogrel Trial With Irbesartan for Prevention of Vascular Events-W; AF, atrial fibrillation; Amadeus, Evaluating the Use of SR34006 Compared to Warfarin or Acenocoumarol in Patients With Atrial Fibrillation; ARISTOTLE, Apixaban Versus Warfarin in Patients With AF; BAFTA, Birmingham Atrial Fibrillation Treatment of the Aged Study; CI, confidence interval; RE-LY, Randomized Evaluation of Long-Term Anticoagulation Therapy; ROCKET AF, Rivaroxaban Versus Warfarin in Nonvalvular Atrial Fibrillation; and SPORTIF, Stroke Prevention Using Oral Thrombin Inhibitor in Atrial Fibrillation. Adapted with permission from Agarwal et al.²²⁴

For the primary outcomes, dabigatran 150 mg twice daily was superior to warfarin, and dabigatran 110 mg twice daily was noninferior to warfarin. When dabigatran was compared with warfarin, the risk of hemorrhagic strokes was also significantly lower (74% lower) with both the 110 mg and 150 mg doses. Major bleeding was significantly decreased with the 110 mg dose but not with the 150 mg dose. Both doses had lower rates of intracranial bleeding and life-threatening bleeding, whereas gastrointestinal bleeding was higher in the 150 mg dose (1.6% versus 1.0% per year) group. Dyspepsia was more frequent for both doses. For secondary prevention of stroke, the results were similar to the primary analysis but statistically weaker because of smaller sample size.²²⁵

Dabigatran is renally excreted, and patients with a CrCl of <30 mL/min were excluded from the RE-LY trial. CKD is associated with increased bleeding risk during both dabigatran therapy and warfarin therapy.²²⁶ The FDA approved the higher dose of 150 mg twice daily but not the lower dose of 110 mg twice daily. The FDA also approved a dose of 75 mg twice daily for those with low CrCl (15 mL/min to 30 mL/min) based on pharmacological modeling, but that dose was never clinically studied.

The RE-LY trial included subjects distributed equally across stroke risk strata (CHADS₂ score 0 to 1 in 31.8% of subjects, 2 in 35.6%, and >2 in 32.5%). For the primary efficacy endpoint and intracranial bleeding, there was similar efficacy across the range of CHADS₂ scores.¹⁷⁷ In patients <75 years of age, both doses of dabigatran were associated with less intracranial and extracranial bleeding than warfarin; in patients ≥75 years of age, both doses reduced intracranial bleeding. However, extracranial bleeding was similar or more frequent compared with warfarin.²²⁷ Higher CHADS₂ scores were associated with increased risks for stroke or systemic embolism, bleeding,

and death in patients with AF receiving oral anticoagulants.²²⁸ The benefits of dabigatran compared with warfarin in terms of efficacy and safety were similar in patient groups with paroxysmal, persistent, and permanent AF.¹⁶⁹ An FDA postmarket analysis of gastrointestinal and intracranial bleeding of dabigatran compared with warfarin indicates that bleeding rates do not appear to be higher for dabigatran.²²⁹

A post hoc analysis of 1989 electrical cardioversions found a very low rate of stroke within 30 days after the procedure (0.6% for warfarin, 0.3% for dabigatran 150 mg twice daily, and 0.8% for dabigatran 110 mg twice daily).²³⁰ Most subjects were treated with their assigned medication for ≥3 weeks before cardioversion. TEE was performed in 25% of subjects. There was no significant difference in the incidence of LAA thrombus (1.1% for warfarin; and for dabigatran, 1.2% for 150 mg twice daily and 1.8% for 110 mg twice daily).²³⁰

In the RE-LY trial, there appeared to be an imbalance of MIs: 0.8%, 0.8%, and 0.6% per year for patients randomized to dabigatran 150 mg twice daily, dabigatran 110 mg twice daily, or warfarin, respectively ($P=0.09$).⁷² Absolute events were low in a population in which 31% of randomized patients had objective evidence of CAD. A meta-analysis of a dabigatran RCT found a statistically significant increase in risk of MI and acute coronary syndromes (ACSs) in patients randomized to dabigatran.²³¹ Interpretation of these results should be made with caution, given the multiple limitations of this type of analysis, which include the use of different controls and different patient populations.

Rivaroxaban is the second new oral anticoagulant approved by the FDA for reduction of risk of stroke and systemic embolism in patients with nonvalvular AF and is a direct factor Xa inhibitor (Figure 4). It should be administered as a single daily

dose with the evening meal to ensure adequate absorption. It is predominantly excreted by the kidneys. The evidence leading to approval was based on the RCT ROCKET AF (Rivaroxaban Versus Warfarin in Nonvalvular Atrial Fibrillation), which compared rivaroxaban (20 mg once daily, 15 mg once daily if CrCl was 30 mL/min to 49 mL/min) with warfarin among 14 264 patients.¹⁷⁸ ROCKET AF differed from RE-LY in that it selected higher-risk patients with AF (≥ 2 risk factors for stroke compared with 1 risk factor). Patients in ROCKET AF were older and had a greater mean CHADS₂ score of 3.47. Similar to other AF trials, the primary outcome was any stroke or systemic embolism, and the primary hypothesis was noninferiority. Although the primary analysis was prespecified as a per-protocol analysis, the intention-to-treat analysis was also presented. The main safety outcome was clinically relevant bleeding events. This was a double-blind trial, and the patients receiving warfarin had a lower mean TTR of 55%. The trial demonstrated noninferiority for rivaroxaban compared with warfarin; however, in the intention-to-treat analysis, superiority was not achieved ($P=0.12$). Major bleeding was similar for rivaroxaban and warfarin, but less fatal bleeding and less intracranial hemorrhage were found for rivaroxaban. At the end of the trial, patients transitioning to open-label therapy had more strokes with rivaroxaban than with warfarin. However, the risk of stroke or noncentral nervous system embolism after elective temporary discontinuation of rivaroxaban compared with warfarin in the ROCKET AF trial did not differ significantly in a post hoc analysis.²³² The risk of stroke was similar for patients assigned to rivaroxaban and warfarin. In ROCKET AF, a decline in renal function was an independent predictor of stroke risk.

Apixaban is the third new oral anticoagulant approved by the FDA for reduction of risk of stroke and systemic embolism with nonvalvular AF and is another direct factor Xa inhibitor (Figure 4). It is predominantly eliminated hepatically and is highly protein bound. It has been investigated in 2 clinical trials. In the ARISTOTLE (Apixaban Versus Warfarin in Patients With Atrial Fibrillation) trial, apixaban (5 mg twice daily) was compared with warfarin in a double-blind RCT of 18 201 patients with AF and a mean CHADS₂ score of 2.1.¹⁷⁹ Apixaban 2.5 mg twice daily was used among patients with ≥ 2 of the following conditions: ≥ 80 years of age, weight ≤ 60 kg, or a serum creatinine level ≥ 1.5 mg/dL. As with the other new anticoagulant trials, the primary outcome was any stroke or systemic embolism and the primary safety outcome was major bleeding. Patients were followed for a mean of 1.8 years, and the mean age was 70 years. For patients treated with warfarin, the TTR was 62%. Apixaban was significantly better than warfarin, with fewer overall strokes (both ischemic and hemorrhagic), systemic emboli, and major bleeding events. Patients treated with apixaban had significantly fewer intracranial bleeds, but gastrointestinal bleeding complications were similar between the 2 study groups. Patients treated with apixaban had fewer deaths than those on warfarin. In ARISTOTLE, the benefit of apixaban was independent of type of AF, risk profile, CHADS₂ or CHA₂DS₂-VASc score, and whether there was a prior stroke.

Apixaban was also compared with aspirin in the AVERROES study, a double-blind study of 5599 patients deemed unsuitable for warfarin therapy¹⁸⁹ (Section 4.2). The mean CHADS₂ score was 2, and 36% of the subjects had a

CHADS₂ score of 0 to 1. After a mean follow-up of 1.1 years, the study was prematurely terminated due to the superiority of apixaban compared with aspirin for preventing the occurrence of any stroke or systemic embolism, whereas bleeding risk between the 2 treatments was similar.

Patients with severe and end-stage CKD (serum creatinine >2.5 mg/dL or CrCl <25 mL/min) were excluded from the ARISTOTLE and AVERROES trials.^{179,189} On the basis of new pharmacokinetic profiles in a limited data set,²³³ prescribing recommendations for apixaban were revised for use in patients with end-stage CKD who were maintained on stable hemodialysis with the recommended dose of 5 mg twice daily with a reduction in dose to 2.5 mg twice daily for either age ≥ 80 years or body weight ≤ 60 kg. For patients with severe or end-stage CKD not on dialysis, a dose recommendation was not provided. There are no published data for the use of apixaban in these clinical settings.

Other factor Xa inhibitors, including edoxaban²³⁴ and betrixaban,²³⁵ are under evaluation but not yet approved by the FDA.

4.2.2.3. Considerations in Selecting Anticoagulants

Selection of agents for antithrombotic therapy depends on a large number of variables, including clinical factors, clinician and patient preference, and, in some circumstances, cost. The new agents are currently considerably more expensive than warfarin. However, dietary limitations and the need for repeated INR testing are eliminated with the new agents. If patients are stable, their condition is easily controlled, and they are satisfied with warfarin therapy, it is not necessary to change to a new agent. However, it is important to discuss this option with patients who are candidates for the new agents.

All 3 new oral anticoagulants represent important advances over warfarin because they have more predictable pharmacological profiles, fewer drug–drug interactions, an absence of major dietary effects, and less risk of intracranial bleeding than warfarin. They have rapid onset and offset of action so that bridging with parenteral anticoagulant therapy is not needed during initiation, and bridging may not be needed in patients on chronic therapy requiring brief interruption of anticoagulation for invasive procedures. However, strict compliance with these new oral anticoagulants is critical. Missing even 1 dose could result in a period without protection from thromboembolism. As a result, the FDA issued black box warnings that discontinuation of these new agents can increase the risk of thromboembolism and that coverage with another anticoagulant may be needed. In addition, reversal agents, while in development, are not available, although the short half-lives lessen the need for an antidote. Although dose adjustments may be warranted for those with CKD or body weight extremes, these new agents do not require regular monitoring of INR or activated partial thromboplastin time.

Importantly, patients with mechanical heart valves or hemodynamically significant mitral stenosis were excluded from all 3 major trials (RE-LY, ROCKET AF, and ARISTOTLE)^{80,89,90}; therefore, these patients should be managed with warfarin. Patients with aortic stenosis or aortic insufficiency who, in the estimation of the local RCT principal investigator, would not need a surgical procedure before the conclusion of the trial were included. The RE-ALIGN (Randomized, Phase II Study to Evaluate the Safety and Pharmacokinetics of Oral Dabigatran

Table 8. Dose Selection of Oral Anticoagulant Options for Patients With Nonvalvular AF and CKD (Based on Prescribing Information for the United States)*

Renal Function	Warfarin ²³⁸	Dabigatran† ¹⁷⁷	Rivaroxaban† ¹⁷⁸	Apixaban† ¹⁷⁹
Normal/mild impairment	Dose adjusted for INR 2.0–3.0	150 mg BID (CrCl >30 mL/min)	20 mg QD with the evening meal (CrCl >50 mL/min)	5.0 or 2.5 mg BID‡
Moderate impairment	Dose adjusted for INR 2.0–3.0	150 mg BID (CrCl >30 mL/min)	15 mg QD with the evening meal (CrCl 30–50 mL/min)	5.0 or 2.5 mg BID‡
Severe impairment	Dose adjusted for INR 2.0–3.0§	75 mg BID (CrCl 15–30 mL/min)	15 mg QD with the evening meal (CrCl 15–30 mL/min)	No recommendation. See Section 4.2.2.2¶
End-stage CKD not on dialysis	Dose adjusted for INR 2.0–3.0§	Not recommended¶ (CrCl <15 mL/min)	Not recommended¶ (CrCl <15 mL/min)	No recommendation. See Section 4.2.2.2¶
End-stage CKD on dialysis	Dose adjusted for INR 2.0–3.0§	Not recommended¶ (CrCl <15 mL/min)	Not recommended¶ (CrCl <15 mL/min)	No recommendation. See Section 4.2.2.2¶#

*Renal function should be evaluated before initiation of direct thrombin or factor Xa inhibitors and should be reevaluated when clinically indicated and at least annually. CrCl should be measured using the Cockcroft-Gault method.

†The concomitant use of P-glycoprotein inducers or inhibitors with dabigatran or the concomitant use of dual P-glycoprotein and strong CYP3A4 inducers or inhibitors with either rivaroxaban or apixaban, particularly in the setting of CKD, may require dosing adjustment or avoidance of concomitant drug use (see the FDA drug label at http://www.accessdata.fda.gov/drugsatfda_docs/label/2014/202155s002lbl.pdf, Section 8.6).

‡Use apixaban 2.5 mg BID if any 2 patient characteristics are present: Cr $\geq$ 1.5 mg/dL, $\geq$ 80 y of age, body weight $\leq$ 60 kg.¹⁷⁹ Apixaban is not recommended in patients with severe hepatic impairment.

§Dose-adjusted warfarin has been used, but observational data on safety and efficacy are conflicting.

|| Modeling studies suggest that dabigatran 75 mg BID might be safe for patients with CrCl 15–30 mL/min, but this has not been validated in a prospective cohort. Some countries outside the United States use 110 mg BID.¹⁷⁷

¶No published studies support a dose for this level of renal function.

#In patients with end-stage CKD on stable hemodialysis, prescribing information indicates the use of apixaban 5 mg BID with dose reduction to 2.5 mg BID if the patient is $\geq$ 80 y of age or body weight is $\leq$ 60 kg.

AF indicates atrial fibrillation; BID, twice daily; CKD, chronic kidney disease; Cr, creatinine; CrCl, creatinine clearance; FDA, Food and Drug Administration; INR, international normalized ratio; and QD, once daily.

Etexilate in Patients After Heart Valve Replacement) trial, a phase 2 dose-range study of the use of dabigatran compared with warfarin in patients with mechanical heart valves, was stopped because dabigatran users were more likely to experience strokes, MI, and thrombus forming on the mechanical heart valves than were warfarin users.^{190,236,237} There was also more bleeding after valve surgery in the dabigatran users than in the warfarin users; thus, dabigatran is contraindicated for use in patients with mechanical heart valves. Similar drug safety and efficacy information is lacking for rivaroxaban and apixaban and mechanical heart valves. Bioprosthetic heart valves have not been studied with any of the new anticoagulants. None of the 3 major trials included pregnant or lactating women, children, patients with reversible causes of AF, or patients with severe hypertension (systolic blood pressure >180 mmHg or diastolic blood pressure >100 mmHg). Patients with a recent stroke (within 7 to 14 days), patients with significant liver disease, and complex patients with multiple chronic conditions were excluded from all trials.

For patients with CKD, dose modifications of the new agents are available (Table 8); however, for those with severe or end-stage CKD, warfarin remains the anticoagulant of choice, as there are no or very limited data for these patients. Among patients on hemodialysis, warfarin has been used with acceptable risks of hemorrhage.¹⁸⁵

The price of an effective anticoagulant is the risk of bleeding, which, if extracranial, is usually not life-threatening. Although INR and activated partial thromboplastin time increase with dabigatran, this is not in a linear fashion and cannot be used to monitor the level of anticoagulation. The Hemoclot (HYPHEN BioMed, Neuville-sur-Oise, France)

thrombin clotting time is a more accurate measure of anti-coagulation levels, but the test is not approved in the United States and is not widely available elsewhere.⁹⁴ If bleeding or overdose occurs, the anticoagulant agent should be discontinued. The use of activated charcoal to reduce absorption may be considered. Dabigatran is dialyzable, but apixaban and rivaroxaban are not, and both are highly plasma protein bound.

Dabigatran, rivaroxaban, and apixaban are substrates for the efflux transporter P-glycoprotein. P-glycoprotein inhibitors, such as ketoconazole, verapamil, amiodarone, dronedarone, quinidine, and clarithromycin, may increase plasma concentrations. In addition, P-glycoprotein inducers (such as phenytoin, carbamazepine, rifampin, and St. John's wort) can decrease levels of these drugs to subtherapeutic blood levels, and coadministration should be avoided. Absorbed dabigatran etexilate is “pumped” back into the intestinal tract. Dabigatran also can cause dyspepsia. Coprescription with a proton pump inhibitor can mildly reduce dabigatran peak concentrations; however, dose adjustment is not warranted.²³⁹ Rivaroxaban and apixaban are contraindicated with drugs that are combined P-glycoprotein and strong *P450 3A4* (CYP3A4) inducers.

Although the new oral anticoagulant trials were similar in design and inclusion/exclusion criteria, it is difficult to make comparisons between the agents to judge differential efficacy in the absence of direct comparisons.

4.2.2.4. Silent AF and Stroke

Clinically unrecognized and asymptomatic AF is a potentially important cause of stroke, supporting efforts for early detection of AF in at-risk individuals. Episodes of asymptomatic

AF are potentially detectable from implantable arrhythmia management devices (pacemakers or defibrillators) that have an atrial lead and can be programmed to record the number, duration, and frequency of atrial rates that exceed a certain threshold and, in some cases, also provide stored electrograms for analysis. These devices typically report “atrial high-rate events.” Whether the high-rate event is AF, atrial flutter, or an atrial tachycardia is not necessarily discernible. Patients receiving arrhythmia management devices often have risk factors for AF. Atrial high-rate episodes have been observed in 10% to 28% of patients who have no prior history of AF.^{61,191}

The ASSERT (Asymptomatic Atrial Fibrillation and Stroke Evaluation in Pacemaker Patients and the Atrial Fibrillation Reduction Atrial Pacing Trial) trial enrolled 2580 patients ≥ 65 years of age with hypertension and no history of AF in whom a pacemaker or defibrillator was recently implanted. During the first 3 months, 10% of subjects had atrial high-rate episodes of >190 bpm for >6 minutes.⁶¹ These high-rate episodes were associated with a >5 -fold increase in subsequent diagnosis of atrial arrhythmia on ECG and a 1.60% per year rate of stroke or systemic embolism compared with a 0.69% per year rate for those without high-rate episodes during the first 3 months. In a subgroup analysis of the MOST (Mode Selection Trial in Sinus Node Dysfunction) trial, patients with atrial high-rate episodes (rate >220 bpm for >10 beats detected by a pacemaker) were more than 2 times as likely to die or have a stroke and 6 times as likely to be subsequently diagnosed with AF as similar patients without atrial high-rate events.¹⁹³ In a prospective study of 2486 patients receiving arrhythmia management devices who had ≥ 1 AF risk factor for stroke—20% of whom had a history of AF—patients with atrial tachycardia/AF burden (defined as the longest total atrial tachycardia/AF duration on any given day during the prior 30-day period) >5.5 hours had a thromboembolism rate of 2.4% per year compared with 1.1% per year for those with less or no atrial tachycardia/AF burden.¹⁹⁴ In a study of 560 patients with HF, the recording of atrial high-rate events lasting >3.8 hours in 1 day was associated with a 9-fold increased thromboembolic event rate.²⁴⁰

Additional studies are needed to further clarify the relationship between stroke risk and atrial high-rate episodes detected by implanted devices and to define key characteristics of atrial high-rate episodes in patients who warrant further investigation or potentially therapy.^{192,194}

4.3. Interruption and Bridging Anticoagulation

Interruption of anticoagulation is often considered for patients with AF who have episodes of bleeding or require surgical or interventional procedures associated with a bleeding risk. There is sparse evidence on which to base specific recommendations on bridging oral anticoagulants among patients with nonvalvular AF with adjusted-dose heparin or LMWH²⁴¹; however, additional studies (eg, BRIDGE [Bridging Anticoagulation in Patients who Require Temporary Interruption of Warfarin Therapy for an Elective Invasive Procedure or Surgery]) are ongoing.²⁴² The duration of interruption and timing of resumption of anticoagulation after the procedure are guided by individualized

consideration of the risk of thrombotic events and the severity of the operative and perioperative bleeding risk. For patients who are treated with warfarin and who are at low risk of thromboemboli or those who are back in normal sinus rhythm and are undergoing surgical or diagnostic procedures that carry a risk of bleeding, stopping warfarin for up to 1 week and allowing the INR to normalize without substituting UFH is a recognized approach. Warfarin is then resumed after adequate hemostasis has been achieved. For patients at higher risk of thromboembolism (mechanical valves, prior stroke, CHA₂DS₂-VASc score ≥ 2), bridging with UFH or LMWH is a common practice, although data for LMWH are limited.²² An increasingly common approach, especially for pacemaker or implantable cardioverter-defibrillator implantation, catheter ablation, coronary angiography, and other vascular interventions, is to perform the procedure without interrupting warfarin.^{241,243–247} Radiofrequency catheter ablation of AF performed with a therapeutic INR does not increase bleeding risk and reduces the risk of emboli.^{243,244} Pacemaker or defibrillator implantation with a therapeutic INR has a lower risk of postoperative bleeding than discontinuing warfarin and initiating bridging anticoagulation with UFH or LMWH and may be considered in those patients requiring device implantation who also have a moderate-to-high thromboembolic risk.^{241,245–250}

For oral factor Xa inhibitors and direct thrombin inhibitors, there is limited experience with drug withdrawal before surgical procedures.²⁴⁴ In the ROCKET AF trial, rivaroxaban was held for 2 days before elective surgery or invasive procedure and for 24 hours before semiurgent procedures.⁶⁰ The increased risk of bleeding should be weighed carefully against the urgency of surgery or an invasive procedure. Interruption of anticoagulation should be guided by the pharmacological properties of the drug. The timing of resumption should take into account the fact that anticoagulation, in contrast to warfarin, is achieved promptly and that reversal agents are not yet available for these agents, which complicates management if bleeding occurs. For elective surgery, holding these agents for 1 day (2 doses for dabigatran and apixaban; 1 dose for rivaroxaban) before the procedure is generally sufficient for patients with normal renal function.²³⁹ Also, for patients with normal renal function, the need for complete hemostasis (eg, for spinal puncture, spinal/epidural catheter, or major surgery) will require discontinuation for ≥ 48 hours. An activated partial thromboplastin time for dabigatran and prothrombin time for apixaban and rivaroxaban may provide useful information; a level close to control suggests a low serum concentration of these agents. For patients undergoing catheter ablation or any procedure in which perforation of the heart chamber is possible, these new agents must be used with caution because of the lack of approved antidotes in the event of cardiac tamponade. In some cases, activated prothrombin complex concentrate and recombinant factor VIIa have been used to reverse the anticoagulant effects of these new agents. Specific reversal agents are not currently available but are in development. Whether hemostasis will be easier and safer for coronary interventions done by a radial artery approach rather than a femoral approach is not known. The use of bare-metal stents

or coronary artery bypass surgery in preference to drug-eluting stents where concomitant long-term use of dual antiplatelet agents is anticipated and might increase bleeding risk is a reasonable consideration when long-term therapy with these anticoagulants is desired.

In patients undergoing percutaneous coronary intervention, dual antiplatelet therapy with aspirin and clopidogrel is indicated to prevent stent thrombosis. The combination of oral anticoagulants and antiplatelet therapy (“triple therapy”) is associated with a high annual risk of fatal and nonfatal bleeding episodes.^{251–254} Recently, in patients taking oral anticoagulants undergoing percutaneous coronary intervention, the efficacy and safety of antiplatelet therapy with aspirin and clopidogrel versus clopidogrel alone were studied.¹⁸⁶ The use of clopidogrel without aspirin was associated with a reduction in bleeding and no increase in the rate of thrombotic events.

4.4. Nonpharmacological Stroke Prevention

4.4.1. Percutaneous Approaches to Occlude the LAA

The LAA is the primary source for thromboembolism in AF.²⁵⁵ Exclusion of the LAA, both surgically and with devices, has been attempted with the goal of reducing thromboembolism in patients with AF. There are 2 general approaches to occlude the LAA using percutaneous strategies. The first involves implantable devices that are inserted percutaneously into the LAA with the goal of occluding or plugging the LAA. Devices for LAA occlusion include the WATCHMAN device (Boston Scientific, Natick, MA) and the Amplatzer cardiac plug (St. Jude Medical, Plymouth, MN). The WATCHMAN device is deployed percutaneously via transseptal puncture and has a polyethylene membrane that covers a self-expanding nitinol cage with barbs to anchor the device in the LAA.²⁵⁶ The early findings for the WATCHMAN device suggest noninferiority to warfarin for the composite endpoint of stroke, systemic embolism, and cardiovascular death; however, early adverse events occur in approximately 10% of patients, including pericardial bleeding. Longer-term follow-up of the WATCHMAN device at 1588 patient-years suggests noninferiority of this device to warfarin.²⁵⁶ A subsequent registry study demonstrated that the WATCHMAN device achieved noninferiority in patients who could not receive warfarin.²⁵⁷ Lastly, data from subsequent experience with the WATCHMAN device suggest that the earlier device-related complications were mitigated with increasing operator experience.²⁵⁸

The Amplatzer cardiac plug, which has Conformité Européenne Mark approval, consists of a small proximal disc, a central polyester patch, and a larger distal disc with hooks to anchor the device in the LAA. It does not require anticoagulation. A European-based trial found a 96% success rate for deployment/implantation but with a 7% incidence of serious complications.²⁵⁹ The second strategy is to tie off the LAA using an epicardial snare, referred to as the LARIAT device (SentreHEART, Redwood City, CA). This device received FDA approval in 2009 for facilitation of suture placement and knot tying for use in surgical applications in which soft tissues are being approximated.

It has been adapted for use in AF and combines a percutaneous epicardial and endocardial approach. The initial experience with this device appeared promising, with 97% acute obliteration of the LAA as confirmed by TEE and a favorable safety profile.²⁶⁰ The long-term outcomes of use of the LARIAT device, requiring RCTs to study reduced stroke risk and safety, are not yet defined. The device requires subxiphoid pericardial access that may not be achievable in the presence of pericardial adhesions, it can provoke pericarditis that can be severe, and it is not suitable for all LAA anatomies. It is not yet clear if occluding the LAA with the LARIAT device lowers stroke risk. Additional devices are in development.

4.4.2. Cardiac Surgery—LAA Occlusion/Excision: Recommendation

Class IIb

1. Surgical excision of the LAA may be considered in patients undergoing cardiac surgery. (*Level of Evidence: C*)

Surgical-based procedures to exclude the LAA during cardiac surgery are controversial for several reasons. What should seem technically simple and reproducible—removal of the LAA—yields inconsistent results, and the anatomy of the LAA is quite variable.²⁶¹ The circumflex coronary artery lies proximate to the base of the LAA, and epicardial and endocardial-based surgical techniques to occlude the LAA are often inadequate because of surgeons’ concern about damage to the circumflex artery during a suture-based closure of the appendage. Epicardial techniques include simple suture ligation, oversewing the base without excision, excising the appendage and oversewing the base, and surgical stapling and excision.²⁶² One device, the Gillinov-Cosgrove clip LAA exclusion system, has FDA approval.²⁶³ Endocardial techniques include inversion of the appendage, amputation, and then oversewing the base from the endocardial aspect.²⁶²

The results of surgical occlusion of the LAA remain suboptimal, with echocardiographic follow-up suggesting incomplete occlusion in $\geq 50\%$ of subjects. In the largest study to examine the success of LAA ligation, 2546 patients undergoing TEE between 1993 and 2004 were retrospectively examined²⁶⁴; 137 patients underwent a surgical attempt at LAA occlusion. Of these 137 patients, 52 underwent excision and 85 underwent exclusion (by either suture or stapling). TEE-defined unsuccessful closures were characterized by either persistent flow into the LAA, a remnant stump of >1.0 cm of the LAA, or color Doppler flow into the LAA. Overall, 55 of 137 closures were successful (40%). Success varied with the technique used: for excision, the success rate was 73%; for suture exclusion, 23%; and for stapling, 0%. Particularly noteworthy is that thrombus was identified in $\geq 25\%$ of patients with unsuccessful LAA occlusion with suture exclusion or stapled LAA remnants. This latter finding constitutes important data guiding the continued need for anticoagulation in patients who have undergone surgical LAA ligation.

It is unclear whether occlusion of the LAA at the time of concomitant heart surgery contributes to the success and

efficacy of occlusion in terms of stroke prevention. The LAAOS (Left Atrial Appendage Occlusion Study) randomized 77 patients with risk factors for stroke to LAA closure or control at the time of coronary artery bypass surgery.²⁶⁵ During this trial, suture-based or staple-based occlusion was permitted; the success of LAA closure in the suture group was 45% versus 72% in the stapled group. Nine appendage tears occurred during the trial (1 control and 8 treatments), but these tears did not contribute to mortality or morbidity. There were 2 thromboembolic events in the occlusion group and none in the control. The authors concluded that LAA occlusion could be performed safely; however, larger randomized studies are needed to determine whether LAA occlusion could reduce stroke risk in patients with risk factors for AF who undergo non-AF-related cardiac surgery. In a retrospective cohort of 205 patients with echocardiography following mitral valve replacement, 58 patients underwent LAA ligation as judged by transthoracic echocardiogram. Of these 58 patients, 52 had a complete ligation of the LAA, as defined by lack of color Doppler flow from the body of the LA into the appendage, and 6 had persistent flow. The principal finding was that lack of and incomplete LAA occlusion were both strongly associated with the occurrence of a thromboembolic event.²⁶⁶

In summary, the current data on LA occlusion at the time of concomitant cardiac surgery reveal a lack of clear consensus because of the inconsistency of techniques used for surgical excision, the highly variable rates of successful LAA occlusion, and the unknown impact of LAA occlusion on future thromboembolic events.

5. Rate Control: Recommendations

See Table 9 for a summary of recommendations for this section.

Class I

1. Control of the ventricular rate using a beta blocker or nondihydropyridine calcium channel antagonist is recommended for patients with paroxysmal, persistent, or permanent AF.^{267–269} (*Level of Evidence: B*)
2. Intravenous administration of a beta blocker or nondihydropyridine calcium channel blocker is recommended to slow the ventricular heart rate in the acute setting in patients without pre-excitation. In hemodynamically unstable patients, electrical cardioversion is indicated.^{270–273} (*Level of Evidence: B*)
3. In patients who experience AF-related symptoms during activity, the adequacy of heart rate control should be assessed during exertion, adjusting pharmacological treatment as necessary to keep the ventricular rate within the physiological range. (*Level of Evidence: C*)

Class IIa

1. A heart rate control (resting heart rate <80 bpm) strategy is reasonable for symptomatic management of AF.^{269,274} (*Level of Evidence: B*)
2. Intravenous amiodarone can be useful for rate control in critically ill patients without pre-excitation.^{275–277} (*Level of Evidence: B*)
3. AV nodal ablation with permanent ventricular pacing is reasonable to control heart rate when

Table 9. Summary of Recommendations for Rate Control

Recommendations	COR	LOE	References
Control ventricular rate using a beta blocker or nondihydropyridine calcium channel antagonist for paroxysmal, persistent, or permanent AF	I	B	267–269
IV beta blocker or nondihydropyridine calcium channel blocker is recommended to slow ventricular heart rate in the acute setting in patients without pre-excitation. In hemodynamically unstable patients, electrical cardioversion is indicated	I	B	270–273
For AF, assess heart rate control during exertion, adjusting pharmacological treatment as necessary	I	C	N/A
A heart rate control (resting heart rate <80 bpm) strategy is reasonable for symptomatic management of AF	IIa	B	269, 274
IV amiodarone can be useful for rate control in critically ill patients without pre-excitation	IIa	B	275–277
AV nodal ablation with permanent ventricular pacing is reasonable when pharmacological therapy is inadequate and rhythm control is not achievable	IIa	B	278–280
A lenient rate-control strategy (resting heart rate <110 bpm) may be reasonable when patients remain asymptomatic and LV systolic function is preserved	IIb	B	274
Oral amiodarone may be useful for ventricular rate control when other measures are unsuccessful or contraindicated	IIb	C	N/A
AV nodal ablation should not be performed without prior attempts to achieve rate control with medications	III: Harm	C	N/A
Nondihydropyridine calcium channel antagonists should not be used in decompensated HF	III: Harm	C	N/A
With pre-excitation and AF, digoxin, nondihydropyridine calcium channel antagonists, or amiodarone should not be administered	III: Harm	B	281
Dronedarone should not be used to control ventricular rate with permanent AF	III: Harm	B	282, 283

AF indicates atrial fibrillation; AV, atrioventricular; bpm, beats per minute; COR, Class of Recommendation; HF, heart failure; IV, intravenous; LOE, Level of Evidence; LV, left ventricular; and N/A, not applicable.

pharmacological therapy is inadequate and rhythm control is not achievable.^{278–280} (*Level of Evidence: B*)

Class IIb

1. A lenient rate-control strategy (resting heart rate <110 bpm) may be reasonable as long as patients remain asymptomatic and LV systolic function is preserved.²⁷⁴ (*Level of Evidence: B*)
2. Oral amiodarone may be useful for ventricular rate control when other measures are unsuccessful or contraindicated. (*Level of Evidence: C*)

Class III: Harm

1. AV nodal ablation with permanent ventricular pacing should not be performed to improve rate control without prior attempts to achieve rate control with medications. (*Level of Evidence: C*)
2. Nondihydropyridine calcium channel antagonists should not be used in patients with decompensated HF as these may lead to further hemodynamic compromise. (*Level of Evidence: C*)
3. In patients with pre-excitation and AF, digoxin, nondihydropyridine calcium channel antagonists, or intravenous amiodarone should not be administered as they may increase the ventricular response and may result in ventricular fibrillation.²⁸¹ (*Level of Evidence: B*)
4. Dronedarone should not be used to control the ventricular rate in patients with permanent AF as it increases the risk of the combined endpoint of stroke, MI, systemic embolism, or cardiovascular death.^{282,283} (*Level of Evidence: B*)

Rate control in AF is an important strategy. It impacts quality of life, reduces morbidity, and decreases the potential for developing tachycardia-induced cardiomyopathy. Multiple agents, including beta blockers, nondihydropyridine calcium channel blockers, digoxin, and certain antiarrhythmic drugs, including amiodarone and sotalol, have been evaluated with regard to efficacy in attaining rate control. This information is summarized in Table 10. When considering which agent(s) to use, clinicians must consider the patient's degree of symptoms, hemodynamic status, presence or absence of HF, and potential precipitants of AF. When evaluating the evidence supporting different agents, clinicians must recognize that most clinical trials were performed in the 1980s and 1990s and have study design limitations that include variable endpoints, small sample sizes, and single-site study and observational trial designs. Issues to consider include the acuity of attaining rate control, which agent(s) to administer, and the degree of rate control required. Over the last 40 years, several themes have emerged. In general, beta blockers are the most common agents used for rate control, followed by nondihydropyridine calcium channel blockers, digoxin, and amiodarone. Patient comorbidities must be understood to avoid medications that may precipitate adverse events such as decompensation of HF, exacerbation of chronic obstructive pulmonary disease, or acceleration of conduction in patients with pre-excitation.

Table 10. Common Medication Dosage for Rate Control of AF

	Intravenous Administration	Usual Oral Maintenance Dose
Beta blockers		
Metoprolol tartrate	2.5–5.0 mg IV bolus over 2 min; up to 3 doses	25–100 mg BID
Metoprolol XL (succinate)	N/A	50–400 mg QD
Atenolol	N/A	25–100 mg QD
Esmolol	500 mcg/kg IV bolus over 1 min, then 50–300 mcg/kg/min IV	N/A
Propranolol	1 mg IV over 1 min, up to 3 doses at 2-min intervals	10–40 mg TID or QID
Nadolol	N/A	10–240 mg QD
Carvedilol	N/A	3.125–25 mg BID
Bisoprolol	N/A	2.5–10 mg QD
Nondihydropyridine calcium channel antagonists		
Verapamil	0.075–0.15 mg/kg IV bolus over 2 min; may give an additional 10.0 mg after 30 min if no response, then 0.005 mg/kg/min infusion	180–480 mg QD (ER)
Diltiazem	0.25 mg/kg IV bolus over 2 min, then 5–15 mg/h	120–360 mg QD (ER)
Digitalis glycosides		
Digoxin	0.25 mg IV with repeat dosing to a maximum of 1.5 mg over 24 h	0.125–0.25 mg QD
Others		
Amiodarone*	300 mg IV over 1 h, then 10–50 mg/h over 24 h	100–200 mg QD

*Multiple dosing schemes exist for the use of amiodarone.

AF indicates atrial fibrillation; BID, twice daily; ER, extended release; IV, intravenous; N/A, not applicable; QD, once daily; QID, 4 times a day; and TID, 3 times a day.

When rapid control of ventricular rate during AF is required, intravenous medications or electrical cardioversion may be used. Electrical cardioversion is preferred in patients with decompensated HF, ongoing myocardial ischemia, or hypotension, although this may carry an increased thromboembolic risk in patients with inadequate anticoagulation or for whom AF is of uncertain duration. In hemodynamically stable patients with a rapid ventricular response, oral medications may be administered.

5.1. Specific Pharmacological Agents for Rate Control

5.1.1. Beta-Adrenergic Receptor Blockers

By blocking sympathetic tone, beta blockers are useful for ventricular rate control in patients with AF. Beta blockers, including esmolol, propranolol, and metoprolol, are effective when administered intravenously in the setting of acute AF.^{270,273,284} Orally administered beta blockers, including atenolol, metoprolol, nadolol, propranolol, as well as the antiarrhythmic drug sotalol, have all been effectively used for ongoing ventricular

rate control in patients with chronic AF. There is less published literature on rate control of AF with additional beta blockers. In the AFFIRM (Atrial Fibrillation Follow-Up Investigation of Rhythm Management) study, beta blockers were the most effective and commonly used drug class for rate control (70% on beta blocker versus 54% on calcium channel blocker).²⁶⁹ In patients with HF, carvedilol had efficacy for heart rate control and, in combination with digoxin, resulted in improved LV function.²⁸⁵ Combination therapy of beta blockers with other agents, including digoxin, is effective in ventricular rate control; however, drugs should be titrated to avoid excessive bradycardia.²⁶⁷

See [Online Data Supplement 6](#) for additional data on beta blockers.

5.1.2. Nondihydropyridine Calcium Channel Blockers

Diltiazem and verapamil have direct AV nodal effects, blocking L-type calcium channels, and are used for ventricular rate control in both acute and chronic AF. In the setting of acute AF, intravenous administration of diltiazem was safe and effective in controlling ventricular response in 83% of patients.²⁷¹ Intravenous verapamil is also effective in establishing acute ventricular rate control.^{273,286,287} Unless immediate rate control is required or an enteral route of administration is not available, oral administration is appropriate. Both verapamil and diltiazem reduce resting and exercise heart rate and can improve exercise tolerance.²⁸⁸ These nondihydropyridine calcium channel blockers should not be used in patients with LV systolic dysfunction and decompensated HF because of their negative inotropic effects, but they may be used in patients with HF with preserved LV systolic function. In addition, these agents should not be used in patients with pre-excitation and AF due to the potential for shortening bypass tract refractoriness, which may accelerate the ventricular rate to precipitate hypotension or ventricular fibrillation^{281,289} (Section 7.8).

See [Online Data Supplement 7](#) for additional data on nondihydropyridine calcium channel blockers.

5.1.3. Digoxin

Digoxin is not usually first-line therapy for ventricular rate control in patients with AF, despite its common use. Although intravenous digoxin does slow the ventricular response, onset of action requires >1 hour and the effect does not peak until approximately 6 hours after initial administration. Therefore, it is not an optimal agent when rapid rate control is desired.²⁹⁰ During chronic oral therapy, digoxin reduces the resting heart rate, but it is ineffective at controlling the ventricular response during exercise.²⁶⁷ Digoxin may be combined with beta blockers or nondihydropyridine calcium channel blockers to improve ventricular rate control during exercise,^{267,291,292} and it has been used in HF as 1 of the few rate control agents that does not have negative inotropic effects. Adverse effects of digoxin include AV block, ventricular arrhythmias, and, infrequently, aggravation of sinus node dysfunction. Dose adjustment is required in patients with renal dysfunction, the elderly, and in the presence of drugs that reduce its excretion, such as amiodarone, propafenone, or nondihydropyridine calcium channel blockers. Therefore, periodic assessment of serum levels is warranted in many patients. Studies finding an association between digoxin therapy and mortality raise further concern about its use, particularly long term.^{293,294} In the AFFIRM trial, digoxin was associated with an

increase in mortality, which in post hoc analysis was irrespective of sex or HF.²⁹⁵ Arrhythmias, which are dose related, are a potential source of mortality; in the DIG (Digitalis Investigation Group) trial, serum levels >0.9 ng/mL were associated with increased mortality.²⁹⁶ However, in another AFFIRM subgroup propensity-matched analysis with paroxysmal and persistent AF, there was no increase in mortality or hospitalization in those taking digoxin as baseline initial therapy.²⁹⁷ Because it can shorten cardiac action potential duration, digoxin should not be used as sole therapy in patients with pre-excitation.

See [Online Data Supplement 8](#) for additional data on digoxin).

5.1.4. Other Pharmacological Agents for Rate Control

Amiodarone exerts sympatholytic and calcium antagonistic properties that can depress AV nodal conduction. Although intravenous amiodarone can be used in critically ill patients without pre-excitation to attain ventricular rate control, it is less effective than nondihydropyridine calcium channel blockers^{272,298} and requires a longer time to achieve rate control (7 hours versus 3 hours for diltiazem). There are limited data on the efficacy of chronic oral therapy with amiodarone for rate control during persistent AF, but in 1 small trial it had similar efficacy to digoxin.²⁹⁹ Amiodarone is uniquely lipid soluble. Its onset of action can be accelerated by a high-dose amiodarone-loading regimen, but there is the potential for worsening hemodynamics in patients with recent decompensated HF or hypotension. Intravenous amiodarone does not have the same electrophysiological effects as oral amiodarone,³⁰⁰ and intravenous amiodarone has the potential to accelerate the ventricular response and precipitate fatal arrhythmias in patients with AF and pre-excitation.^{301,302} Amiodarone has many potential toxicities and drug interactions that limit its long-term use for control of ventricular rate.

Dronedarone, which lacks iodine moieties of amiodarone, slows the resting rate in AF by an average of 12 bpm and also improves exercise heart rate control³⁰³; however, it should not be used for rate control in permanent AF, as it was found to increase rates of HF, stroke, cardiovascular death, and unplanned hospitalization.²⁸² Furthermore, dronedarone should not be used for ventricular rate control in patients with HF and LV systolic dysfunction because it increases the likelihood of the combined endpoint of stroke, MI, systemic embolism, or cardiovascular death.^{282,283}

See [Online Data Supplement 9](#) for additional data on pharmacological agents for rate control.

5.2. AV Nodal Ablation

AV nodal ablation with permanent pacemaker implantation effectively controls and regularizes ventricular heart rate and, in selected patients, improves symptoms. Patients most likely to benefit include those with tachycardia-induced cardiomyopathy with ventricular rate control refractory to medical therapy.^{280,304–307} AV nodal ablation is usually reserved for elderly patients, because it leads to pacemaker dependency. Patients with symptoms refractory to medical therapy who are treated with AV nodal ablation and permanent pacemaker implantation have an improvement in cardiac symptoms, quality of life, and health care utilization. With this approach, no

rate-control medications are necessary, but anticoagulation to prevent thromboembolism is required, based on the patient's stroke risk as assessed by the CHA₂DS₂-VASc system. When this approach is under consideration, the patient must receive counseling to understand that this is an irreversible measure that results in a lifelong pacemaker dependency with its potential complications. Time permitting, pacemaker implantation may be performed 4 to 6 weeks before the AV node ablation to ensure proper pacemaker function, because malfunction due to lead dislodgment can be catastrophic. Sudden death secondary to torsades de pointes or ventricular fibrillation has been reported after AV junction ablation. This outcome is possibly related to increased dispersion of ventricular refractoriness produced by sudden heart rate slowing and ventricular pacing.³⁰⁸ After ablation, the ventricular pacing rate is usually set between 90 bpm and 100 bpm and then gradually tapered over several months.^{309,310} RV apical pacing also creates a ventricular activation sequence that can lead to depressed ventricular function. In patients with left ventricular ejection fraction (LVEF) <35% and symptoms of HF, implantation of a biventricular pacing system is recommended. This procedure should also be considered for patients with less severe ventricular dysfunction.¹⁸ In the BLOCK HF (Biventricular Versus Right Ventricular Pacing in Heart Failure Patients With Atrioventricular Block) trial, patients with advanced AV block with LVEF <50% had improved clinical outcomes when treated with a biventricular pacemaker compared with

RV apical pacing.³¹¹ An upgrade to a biventricular pacing system should be considered for patients who have undergone AV nodal ablation coupled with an RV pacing system who develop moderate-to-severe LV systolic dysfunction.³¹²

See [Online Data Supplement 10](#) for additional data on AV junction ablation.

5.3. Selecting and Applying a Rate-Control Strategy

5.3.1. Broad Considerations in Rate Control

The optimal heart rate targets for rate control are controversial. The target used in the AFFIRM trial was a resting heart rate of either ≤80 bpm or averaging ≤100 bpm on ambulatory monitoring, without a rate >100% of the maximum age-adjusted predicted exercise heart rate. These conditions were achieved in 58% of patients during initial drug therapy.²⁶⁹ One RCT, the RACE (Rate Control Efficacy in Permanent Atrial Fibrillation)-II trial, assessed lenient versus strict rate control.²⁷⁴ In this trial, 614 patients with permanent AF were randomized to a lenient rate-control (resting heart rate <110 bpm) strategy or a strict rate-control (resting heart rate <80 bpm) strategy. At 3 years, the primary composite endpoint of cardiovascular death, hospitalization for HF, stroke, embolism, bleeding, or life-threatening arrhythmic events was similar between the 2 groups (12.9% lenient rate control versus 14.9% strict rate control); thus, a strict rate-control strategy did not improve outcomes. Several considerations warrant a

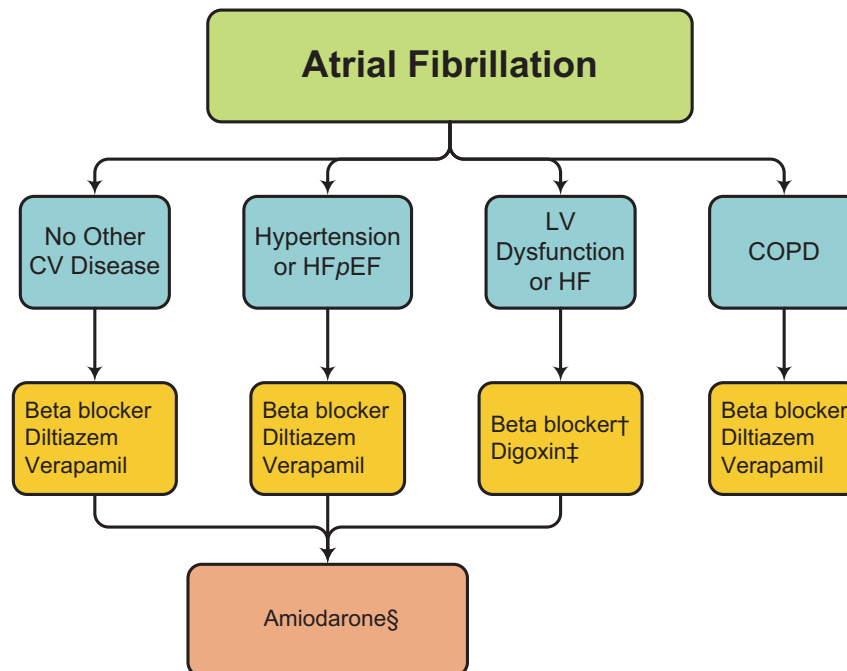


Figure 6. Approach to selecting drug therapy for ventricular rate control.*

*Drugs are listed alphabetically. †Beta blockers should be instituted following stabilization of patients with decompensated HF. The choice of beta blocker (eg, cardioselective) depends on the patient's clinical condition. ‡Digoxin is not usually first-line therapy. It may be combined with a beta blocker and/or a nondihydropyridine calcium channel blocker when ventricular rate control is insufficient and may be useful in patients with HF. §In part because of concern over its side-effect profile, use of amiodarone for chronic control of ventricular rate should be reserved for patients who do not respond to or are intolerant of beta blockers or nondihydropyridine calcium antagonists. COPD indicates chronic obstructive pulmonary disease; CV, cardiovascular; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; and LV, left ventricular.

cautious approach to extrapolating these findings to the general AF population. The majority of patients in the RACE-II trial had preserved LV systolic function. RACE-II was a single noninferiority trial with a 90% CI for a composite endpoint. The resting heart rate achieved in both groups only differed by 10 bpm, and 78% of patients in the lenient control group had resting rates <100 bpm. This single RCT does not provide sufficient evidence to assess definitive results of the impact on all-cause mortality, HF symptoms, hospitalizations, or quality of life. The degree of rate control, however, remains an area of uncertainty and controversy that requires further study.

See [Online Data Supplement 11](#) for additional data on rate control.

5.3.2. Individual Patient Considerations

Optimal ventricular rate control may differ and is impacted by the degree of patient symptoms and comorbidities, including the presence of valvular heart disease, LV systolic dysfunction, HF, and presence of pre-excitation. Figure 6 provides a brief outline of the approach(es) to rate control in different patient populations.

6. Rhythm Control

Long-term AF management may attempt to restore and maintain sinus rhythm, commonly referred to as “a rhythm-control strategy,” using a combination of approaches, including cardioversion, antiarrhythmic drugs, and radiofrequency catheter ablation in the setting of appropriate anticoagulation and rate control. RCTs comparing outcomes of a rhythm-control strategy using antiarrhythmic drugs with a rate-control strategy in patients with AF failed to show a superiority of rhythm control on mortality for either strategy.^{269,313} Furthermore, when applied in patients who are candidates for both treatment strategies (rhythm or rate control), a rhythm-control strategy results in more hospitalizations. Therefore, the routine use of a rhythm-control strategy is not warranted for some patients. Catheter ablation has not been studied in this context.

Although an initial rate-control strategy is reasonable for many patients, several considerations favor pursuing a rhythm-control strategy. Successful sinus rhythm maintenance is associated with improvements in symptoms and quality of life for some patients.^{314,315} Persistent symptoms associated with AF remain the most compelling indication for a rhythm-control strategy. Other factors that may favor attempts at rhythm control include difficulty in achieving adequate rate control, younger patient age, tachycardia-mediated cardiomyopathy, first episode of AF, AF precipitated by an acute illness, and patient preference. AF progresses from paroxysmal to persistent in many patients and subsequently results in electrical and structural remodeling that becomes irreversible with time.^{126,316} For this reason, acceptance of AF as permanent in a patient may render future rhythm-control therapies less effective. This may be more relevant for a younger patient who wishes to remain a candidate for future developments in rhythm-control therapies. Early intervention with a rhythm-control strategy to prevent progression of AF may be beneficial.^{317–319}

6.1. Electrical and Pharmacological Cardioversion of AF and Atrial Flutter

See Table 11 for a summary of recommendations from this section.

6.1.1. Prevention of Thromboembolism: Recommendations Class I

1. For patients with AF or atrial flutter of 48 hours' duration or longer, or when the duration of AF is unknown, anticoagulation with warfarin (INR 2.0 to 3.0) is recommended for at least 3 weeks before and 4 weeks after cardioversion, regardless of the CHA₂DS₂-VASc score and the method (electrical or pharmacological) used to restore sinus rhythm.^{320–323} (Level of Evidence: B)
2. For patients with AF or atrial flutter of more than 48 hours' duration or unknown duration that requires immediate cardioversion for hemodynamic instability, anticoagulation should be initiated as soon as possible and continued for at least 4 weeks after cardioversion unless contraindicated. (Level of Evidence: C)
3. For patients with AF or atrial flutter of less than 48 hours' duration and with high risk of stroke, intravenous heparin or LMWH, or administration of a factor Xa or direct thrombin inhibitor, is recommended as soon as possible before or immediately after cardioversion, followed by long-term anticoagulation therapy. (Level of Evidence: C)
4. Following cardioversion for AF of any duration, the decision about long-term anticoagulation therapy should be based on the thromboembolic risk profile (Section 4). (Level of Evidence: C)

Class IIa

1. For patients with AF or atrial flutter of 48 hours' duration or longer or of unknown duration who have not been anticoagulated for the preceding 3 weeks, it is reasonable to perform TEE before cardioversion and proceed with cardioversion if no LA thrombus is identified, including in the LAA, provided that anticoagulation is achieved before TEE and maintained after cardioversion for at least 4 weeks.¹⁶⁴ (Level of Evidence: B)
2. For patients with AF or atrial flutter of 48 hours' duration or longer or when duration of AF is unknown, anticoagulation with dabigatran, rivaroxaban, or apixaban is reasonable for at least 3 weeks before and 4 weeks after cardioversion.^{230,324,325} (Level of Evidence: C)

Class IIb

1. For patients with AF or atrial flutter of less than 48 hours' duration who are at low thromboembolic risk, anticoagulation (intravenous heparin, LMWH, or a new oral anticoagulant) or no antithrombotic therapy may be considered for cardioversion, without the need for postcardioversion oral anticoagulation.³²⁶ (Level of Evidence: C)

Table 11. Summary of Recommendations for Electrical and Pharmacological Cardioversion of AF and Atrial Flutter

Recommendations	COR	LOE	References
Prevention of thromboembolism			
With AF or atrial flutter for ≥ 48 h, or unknown duration, anticoagulate with warfarin for at least 3 wk before and 4 wk after cardioversion	I	B	320–323
With AF or atrial flutter for > 48 h or unknown duration, requiring immediate cardioversion, anticoagulate as soon as possible and continue for at least 4 wk	I	C	N/A
With AF or atrial flutter < 48 h and high stroke risk, IV heparin or LMWH, or factor Xa or direct thrombin inhibitor, is recommended before or immediately after cardioversion, followed by long-term anticoagulation	I	C	N/A
Following cardioversion of AF, long-term anticoagulation should be based on thromboembolic risk	I	C	N/A
With AF or atrial flutter for ≥ 48 h or unknown duration and no anticoagulation for preceding 3 wk, it is reasonable to perform TEE before cardioversion and then cardiovert if no LA thrombus is identified, provided anticoagulation is achieved before TEE and maintained after cardioversion for at least 4 wk	IIa	B	164
With AF or atrial flutter ≥ 48 h or unknown duration, anticoagulation with dabigatran, rivaroxaban, or apixaban is reasonable for ≥ 3 wk before and 4 wk after cardioversion	IIa	C	230, 324, 325
With AF or atrial flutter < 48 h and low thromboembolic risk, IV heparin, LMWH, a new oral anticoagulant, or no antithrombotic may be considered for cardioversion	IIb	C	326
Direct-current cardioversion			
Cardioversion is recommended for AF or atrial flutter to restore sinus rhythm. If unsuccessful, cardioversion attempts may be repeated.	I	B	327
Cardioversion is recommended for AF or atrial flutter with RVR, that does not respond to pharmacological therapies	I	C	N/A
Cardioversion is recommended for AF or atrial flutter and pre-excitation with hemodynamic instability	I	C	N/A
It is reasonable to repeat cardioversion in persistent AF when sinus rhythm can be maintained for a clinically meaningful time period between procedures	IIa	C	N/A
Pharmacological cardioversion			
Flecainide, dofetilide, propafenone, and IV ibutilide are useful for cardioversion of AF or atrial flutter, provided contraindications to the selected drug are absent	I	A	328–333
Amiodarone is reasonable for pharmacological cardioversion of AF	IIa	A	334, 335
Propafenone or flecainide (“pill-in-the-pocket”) to terminate AF out of hospital is reasonable once observed to be safe in a monitored setting	IIa	B	328
Dofetilide should not be initiated out of hospital	III: Harm	B	332, 336

AF indicates atrial fibrillation; COR, Class of Recommendation; IV, intravenous; LA, left atrial; LMWH, low-molecular-weight heparin; LOE, Level of Evidence; N/A, not applicable; RVR, rapid ventricular response; and TEE, transesophageal echocardiography.

6.1.2. Direct-Current Cardioversion: Recommendations

Class I

1. In pursuing a rhythm-control strategy, cardioversion is recommended for patients with AF or atrial flutter as a method to restore sinus rhythm. If cardioversion is unsuccessful, repeated attempts at direct-current cardioversion may be made after adjusting the location of the electrodes, applying pressure over the electrodes or following administration of an antiarrhythmic medication.³²⁷ (Level of Evidence: B)
2. Cardioversion is recommended when a rapid ventricular response to AF or atrial flutter does not respond promptly to pharmacological therapies and contributes to ongoing myocardial ischemia, hypotension, or HF. (Level of Evidence: C)
3. Cardioversion is recommended for patients with AF or atrial flutter and pre-excitation when tachycardia is associated with hemodynamic instability. (Level of Evidence: C)

Class IIa

1. It is reasonable to perform repeated cardioversions in patients with persistent AF, provided that sinus rhythm can be maintained for a clinically meaningful period between cardioversion procedures. Severity of AF symptoms and patient preference should be considered when embarking on a strategy requiring serial cardioversion procedures. (Level of Evidence: C)

6.1.3. Pharmacological Cardioversion: Recommendations

Class I

1. Flecainide, dofetilide, propafenone, and intravenous ibutilide are useful for pharmacological cardioversion of AF or atrial flutter, provided contraindications to the selected drug are absent.^{328–333} (Level of Evidence: A)

Class IIa

1. Administration of oral amiodarone is a reasonable option for pharmacological cardioversion of AF.^{334,335} (Level of Evidence: A)

2. Propafenone or flecainide (“pill-in-the-pocket”) in addition to a beta blocker or nondihydropyridine calcium channel antagonist is reasonable to terminate AF outside the hospital once this treatment has been observed to be safe in a monitored setting for selected patients.³²⁸ (Level of Evidence: B)

Class III: Harm

1. Dofetilide therapy should not be initiated out of hospital because of the risk of excessive QT prolongation that can cause torsades de pointes.^{332,336} (Level of Evidence: B)

Direct-current cardioversion involves the delivery of an electrical shock synchronized with the QRS complex to avoid inducing ventricular fibrillation that can occur when a shock is delivered during ventricular repolarization on the T wave. It is clinically relevant to differentiate between a cardioversion in which sinus rhythm was not restored, even transiently, and a cardioversion in which sinus rhythm was restored but AF recurs. In the former scenario, approaches that improve energy delivery and may allow for successful cardioversion include increasing shock strength, delivering a biphasic rather than monophasic waveform, changing the shock vector by altering the electrode pad position, improving energy transfer via pressure on the anterior electrode pad, or using a drug such as ibutilide to lower the defibrillation threshold. In the latter scenario, when sinus rhythm is restored but AF returns, pretreatment with selected antiarrhythmic drugs may increase the likelihood of maintenance of sinus rhythm.^{327,337}

A number of technical factors influence cardioversion efficacy, including energy, waveform, and electrode placement.^{7a,7b} A biphasic waveform is more effective than a monophasic waveform.³³⁸ Anteroposterior electrode placement is superior to anterolateral placement in some but not all studies.^{7a,7b,339} If an attempt at cardioversion using 1 electrode placement fails, another attempt using the alternative placement is recommended. The initial use of a higher-energy shock is more effective and may minimize the number of shocks required as well as the duration of sedation.³⁴⁰ Risks associated with cardioversion include thromboembolism, sedation-related complications, ventricular tachycardia and fibrillation, bradyarrhythmias, skin burn or irritation from electrodes, muscle soreness, and reprogramming or altering implanted cardiac device function. Elective cardioversion should not be performed in patients with evidence of digoxin toxicity, severe hypokalemia, or other electrolyte imbalances until these factors are corrected.

Appropriate anticoagulation management around the time of cardioversion is essential for reducing thromboembolic risk. Results of observational studies suggest that thromboembolic risk after cardioversion is highest in the first 72 hours and that the majority of events occur within 10 days.^{341,342} Thromboembolism after cardioversion can be due to migration of thrombi present at the time of cardioversion or to the formation and subsequent migration of de novo thrombi that form while atrial function is still depressed in the postcardioversion period. In this guideline, the Class I recommendation for anticoagulation with warfarin for ≥ 3 weeks before and continuing for ≥ 4 weeks after cardioversion

is based on pathophysiological and observational data.^{322,323} For new oral anticoagulants, available data supporting similar use at cardioversion consist of subgroup analyses of dabigatran from RE-LY, rivaroxaban from ROCKET AF, and apixaban from ARISTOTLE in patients who were receiving long-term anticoagulation (>3 weeks) around the time of cardioversion.^{230,324,325}

TEE guidance is an alternative to 3 weeks of anticoagulation before cardioversion.^{164,343} Therapeutic anticoagulation is achieved and followed by TEE; if no thrombus is seen (including in the LAA), cardioversion is performed and anticoagulation is continued for ≥ 4 weeks. The absence of LA thrombus on TEE does not preclude the need for anticoagulation during and after cardioversion. In the ACUTE (Assessment of Cardioversion Using Transesophageal Echocardiography) trial, hospitalized patients were typically started on intravenous heparin before cardioversion, whereas outpatients were typically started on warfarin 5 days before cardioversion and anticoagulation status was verified at the time of cardioversion.¹⁶⁴ Alternative strategies for achieving rapid anticoagulation include administration of LMWH³⁴⁴ or a new oral anticoagulant. If a thrombus is identified on TEE, the cardioversion should be postponed, followed by ≥ 3 to 4 weeks of anticoagulation. Repeat TEE to ensure thrombus resolution is an option before another cardioversion attempt.³²² If a thrombus remains on repeat TEE, an alternative strategy such as rate control in conjunction with appropriate anticoagulation may be considered.

Data on cardioversion risks for atrial flutter are limited. Atrial flutter, however, can be associated with thrombi and episodes of AF. Therefore, it is recommended that the anticoagulation management strategy for cardioversion of atrial flutter be the same as for AF.

In patients with AF clearly of <48 hours' duration, it is common practice to perform cardioversion without TEE or antecedent anticoagulation.³⁴⁵ No RCTs comparing anticoagulation strategies in patients with AF duration of <48 hours exist.³⁴² If high-risk features are present, such as mitral stenosis or prior history of thromboembolism, long-term anticoagulation should be considered. Decisions about whether to initiate long-term systemic anticoagulation at the time of cardioversion in a patient with AF of <48 hours' duration should be based on the patient's long-term risk of stroke using the CHA₂DS₂-VASc risk score discussed in Section 4.1.

For patients with AF requiring emergency cardioversion because of hemodynamic instability, initiation of anticoagulation should not delay interventions to stabilize the patient. No RCTs have evaluated optimal anticoagulation strategies in this patient population. It is reasonable to administer heparin (intravenous bolus of UFH followed by infusion, or LMWH) or new anticoagulant and to continue this after cardioversion unless contraindicated. For patients with AF or atrial flutter of ≥ 48 hours' duration or uncertain duration, oral anticoagulation is recommended for ≥ 4 weeks after emergency cardioversion (similar to patients undergoing elective cardioversion). If warfarin is used, bridging with UFH or LMWH is indicated until the INR is therapeutic. For patients with AF and thromboembolic risk factors, oral long-term anticoagulation is recommended.

Antiarrhythmic drugs can be administered for attempted conversion of AF to sinus rhythm or to facilitate electrical cardioversion. Pharmacological cardioversion is most likely

Table 12. Recommended Drug Doses for Pharmacological Cardioversion of AF

Drug	Route of Administration	Dosage		Potential Adverse Effects	References
Amiodarone*	Oral	600–800 mg daily in divided doses to a total load of up to 10 g, then 200 mg QD as maintenance		Phlebitis (IV), hypotension, bradycardia, QT prolongation, torsades de pointes (rare), GI upset, constipation, increased INR	334, 335
	IV	150 mg over 10 min, then 1 mg/min for 6 h, then 0.5 mg/min for 18 h or change to oral dosing			
Dofetilide	Oral	CrCl (mL/min)	Dose (mcg BID)	QT prolongation, torsades de pointes; adjust dose for renal function, body size, and age	332
		>60	500		
		40–60	250		
		20–40	125		
		<20	Not recommended		
Flecainide	Oral	200–300 mg × 1†		Hypotension, atrial flutter with 1:1 AV conduction, ventricular proarrhythmia; avoid in patients with CAD and significant structural heart disease	328
Ibutilide	IV	1 mg over 10 min; may repeat 1 mg once if necessary (if weight <60 kg, use 0.01 mg/kg)		QT prolongation, torsades de pointes, hypotension	329, 333, 346
Propafenone	Oral	450–600 mg × 1†		Hypotension, atrial flutter with 1:1 AV conduction, ventricular proarrhythmia; avoid in patients with CAD and significant structural heart disease	328, 330

*Multiple dosing schemes exist for the use of amiodarone.

†Recommended given in conjunction with a beta blocker or nondihydropyridine calcium channel antagonist administered ≥30 minutes before administering the Vaughan Williams class IC agent.³²⁸

AF indicates atrial fibrillation; AV, atrioventricular; BID, twice a day; CAD, coronary artery disease; CrCl, creatinine clearance; GI, gastrointestinal; INR, international normalized ratio; IV, intravenous; and QD, once daily.

Adapted with permission from Fuster et al.⁴

effective when initiated within 7 days after onset of an episode of AF. The most commonly effective antiarrhythmic drugs are specified in Table 12. In patients with recent-onset AF, intravenous administration of ibutilide restored sinus rhythm in about 50% of patients, with an average conversion time of <30 minutes. The rates of successful termination were higher in patients with atrial flutter than in those with AF.³⁴⁶ Pretreatment with ibutilide also improves the efficacy of transthoracic electrical cardioversion of AF.³²⁷ The major risk is excessive QT prolongation, which can cause polymorphic ventricular tachycardia/torsades de pointes. The latter occurs in up to 3% to 4% of patients. Electrocardiographic monitoring should be continued for ≥4 hours after administration, and resuscitation equipment must be immediately available. Ibutilide should be avoided in patients with QT prolongation, marked hypokalemia, or a very low ejection fraction (EF) (<30%) because of the risk of ventricular proarrhythmia.³²⁷ Some experts administer magnesium sulfate intravenously before administering ibutilide in an attempt to lower this risk.³³¹ Intravenous amiodarone may facilitate slowing of the ventricular rate in AF, but the effect to restore sinus rhythm is often delayed. In 1 study, oral amiodarone loaded over the course of several weeks resulted in conversion of persistent AF to sinus rhythm in about 25% of patients.³¹⁴ An oral dose of flecainide or propafenone can be used as a “pill-in-the-pocket” strategy to attempt to restore sinus rhythm shortly after the onset of symptomatic AF.^{328,330} Because termination of AF may be associated with bradycardia due to sinus node or AV node dysfunction or a proarrhythmic

response, an initial conversion trial in a monitored setting is recommended before this approach is used in the unmonitored outpatient setting. A beta blocker or nondihydropyridine calcium channel antagonist should be administered ≥30 minutes before administering the Vaughan Williams class IC agent to prevent a rapid ventricular response due to 1:1 AV conduction during atrial flutter.³²⁸

6.2. Pharmacological Agents for Preventing AF and Maintaining Sinus Rhythm

6.2.1. Antiarrhythmic Drugs to Maintain Sinus Rhythm: Recommendations

Class I

1. Before initiating antiarrhythmic drug therapy, treatment of precipitating or reversible causes of AF is recommended. (*Level of Evidence: C*)
2. The following antiarrhythmic drugs are recommended in patients with AF to maintain sinus rhythm, depending on underlying heart disease and comorbidities (*Level of Evidence: A*):
 - a. Amiodarone^{314,347–349}
 - b. Dofetilide^{332,336}
 - c. Dronedaron^{350–352}
 - d. Flecainide^{347,353}
 - e. Propafenone^{347,354–357}
 - f. Sotalol^{347,355,358}

3. **The risks of the antiarrhythmic drug, including proarrhythmia, should be considered before initiating therapy with each drug. (Level of Evidence: C)**
4. **Because of its potential toxicities, amiodarone should only be used after consideration of risks and when other agents have failed or are contraindicated.^{314,354,359–362} (Level of Evidence: C)**

Class IIa

1. **A rhythm-control strategy with pharmacological therapy can be useful in patients with AF for the treatment of tachycardia-induced cardiomyopathy. (Level of Evidence: C)**

Class IIb

1. **It may be reasonable to continue current antiarrhythmic drug therapy in the setting of infrequent, well-tolerated recurrences of AF when the drug has reduced the frequency or symptoms of AF. (Level of Evidence: C)**

Class III: Harm

1. **Antiarrhythmic drugs for rhythm control should not be continued when AF becomes permanent (Level of Evidence: C), including dronedarone.²⁸² (Level of Evidence: B)**
2. **Dronedarone should not be used for treatment of AF in patients with New York Heart Association (NYHA) class III and IV HF or patients who have had an episode of decompensated HF in the past 4 weeks.²⁸³ (Level of Evidence: B)**

When a rhythm-control strategy is desired, antiarrhythmic drug therapy may be selected to reduce the frequency and duration of AF and improve quality of life. Before antiarrhythmic drug treatment is initiated, reversible precipitants of AF should be identified and corrected. After the first episode of AF that resolves, it is reasonable to address the underlying causes of AF and need for anticoagulation, and to not initiate antiarrhythmic drug treatment until warranted by recurrences of AF. Decisions about anticoagulation should be based on the individual patient's stroke risk profile and not on the response to antiarrhythmic drug therapy. Antiarrhythmic drug efficacy is modest, and asymptomatic AF recurrences are common. Therefore, a rhythm-control strategy should not result in cessation of antithrombotic therapy, rate-control therapy, or treatment of underlying heart disease.

Drug selection is guided to a greater extent by safety concerns than by drug efficacy. A common approach is to identify available drug choices by first eliminating, on the basis of clinical parameters, drugs that have absolute or relative contraindications. Patients with CAD, significant LV hypertrophy, and HF have more restricted options than those with minimal or no structural heart disease. Several other important factors must be considered, including the risk for bradyarrhythmias, risk factors for excessive QT prolongation and torsades de pointes (eg, baseline QT prolongation, history of torsades de pointes during therapy with a QT interval-prolonging drug,

potassium-wasting syndromes), and factors that influence drug disposition, such as patient age and renal or hepatic dysfunction. Because of its toxicity profile, amiodarone should only be used after consideration of risks and when other agents have failed or are contraindicated.

Table 13 summarizes antiarrhythmic drugs useful in the maintenance of sinus rhythm along with toxicity profiles. In general, antiarrhythmic drugs have the potential to precipitate or worsen bradycardia due to sinus node dysfunction or abnormal AV conduction. A history of syncope, sinus bradycardia, PR interval prolongation, and bundle-branch block raise concerns for a risk of bradyarrhythmia during antiarrhythmic drug therapy. Depending on the specific agent selected, a pacemaker may be required for patients with significant bradyarrhythmias.

In selecting a strategy of rhythm control with an antiarrhythmic drug, providing for adequate rate control in the event of AF recurrence should also be considered. Once antiarrhythmic drug therapy is initiated, patient symptoms may improve without complete AF suppression. The transition from frequent AF to infrequent, well-tolerated recurrence of AF is a reasonable outcome and does not necessarily indicate that the therapy should be discontinued. However, if attempts at rhythm control are abandoned (eg, after AF has been declared permanent), the antiarrhythmic drug should be discontinued.

Several systematic reviews have summarized the data on the efficacy and safety of antiarrhythmic drugs for treating AF.^{347,359,363,364} In a meta-analysis of 44 trials, antiarrhythmic drug therapy significantly reduced recurrence of AF (with a number needed to treat ranging from 2 to 9). All drugs may require discontinuation of therapy because of adverse effects (number needed to harm ranging from 9 to 27), and all but amiodarone and propafenone increased proarrhythmia in this analysis (number needed to harm ranging from 17 to 119). Vaughan Williams class IA drugs (quinidine and disopyramide, pooled data) were associated with increased mortality compared with controls, whereas no other antiarrhythmic drug showed a significant effect on mortality.³⁶⁵ Most of the trials in this meta-analysis had relatively short duration of follow-up and enrolled relatively healthy patients; therefore, it is difficult to extrapolate these data to other patient populations. Conclusions about other important clinical outcomes such as stroke and HF were not analyzed, and dronedarone was not included.

Antiarrhythmic drugs that prolong the QT interval, notably sotalol, dofetilide, and disopyramide (all of which block the rapidly activating delayed rectifier potassium current I_{Kr}), have a risk of causing torsades de pointes and should be avoided in patients at increased risk of this form of proarrhythmia. Amiodarone and dronedarone have rarely been associated with prolongation of the QT interval and torsades de pointes.^{366,367} General risk factors associated with increased risk of torsades de pointes include bradycardia, advanced age, hypokalemia, hypomagnesemia, female sex, baseline prolonged QT interval, congenital long-QT syndrome, concomitant use of other QT-prolonging therapies, HF, and possibly LV hypertrophy.

Structural heart disease has been associated with an increased risk of drug-induced proarrhythmia that may manifest as life-threatening ventricular arrhythmias. Manifestations of heart disease sufficient to warrant consideration include

Table 13. Dosage and Safety Considerations for Maintenance of Sinus Rhythm in AF

Drug	Usual Doses	Exclude/Use With Caution	Major Pharmacokinetic Drug Interactions
Vaughan Williams class IA			
Disopyramide	• Immediate release: 100–200 mg once every 6 h • Extended release: 200–400 mg once every 12 h	• HF • Prolonged QT interval • Prostatism, glaucoma • Avoid other QT interval–prolonging drugs	• Metabolized by CYP3A4: caution with inhibitors (eg, verapamil, diltiazem, ketoconazole, macrolide antibiotics, protease inhibitors, grapefruit juice) and inducers (eg, rifampin, phenobarbital, phenytoin)
Quinidine	• 324–648 mg every 8 h	• Prolonged QT interval • Diarrhea	• Inhibits CYP2D6: ↑concentrations of tricyclic antidepressants, metoprolol, antipsychotics; ↓efficacy of codeine • Inhibits P-glycoprotein: ↑digoxin concentration
Vaughan Williams class IC			
Flecainide	• 50–200 mg once every 12 h	• Sinus or AV node dysfunction • HF • CAD • Atrial flutter • Infranodal conduction disease • Brugada syndrome • Renal or liver disease	• Metabolized by CYP2D6 (inhibitors include quinidine, fluoxetine, tricyclics; also genetically absent in 7%–10% of population) and renal excretion (dual impairment can ↑↑plasma concentration)
Propafenone	• Immediate release: 150–300 mg once every 8 h • Extended release: 225–425 mg once every 12 h	• Sinus or AV node dysfunction • HF • CAD • Atrial flutter • Infranodal conduction disease • Brugada syndrome • Liver disease • Asthma	• Metabolized by CYP2D6 (inhibitors include quinidine, fluoxetine, tricyclics; also genetically absent in 7%–10% of population)—poor metabolizers have ↑beta blockade • Inhibits P-glycoprotein: ↑digoxin concentration • Inhibits CYP2C9: ↑warfarin concentration (↑INR 25%)
Vaughan Williams class III			
Amiodarone	• Oral: 400–600 mg daily in divided doses for 2–4 wk; maintenance typically 100–200 mg QD • IV: 150 mg over 10 min; then 1 mg/min for 6 h; then 0.5 mg/min for 18 h or change to oral dosing; after 24 h, consider decreasing dose to 0.25 mg/min	• Sinus or AV node dysfunction • Infranodal conduction disease • Lung disease • Prolonged QT interval	• Inhibits most CYPs to cause drug interaction: ↑concentrations of warfarin (↑INR 0%–200%), statins, many other drugs • Inhibits P-glycoprotein: ↑digoxin concentration
Dofetilide	• 125–500 mcg once every 12 h	• Prolonged QT interval • Renal disease • Hypokalemia • Hypomagnesemia • Diuretic therapy • Avoid other QT interval–prolonging drugs	• Primary renal elimination involving glomerular filtration and active tubular secretion: verapamil, HCTZ, cimetidine, ketoconazole, trimethoprim, prochlorperazine, and megestrol are contraindicated; discontinue amiodarone at least 3 mo before initiation
Dronedarone	• 400 mg once every 12 h	• Bradycardia • HF • Long-standing persistent AF/flutter • Liver disease • Prolonged QT interval	• Metabolized by CYP3A: caution with inhibitors (eg, verapamil, diltiazem, ketoconazole, macrolide antibiotics, protease inhibitors, grapefruit juice) and inducers (eg, rifampin, phenobarbital, phenytoin) • Inhibits CYP3A, CYP2D6, P-glycoprotein: ↑concentrations of some statins, sirolimus, tacrolimus, beta blockers, digoxin
Sotalol	• 40–160 mg once every 12 h	• Prolonged QT interval • Renal disease • Hypokalemia • Hypomagnesemia • Diuretic therapy • Avoid other QT interval–prolonging drugs • Sinus or AV nodal dysfunction • HF • Asthma	• None (renal excretion)

AF indicates atrial fibrillation; AV, atrioventricular; CAD, coronary artery disease; HCTZ, hydrochlorothiazide; HF, heart failure; INR, international normalized ratio; IV, intravenous; and QD, once daily.

Adapted with permission from Roden et al.³⁶⁹

prior MI, HF, and significant LV hypertrophy. Drugs that have prominent sodium channel-blocking effects (eg, flecainide, Vaughan Williams class IC drug) increase mortality in patients with MI from CAD.³⁶⁸ This consideration has been inferred for propafenone (Vaughan Williams class IC agents), and these drugs should be avoided in patients with MI from CAD.

6.2.1.1. Specific Drug Therapy

Amiodarone is an iodinated compound that, along with its metabolites, blocks multiple ion channels (eg, I_{Kr} , I_{Na} , I_{Kur} , I_{to} , I_{CaL} , I_{KACh} , I_{Ks}). It is a noncompetitive beta-adrenergic antagonist. It has a long half-life of weeks and large volume of distribution into adipose tissue. Although suppression of sinus and AV nodal function can occur early within the first few days of oral therapy, the antiarrhythmic effect and QT prolongation can be delayed for days or weeks. A loading phase accelerates the onset of its antiarrhythmic activity, and administration in divided doses and with food minimizes the gastrointestinal symptoms associated with large doses (≥ 600 mg) during the loading phase. Administration with food also significantly increases the rate and extent of amiodarone absorption. Use of oral amiodarone for AF is associated with the added benefit of effective rate control, frequently eliminating the need for other drugs to control the ventricular rate for AF recurrences. Drug interactions and toxicities, however, are sufficient to preclude its routine use as a rate-controlling agent.

Amiodarone is known to inhibit CYP3A, CYP2C9, P-glycoprotein, and, consequently, the elimination of multiple other medications. In patients also taking warfarin or digoxin, dose reduction in these drugs may be needed upon amiodarone initiation in anticipation of a rise in INR (that can be variable) and serum digoxin level. Doses of other medications for rate control should be reduced when the rate slows after initiation of amiodarone and stopped if the rate slows excessively.

Amiodarone is the most effective available antiarrhythmic drug for maintenance of sinus rhythm in patients with paroxysmal or persistent AF. In direct comparisons, it is more effective than dronedarone, sotalol, or propafenone.^{314,360,362,370} A mixed treatment comparison of amiodarone, dronedarone, flecainide, propafenone, and sotalol for the treatment of AF or atrial flutter found that amiodarone had the largest reduction of AF recurrence (OR: 0.22; 95% CI: 0.16 to 0.29) but was associated with the highest rate of patients experiencing ≥ 1 serious adverse event (OR: 2.41; 95% CI: 0.96 to 6.06) and treatment withdrawals due to adverse events (OR: 2.91; 95% CI: 1.66 to 5.11).³⁵⁹ Trends for increased mortality (OR: 2.17; 95% CI: 0.63 to 7.51) were found that were stronger when small studies randomizing <100 subjects per group were excluded from the analysis. Amiodarone therapy was associated with an increase in noncardiac mortality in patients with NYHA class III HF in SCD-HeFT (Sudden Cardiac Death in Heart Failure Trial).³⁷¹

The major cardiovascular side effect of amiodarone is bradycardia. Marked QT prolongation can occur, but it is very rarely associated with torsades de pointes.³⁶⁶ Extracardiac toxicities, including thyroid, liver, pulmonary, and ocular and skin discoloration, are a major problem with amiodarone, so it is not a first-choice agent (especially in younger patients) when other antiarrhythmic drugs are an option. The risk of many toxicities, including pulmonary toxicity, is dose related

and can be fatal. Chronic oral doses of ≤ 200 mg daily may be effective and result in fewer side effects than higher-dose regimens. In patients with LV hypertrophy, HF, CAD, and/or previous MI, amiodarone is associated with a low risk of proarrhythmia, making it an appropriate initial choice to prevent recurrent AF in these clinical settings. Appropriate surveillance for lung, liver, and thyroid toxicity is warranted.

Flecainide and Propafenone are Vaughan Williams class IC drugs that may be considered for rhythm control in patients with AF without structural heart disease. Flecainide, along with other potent sodium channel-blocking drugs, increased mortality in patients with prior MI and therefore should be avoided in patients with ischemic heart disease.³⁶⁸ In addition, both drugs are negative inotropes and should be avoided in patients with LV dysfunction.

These medications can cause slowing of the atrial rate in atrial flutter, resulting in 1:1 AV conduction and an increased ventricular rate; therefore, concomitant AV nodal blocking medication is recommended. Drug-induced, use-dependent increases in PR and QRS durations of up to 25% compared with baseline can also occur during sinus rhythm. However, a greater increase in the QRS duration may be a marker for proarrhythmia risk.³⁷² These agents should be used with caution in the presence of significant conduction system disease, including intraventricular conduction delay or bundle-branch block in the absence of a pacing system. Noncardiac side effects are uncommon and include dizziness and visual disturbance, and propafenone can cause a metallic taste. The parent compound has beta-blocker properties, and its metabolites are electrophysiologically active with weak beta-blocking activity. Propafenone is a substrate for CYP2D6, which is genetically absent in approximately 7% of patients (poor metabolizers) and is inhibited by quinidine, fluoxetine, and tricyclic antidepressants, among others. Thus, drug interactions and genetic susceptibility can cause abnormally increased plasma concentrations of propafenone, resulting in significant beta blockade.

Sotalol, an I_{Kr} inhibitor and beta blocker, is not effective for conversion of AF to sinus rhythm but may be used to prevent recurrent AF. Much like with other antiarrhythmic drugs, with the exception of amiodarone, the rates of maintaining sinus rhythm at 1 year for sotalol are 30% to 50%.³⁴⁷ Sotalol is renally cleared and should be used with caution or avoided in patients with CKD or unstable renal function. Sotalol causes drug-induced QT-interval prolongation, so it should be administered with caution or avoided when administered with other drugs known to prolong the QT interval. During follow-up, serum potassium and magnesium levels and renal function should be checked periodically. Trends toward increased mortality for sotalol (OR: 3.44; 95% CI: 1.02 to 11.59) were observed in a comparison study,³⁵⁹ and it is likely that proarrhythmia is a contributing mechanism. Some experts initiate sotalol in hospital with electrocardiographic monitoring to observe for QT prolongation and proarrhythmia in the absence of an implanted cardioverter-defibrillator.

Dofetilide is a potent and selective inhibitor of I_{Kr} that may be considered for rhythm control in patients who are at low risk for torsades de pointes induced by QT-interval prolongation. Dofetilide has minimal noncardiac side effects. In the SAFIRE-D (Symptomatic Atrial Fibrillation Investigative Research on Dofetilide) trial, dofetilide (500 mcg twice daily)

exhibited 58% efficacy in maintaining sinus rhythm at 1 year after cardioversion, compared with only 25% in the placebo group.³³² Torsades de pointes occurred with an incidence of 0.8%. Dofetilide was discontinued because of excessive QT prolongation in 5% of patients. In the DIAMOND (Danish Investigations of Arrhythmia and Mortality On Dofetilide) study of patients with reduced LV function, sinus rhythm was maintained at 1 year in 79% of the dofetilide group compared with 42% of the placebo group.³³⁶ In the United States, for initiation or dose escalation of therapy, inpatient electrocardiographic monitoring is mandatory, as was the case in clinical trials. Under these circumstances, dofetilide does not increase mortality in HF and post-MI populations.³⁷³ It is renally cleared, dosed according to CrCl, and adjusted or discontinued depending on the degree of QT prolongation. It should not be administered concomitantly with multiple other drugs that influence dofetilide disposition (Table 13) or can prolong the QT interval.

Dronedarone may be considered for rhythm control in patients who do not have HF. Dronedarone is a structural analogue of amiodarone but lacks the iodine moieties of amiodarone. It is associated with a lower incidence of adverse events than amiodarone but is also less efficacious.³⁶⁰ Its multiple electrophysiological actions include sympatholytic effects as well as blocking of calcium, sodium, and potassium currents. Dronedarone reduced the combined endpoint of death and cardiovascular complications (largely by reducing hospitalizations for AF) in patients with paroxysmal or persistent AF or atrial flutter and risk factors for thromboembolism.³⁵⁰

Dronedarone increases mortality in patients with recently decompensated HF and depressed LV function²⁸³ and is contraindicated in patients with NYHA class III or IV HF and in patients who have had an episode of decompensated HF in the past 4 weeks, especially if they have depressed LV function. In patients with permanent AF, dronedarone increases the combined endpoint of stroke, cardiovascular death, and hospitalization.²⁸² Therefore, dronedarone is contraindicated in patients whose sinus rhythm is not restored.

The major cardiac adverse effects of dronedarone are bradycardia and QT prolongation. Torsades de pointes is rare but has been reported. Like amiodarone, dronedarone inhibits renal tubular secretion of creatinine, which can increase plasma creatinine levels. However, there is no reduction in the glomerular filtration rate. Dronedarone is metabolized by CYP3A4 and is a moderate inhibitor of CYP2D6 and P-glycoprotein. Consequently, it increases levels of digoxin and dabigatran and should not be administered with strong inhibitors of CYP3A4 (eg, ketoconazole and macrolide antibiotics), which may potentiate its effects. Dronedarone can be administered with verapamil or diltiazem, which are moderate CYP3A4 inhibitors, but low doses of these agents should be used initially and titrated according to response and tolerance. Dronedarone does not alter the INR when used with warfarin. Dronedarone has been associated with rare case reports of severe hepatotoxicity occurring within 6 months of initiation; therefore, hepatic serum enzymes should be monitored, especially during the first 6 months of treatment.

Disopyramide is a sodium channel-blocking drug with potent anticholinergic and negative inotropic effects that can be considered for rhythm control in patients with AF. Disopyramide can reduce recurrence of AF after direct-current

cardioversion.³⁷⁴ Because of its prominent vagolytic pharmacological effects, disopyramide is useful in AF that occurs in the setting of high vagal tone (“vagal mediated AF”), such as sleep and in response to stimuli that elicit a vagal response, but there is little supporting evidence for this approach. Its negative inotropic effects may be desirable in patients with HCM associated with dynamic outflow tract obstruction.³⁷⁵ Otherwise, it is avoided in structural heart disease. Disopyramide can also prolong the QT interval.

Quinidine has a sodium channel-blocking effect at rapid heart rates and a potassium channel-blocking effect at slower heart rates, as well as vagolytic and alpha-adrenergic receptor-blocking effects, and was among the first antiarrhythmic drugs used to treat AF. It prolongs the QT interval, can cause torsades de pointes, and is used infrequently. Cumulative evidence from a systematic review suggests that quinidine and disopyramide may increase mortality slightly.³⁶⁵ Quinidine has no negative inotropic effects and can be used when there is advanced renal dysfunction. Quinidine requires close electrocardiographic monitoring at initiation and may be an alternative treatment for AF when other, newer antiarrhythmic drugs cannot be used.

Beta blockers are usually not considered effective for maintaining sinus rhythm in patients with AF. One placebo-controlled study of 394 patients with persistent AF found a lower risk of early recurrence after cardioversion and slower ventricular response with sustained-release metoprolol than with placebo.³⁷⁶ Combining an antiarrhythmic drug with a beta blocker may be helpful in some patients. These agents are useful to prevent AF in patients following cardiac surgery and during a high adrenergic state, such as exercise and thyrotoxicosis-related AF. At least theoretically, they can aggravate vagally mediated AF.

See Online Data Supplement 12 for additional data on antiarrhythmic drug therapy.

6.2.1.2. Outpatient Initiation of Antiarrhythmic Drug Therapy

Drug-related proarrhythmia is most common during the initiation phase of drug therapy. Serial ECGs are important to detect excessive QT prolongation (such as with dofetilide or sotalol), the appearance of “giant” U waves, or QRS prolongation >25% (such as with flecainide or propafenone), and should be performed near the time of peak drug concentration.³⁷⁷ Inpatient initiation or dose escalation of dofetilide in an electrocardiographic-monitored environment is required because of the risk of untoward QT-interval prolongation and arrhythmia provocation.^{332,336} Sotalol also results in QT prolongation and may cause proarrhythmia. Its initiation and dose escalation during hospitalization with electrocardiographic monitoring should be considered; the package insert has a corresponding black box warning. There is considerable experience, however, initiating sotalol in an outpatient setting. Some experts allow outpatient initiation when sotalol is started with the patient in sinus rhythm, provided that the QT interval and serum potassium level are normal and no other QT interval-prolonging medications are present, but require inpatient hospitalization when sotalol is initiated while a patient is in AF.³¹⁶ Other experts always initiate sotalol in an inpatient-monitored setting.

Practice patterns vary widely both in terms of which patients are hospitalized for initiation of antiarrhythmic drug therapy and in length of hospitalization. The decision about whether to initiate other antiarrhythmic drugs in an inpatient or outpatient setting should be carefully individualized.³⁷⁸ Data supporting the outpatient initiation of antiarrhythmic drug therapy are best established for amiodarone and dronedarone (Table 13).

See *Online Data Supplement 13* for additional data on antiarrhythmic drug therapy.

6.2.2. Upstream Therapy: Recommendations

Class IIa

1. An ACE inhibitor or angiotensin-receptor blocker (ARB) is reasonable for primary prevention of new-onset AF in patients with HF with reduced LVEF.^{130,379,380} (*Level of Evidence: B*)

Class IIb

1. Therapy with an ACE inhibitor or ARB may be considered for primary prevention of new-onset AF in the setting of hypertension.³⁸¹ (*Level of Evidence: B*)
2. Statin therapy may be reasonable for primary prevention of new-onset AF after coronary artery surgery.^{136,382} (*Level of Evidence: A*)

Class III: No Benefit

1. Therapy with an ACE inhibitor, ARB, or statin is not beneficial for primary prevention of AF in patients without cardiovascular disease.³⁸³ (*Level of Evidence: B*)

The goal of “upstream” therapy (ie, ACE inhibitors, ARBs, statins, and n-3 polyunsaturated fatty acids) is to modify the atrial substrate to reduce susceptibility to, or progression of, AF. Agents delivered as upstream drug therapy might have the ability to halt or delay the cellular processes leading to AF either before (primary prevention) or after (secondary prevention) the development of AF.

A number of prospective trials investigating ARBs and polyunsaturated fatty acids for prevention of recurrent AF have been disappointing.^{81,383–388} Although upstream therapies may be valuable strategies for primary prevention of cardiac changes leading to AF in selected patients, reversal of AF substrate has not been demonstrated, and such therapy is not recommended for the prevention of recurrence of AF in patients without another indication. In retrospective studies and studies in which AF was a prespecified secondary endpoint, ACE inhibitors or ARBs slightly reduce the development of AF in patients with HF and LV dysfunction and possibly those with hypertension and LV hypertrophy.⁸¹ Several systematic reviews of statin therapy to prevent AF have been performed.^{136,384,389,390} The administration of statins may reduce postoperative AF in patients undergoing coronary artery bypass graft surgery.^{136,382,391}

See *Online Data Supplement 14* for additional data on upstream therapy.

6.3. AF Catheter Ablation to Maintain Sinus Rhythm: Recommendations

Class I

1. AF catheter ablation is useful for symptomatic paroxysmal AF refractory or intolerant to at least 1 class I or III antiarrhythmic medication when a rhythm-control strategy is desired.^{363,392–397} (*Level of Evidence: A*)
2. Before consideration of AF catheter ablation, assessment of the procedural risks and outcomes relevant to the individual patient is recommended. (*Level of Evidence: C*)

Class IIa

1. AF catheter ablation is reasonable for some patients with symptomatic persistent AF refractory or intolerant to at least 1 class I or III antiarrhythmic medication.^{394,398–400} (*Level of Evidence: A*)
2. In patients with recurrent symptomatic paroxysmal AF, catheter ablation is a reasonable initial rhythm-control strategy before therapeutic trials of antiarrhythmic drug therapy, after weighing the risks and outcomes of drug and ablation therapy.^{401–403} (*Level of Evidence: B*)

Class IIb

1. AF catheter ablation may be considered for symptomatic long-standing (>12 months) persistent AF refractory or intolerant to at least 1 class I or III antiarrhythmic medication when a rhythm-control strategy is desired.^{363,404} (*Level of Evidence: B*)
2. AF catheter ablation may be considered before initiation of antiarrhythmic drug therapy with a class I or III antiarrhythmic medication for symptomatic persistent AF when a rhythm-control strategy is desired. (*Level of Evidence: C*)

Class III: Harm

1. AF catheter ablation should not be performed in patients who cannot be treated with anticoagulant therapy during and after the procedure. (*Level of Evidence: C*)
2. AF catheter ablation to restore sinus rhythm should not be performed with the sole intent of obviating the need for anticoagulation. (*Level of Evidence: C*)

The role of catheter ablation in the management of AF continues to evolve rapidly, with improvements in the efficacy and safety of the procedure.²⁸ The efficacy of radiofrequency catheter ablation for maintaining sinus rhythm is superior to current antiarrhythmic drug therapy for maintenance of sinus rhythm in selected patient populations. A number of systematic reviews of the efficacy of AF catheter ablation versus antiarrhythmic drug therapy have been performed.^{363,392–395,405,406} Cryoballoon ablation is an alternative to point-by-point radiofrequency ablation to achieve pulmonary vein isolation.⁴⁰⁷ The evidence supporting the efficacy of catheter ablation is strongest for paroxysmal AF in younger patients with little to no structural heart

disease⁴⁰⁸ and in procedures performed in highly experienced centers. Studies have demonstrated a reduction of AF-related symptoms in these contexts.⁴⁰⁹ Evidence is insufficient to determine whether AF catheter ablation reduces all-cause mortality, stroke, and HF.^{7a,7b} Ongoing clinical trials (CABANA [Catheter Ablation Versus Antiarrhythmic Drug Therapy for Atrial Fibrillation] and EAST [Early Therapy of Atrial Fibrillation for Stroke Prevention Trial]) should provide new information for assessing whether AF catheter ablation is superior to standard therapy with either rate- or rhythm-control drugs for reducing total mortality and other secondary outcome measures and whether early application of a rhythm-control therapy involving ablation, antiarrhythmic drugs, or both can impact endpoints of stroke, cardiovascular death, or HF compared with usual care. These important trials will help address whether catheter ablation provides benefit beyond improvements in quality of life.

See [Online Data Supplements 15 and 16](#) for additional data on maintaining sinus rhythm and AF catheter ablation.

6.3.1. Patient Selection

The decision whether to pursue catheter ablation depends on a large number of variables, including the type of AF (paroxysmal versus persistent versus long-standing persistent), degree of symptoms, presence of structural heart disease, candidacy for alternative options such as rate control or antiarrhythmic drug therapy, likelihood of complications, and patient preference.²⁸ It is important to recognize that most patients enrolled in trials of AF catheter ablation have generally been younger, healthy persons with symptomatic paroxysmal AF refractory to 1 or more antiarrhythmic medications. The safety and efficacy of catheter ablation are less well established for other populations of patients, especially patients with long-standing persistent AF, very elderly patients, and patients with

significant HF, including tachycardia-induced cardiomyopathy²⁸ (Section 6.3). Figure 7 shows an approach to the integration of antiarrhythmic drugs and catheter ablation of AF in patients without and with structural heart disease.

Two RCTs compared radiofrequency catheter ablation with antiarrhythmic drug therapy as a first-line rhythm-control treatment. The RAAFT (Radiofrequency Ablation Versus Antiarrhythmic Drugs for Atrial Fibrillation Treatment)-2 trial compared the efficacy of AF catheter ablation with that of antiarrhythmic drug therapy as first-line therapy for rhythm control in 127 patients (88% with paroxysmal AF) with a higher 1-year freedom from AF (45% versus 28%; $P=0.02$).⁴⁰² The MANTRA-PAF (Medical Antiarrhythmic Treatment or Radiofrequency Ablation in Paroxysmal Atrial Fibrillation) trial compared AF catheter ablation with antiarrhythmic drug therapy as first-line therapy in 294 patients.⁴¹⁰ At the 24-month follow-up, more patients in the ablation group were free from any AF or symptomatic AF, and quality of life was significantly better.⁴⁰³ However, total AF burden was not significantly different between the 2 groups, and major complications requiring intervention were more common in the ablation group. On the basis of these data, radiofrequency catheter ablation may be considered as first-line therapy in select patients before a trial of antiarrhythmic drug therapy when a rhythm-control strategy is desired.

6.3.2. Recurrence After Catheter Ablation

Recurrences of AF after catheter ablation are common during the first 3 months and do not preclude long-term success, although they are associated with an increased risk of procedural failure and rehospitalization. Therefore, when AF occurs early after catheter ablation, a pharmacological rhythm-control approach rather than early repeat ablation should be considered.²⁸ Patients who have had AF catheter ablation and

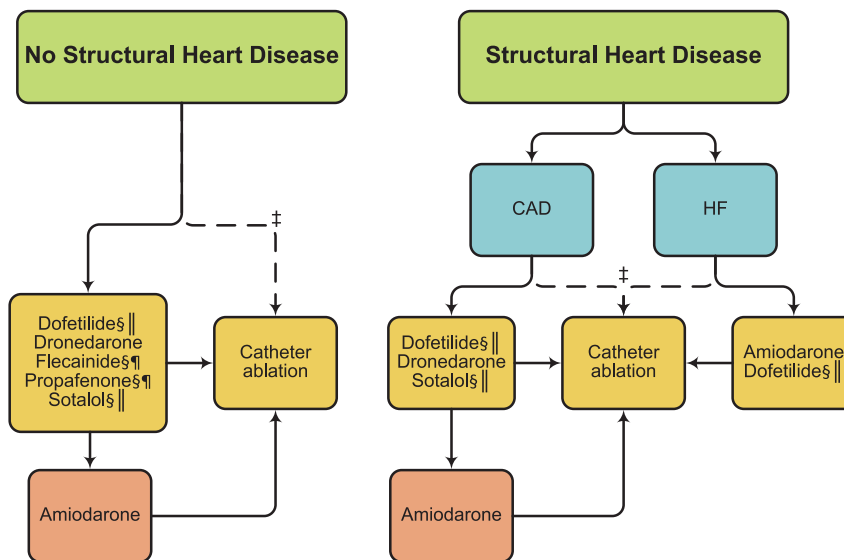


Figure 7. Strategies for rhythm control in patients with paroxysmal* and persistent AF. †Catheter ablation is only recommended as first-line therapy for patients with paroxysmal AF (Class IIa recommendation). ‡Drugs are listed alphabetically. ‡Depending on patient preference when performed in experienced centers. §Not recommended with severe LVH (wall thickness >1.5 cm). ||Should be used with caution in patients at risk for torsades de pointes ventricular tachycardia. ¶Should be combined with AV nodal blocking agents. AF indicates atrial fibrillation; AV, atrioventricular; CAD, coronary artery disease; HF, heart failure; and LVH, left ventricular hypertrophy.

develop persistent AF within the 3 months following ablation may require cardioversion. Recurrent AF after 3 months is usually an indication of recovery of pulmonary vein conduction and may respond to repeat ablation or initiation of an antiarrhythmic drug.⁴¹¹ A number of centers have reported late AF recurrences >1 year after catheter ablation.^{78,412–415}

6.3.3. Anticoagulation Therapy Periaction

Because of the well-established risk of periprocedure stroke or TIA associated with AF catheter ablation, there is consensus that anticoagulation is indicated to prevent thromboembolism around the time of radiofrequency catheter ablation regardless of the patient's baseline thromboembolic risk. Detailed consensus recommendations have been published about the approach to anticoagulation before, during, and after catheter ablation.²⁸ Both intraprocedural heparin and oral anticoagulation are recommended for ≥ 2 months postprocedure. AF catheter ablation should not be performed in patients who cannot be treated with anticoagulant therapy during and after the procedure.

Several reports indicate that AF catheter ablation may be performed with fewer complications when oral warfarin anticoagulation is continued as an alternative to a bridging approach with UFH or LMWH.^{243,416–418} Several centers reported their experience with the use of direct thrombin and factor Xa inhibitors (mainly dabigatran) around the time of AF catheter ablation.^{244,324,419–422} Typically, dabigatran was held for 1 or 2 doses before the ablation procedure, in part reflecting the lack of a reversal agent. These reports suggest that the use of dabigatran is associated with a similar risk of bleeding and thromboembolic complications compared with uninterrupted warfarin; however, this is not a uniform finding.²⁴⁴

Continuation of anticoagulation >2 months after AF catheter ablation, if the procedure is perceived as successful, should be based on consideration of the patient's thromboembolic risk profile (Section 4.1), bleeding risk, and patient choice. Recurrence of AF after ablation is 3 to 7 times more likely to be asymptomatic compared with before ablation,^{423,424} and AF can recur late. Several large case series have reported a low risk of stroke after AF ablation.^{425–428} Although the stroke rate is low in these series, few patients at high risk of stroke were monitored after anticoagulation was stopped for a significant period of time.

6.3.4. Catheter Ablation in HF

A number of smaller clinical trials have evaluated the role of AF catheter ablation in selected patients with LV dysfunction and HF and demonstrate a reasonable rate of successful sinus rhythm maintenance with improvements in LVEF and symptoms.^{47,307,429} The degree to which LVEF improves varies according to patient characteristics.⁴³⁰ In cases where LV dysfunction is thought to be due to AF itself, AF catheter ablation and maintenance of sinus rhythm may result in a marked improvement. It may be difficult to determine in this population whether symptoms are related to AF or the underlying HF and whether AF itself has contributed to the decline in LVEF. Improved rate control or cardioversion with antiarrhythmic drug therapy may help determine the causality. Because of the extent of remodeling and underlying heart disease, recurrence rates⁴³¹ and complication rates are higher in this population. A meta-analysis reported that the single-procedure efficacy of AF catheter ablation was lower in patients with systolic

dysfunction, but a similar success rate could be achieved among patients with and without systolic dysfunction with repeat procedures.⁴³² Patient selection biases likely influence reported outcomes. Taken as a whole, catheter ablation may be reasonable to treat symptomatic AF in selected patients with significant LV dysfunction and HF.

6.3.5. Complications Following AF Catheter Ablation

AF catheter ablation is associated with important risks of major complications. A 2010 international survey of radiofrequency catheter ablation procedures reported a 4.5% incidence of major complications, including a 1.3% rate of cardiac tamponade, a 0.94% rate of stroke or TIA, a 0.04% rate of atrial-esophageal fistula, and a 0.15% rate of death.⁴³³ A European observational multinational registry reported a complication rate of 7.7%, of which 1.7% were major complications.⁴³⁴ A report from a statewide inpatient database described a complication rate of 5% with a 9% readmission rate.⁴³⁵ Much of the data on rates of complications is derived from experienced centers or voluntary registries.

Table 14 lists the complications associated with radiofrequency catheter ablation for AF. A detailed summary of definitions and prevention of specific complications is covered elsewhere.²⁸ Factors associated with complication rates include older age, female sex, and a CHADS₂ score ≥ 2 .^{435–437} Also, LA catheter ablation results in a small incidence of asymptomatic cerebral embolism detectable on cranial magnetic resonance imaging. Most of these lesions resolve or disappear over time. Further research is needed to better define the relationship between ablation strategy and risk and to determine methods to eliminate them.^{28,438,439}

6.4. Pacemakers and Implantable Cardioverter-Defibrillators for Prevention of AF

The primary role of pacemakers in the treatment of patients with AF is treatment of symptomatic bradycardia, which is often related to underlying sick sinus syndrome. Antiarrhythmic therapy may exacerbate sick sinus syndrome and require pacemaker implantation. For patients with sick sinus syndrome who need pacing, atrial or dual-chamber pacing significantly decreases the incidence of subsequent AF compared with RV pacing.¹⁸ Attempts to prevent AF episodes by proprietary overdrive atrial pacing algorithms that react to premature atrial complexes are inconsistent.¹⁸ Therefore, permanent pacing is not indicated for the prevention of AF in patients without other indications for pacemaker implantation. Atrial defibrillators to automatically cardiovert AF do not have clinical value; most patients find discharge energies >1 J uncomfortable, and early recurrence of AF following a shock is common. Implanted defibrillators are not indicated for rhythm control of AF.

6.5. Surgical Maze Procedures: Recommendations

Class IIa

1. An AF surgical ablation procedure is reasonable for selected patients with AF undergoing cardiac surgery for other indications. (*Level of Evidence: C*)

Table 14. Complications of Radiofrequency Catheter Ablation for AF

Complication	Symptoms/Signs	Treatment
Air embolism	Acute ischemia, cardiac arrest, AV block, hypotension	Supplemental oxygen, fluids, CPR, or pacing if indicated
Atrial-esophageal fistula	Usually 1–4 wk after ablation, dysphagia, unexplained fever, chills, sepsis, neurological events (septic emboli)	CT or MRI of esophagus, avoiding endoscopy, immediate surgical correction
Cardiac tamponade/perforation	Abrupt or gradual fall in BP	Pericardiocentesis, emergent surgical drainage if pericardiocentesis fails
Phrenic nerve injury resulting in diaphragmatic paralysis	Shortness of breath, elevated hemidiaphragm	None; usually resolves spontaneously
Iatrogenic atrial flutter	Tachycardia	Cardioversion, antiarrhythmic drugs, or repeat ablation
Gastric motility disorder	Nausea, vomiting, bloating, abdominal pain	Depends on severity of symptoms
Mitral valve injury requiring surgery	Entrapment of catheter	Advance sheath with gentle catheter retraction, surgical removal
MI	Chest pain, ST changes, hypotension	Standard therapy
Pericarditis	Chest pain, typical quality	NSAIDs, colchicine, steroids
Pulmonary vein stenosis	Shortness of breath, cough, hemoptysis	PV dilation/stent or no therapy
Radiation injury	Pain and reddening at radiation site; can present late	Treat as burn injury
Stroke or TIA	Neurological deficit	Consider lysis therapy
Vascular access complications		
• Femoral pseudoaneurysm	Pain or pulsatile mass at groin	Observation, compression, thrombin injection, possible surgery
• Arteriovenous fistula	Pain, bruit at groin site	Observation, compression, possible surgery
• Hematoma	Pain, swelling	Compression
Death	N/A	N/A

AF indicates atrial fibrillation; AV, atrioventricular; BP, blood pressure; CPR, cardiopulmonary resuscitation; CT, computed tomography; MI, myocardial infarction; MRI, magnetic resonance imaging; N/A, not applicable; NSAIDs, nonsteroidal anti-inflammatory drugs; PV, pulmonary valve; and TIA, transient ischemic attack.

Class IIb

1. A stand-alone AF surgical ablation procedure may be reasonable for selected patients with highly symptomatic AF not well managed with other approaches.⁴⁴⁰ (Level of Evidence: B)

The surgical maze procedure was introduced in 1987. The initial 2 iterations were associated with high rates of pacemaker implantation and are no longer performed. The third version (Cox maze III) became the standard surgical procedure to restore sinus rhythm in patients with AF⁴⁴¹ but is not widely performed because of surgeons' reluctance to perform this complicated "cut and sew" atrial lines of ablation operation approach in association with valve or coronary artery bypass procedures or as a stand-alone procedure. The Cox maze intravenous operation is less invasive, using radiofrequency or cryoablation to replicate surgical lines of ablation.⁴⁴²

Data on long-term outcomes in patients undergoing stand-alone AF surgery are limited. Of 282 patients prospectively studied from 2002 to 2009 who underwent the Cox maze IV procedure, 42% had paroxysmal AF and 58% had either persistent or long-standing persistent AF.⁴⁴² Ninety-five of 282 patients (34%) had a stand-alone procedure, and 187 of 282

patients (66%) had a concomitant AF procedure. Overall operative mortality was 2% (1% in stand-alone maze procedures), and freedom from atrial tachyarrhythmias was 89%, 93%, and 89% at 3, 6, and 12 months, respectively. Freedom from atrial tachyarrhythmias off all antiarrhythmic drugs was 63%, 79%, and 78% at 3, 6, and 12 months, respectively. In the period of the study subsequent to 2006, 24-hour Holter monitoring or pacemaker interrogation was performed in these patients. In this cohort, 92% were free of atrial tachyarrhythmias, and 78% were not taking antiarrhythmic drugs.⁴⁴²

Nine RCTs comparing patients who undergo concomitant AF surgery with patients who undergo mitral valve surgery alone suggest greater freedom from AF in treated patients^{443–451}; however, in the composite body of evidence, there was no consistent surgical technique, patient populations in the trial were quite varied, a consistent endpoint defining procedural success was lacking, and long-term clinical endpoints were often missing as well.

The Society of Thoracic Surgeons Adult Cardiac Surgery Database from 2005 to 2010 recorded 91 801 AF surgical ablations, of which 4893 (5.3%) were stand-alone procedures.⁴⁵² Propensity matching of 1708 patients with and without cardiopulmonary bypass showed no difference in mortality risk between groups, but the "off bypass group" had fewer

Table 15. Summary of Recommendations for Specific Patient Groups and AF

Recommendations	COR	LOE	References
Hypertrophic cardiomyopathy			
Anticoagulation is indicated in HCM with AF independent of the CHA ₂ DS ₂ -VASc score	I	B	50, 457
Antiarrhythmic drugs can be useful to prevent recurrent AF in HCM. Amiodarone or disopyramide combined with a beta blocker or nondihydropyridine calcium channel antagonist are reasonable	IIa	C	N/A
AF catheter ablation can be beneficial for HCM to facilitate a rhythm-control strategy when antiarrhythmics fail or are not tolerated	IIa	B	458–461
Sotalolol, dofetilide, and dronedarone may be considered for a rhythm-control strategy in HCM	IIb	C	11
AF complicating ACS			
Urgent cardioversion of new-onset AF in the setting of ACS is recommended for patients with hemodynamic compromise, ongoing ischemia, or inadequate rate control	I	C	N/A
IV beta blockers are recommended to slow RVR with ACS and no HF, hemodynamic instability, bronchospasm	I	C	N/A
With ACS and AF with CHA ₂ DS ₂ -VASc score ≥ 2 , anticoagulation with warfarin is recommended unless contraindicated	I	C	N/A
Amiodarone or digoxin may be considered to slow RVR with ACS and AF and severe LV dysfunction and HF or hemodynamic instability	IIb	C	N/A
Nondihydropyridine calcium antagonists might be considered to slow RVR with ACS and AF only in the absence of significant HF or hemodynamic instability	IIb	C	N/A
Hyperthyroidism			
Beta blockers are recommended to control ventricular rate with AF complicating thyrotoxicosis unless contraindicated	I	C	N/A
When beta blockers cannot be used, a nondihydropyridine calcium channel antagonist is recommended to control ventricular rate	I	C	N/A
Pulmonary diseases			
A nondihydropyridine calcium channel antagonist is recommended to control ventricular rate with AF and COPD	I	C	N/A
Cardioversion should be attempted for patients with pulmonary disease who become hemodynamically unstable with new-onset AF	I	C	N/A
WPW and pre-excitation syndromes			
Cardioversion is recommended for patients with AF, WPW syndrome, and RVR who are hemodynamically compromised	I	C	64
IV procainamide or ibutilide to restore sinus rhythm or slow ventricular rate is recommended for patients with pre-excited AF and RVR who are not hemodynamically compromised	I	C	64
Catheter ablation of the accessory pathway is recommended in symptomatic patients with pre-excited AF, especially if the accessory pathway has a short refractory period	I	C	64
IV amiodarone, adenosine, digoxin, or nondihydropyridine calcium channel antagonists in patients with WPW syndrome who have pre-excited AF is potentially harmful	III: Harm	B	493–495
Heart failure			
A beta blocker or nondihydropyridine calcium channel antagonist is recommended for persistent or permanent AF in patients with HFpEF	I	B	269
In the absence of preexcitation, an IV beta blocker (or a nondihydropyridine calcium channel antagonist with HFpEF) is recommended to slow ventricular response to AF in the acute setting, with caution in patients with overt congestion, hypotension, or HF/EF	I	B	502–505
In the absence of pre-excitation, IV digoxin or amiodarone is recommended to control heart rate acutely	I	B	277, 503, 506, 507
Assess heart rate during exercise and adjust pharmacological treatment in symptomatic patients during activity	I	C	N/A
Digoxin is effective to control resting heart rate with HF/EF	I	C	N/A
A combination of digoxin and beta blocker (or a nondihydropyridine calcium channel antagonist with HFpEF) is reasonable to control resting and exercise heart rate with AF	IIa	B	267, 503
It is reasonable to perform AV node ablation with ventricular pacing to control heart rate when pharmacological therapy is insufficient or not tolerated	IIa	B	269, 508, 509
IV amiodarone can be useful to control heart rate with AF when other measures are unsuccessful or contraindicated	IIa	C	N/A

(Continued)

Table 15. Continued

Recommendations	COR	LOE	References
With AF and RVR causing or suspected of causing tachycardia-induced cardiomyopathy, it is reasonable to achieve rate control by AV nodal blockade or a rhythm-control strategy	Ila	B	51, 307, 510
In patients with chronic HF who remain symptomatic from AF despite a rate-control strategy, it is reasonable to use a rhythm-control strategy	Ila	C	N/A
Amiodarone may be considered when resting and exercise heart rate cannot be controlled with a beta blocker (or a nondihydropyridine calcium channel antagonist with HFpEF) or digoxin, alone or in combination	Ilb	C	N/A
AV node ablation may be considered when rate cannot be controlled and tachycardia-mediated cardiomyopathy is suspected	Ilb	C	N/A
AV node ablation should not be performed without a pharmacological trial to control ventricular rate	III: Harm	C	N/A
For rate control, IV nondihydropyridine calcium channel antagonists, IV beta blockers, and dronedarone should not be given with decompensated HF	III: Harm	C	N/A
Familial (genetic) AF			
For patients with AF and multigenerational family members with AF, referral to a tertiary care center for genetic counseling and testing may be considered	Ilb	C	N/A
Postoperative cardiac and thoracic surgery			
A beta blocker is recommended to treat postoperative AF unless contraindicated	I	A	533–536
A nondihydropyridine calcium channel blocker is recommended when a beta blocker is inadequate to achieve rate control with postoperative AF	I	B	537
Preoperative amiodarone reduces AF with cardiac surgery and is reasonable as prophylactic therapy for patients at high risk of postoperative AF	Ila	A	538–540
It is reasonable to restore sinus rhythm pharmacologically with ibutilide or direct-current cardioversion with postoperative AF	Ila	B	541
It is reasonable to administer antiarrhythmic medications to maintain sinus rhythm with recurrent or refractory postoperative AF	Ila	B	537
It is reasonable to administer antithrombotic medications for postoperative AF	Ila	B	542
It is reasonable to manage new-onset postoperative AF with rate control and anticoagulation with cardioversion if AF does not revert spontaneously to sinus rhythm during follow-up	Ila	C	N/A
Prophylactic sotalol may be considered for patients with AF risk after cardiac surgery	Ilb	B	536, 543
Colchicine may be considered postoperatively to reduce AF after cardiac surgery	Ilb	B	544

ACS indicates acute coronary syndromes; AF, atrial fibrillation; AV, atrioventricular; CHA₂DS₂-VASC, Congestive heart failure, Hypertension, Age ≥ 75 years (doubled), Diabetes mellitus, Prior Stroke or TIA or thromboembolism (doubled), Vascular disease, Age 65 to 74 years, Sex category; COPD, chronic obstructive pulmonary disease; COR, Class of Recommendation; HCM, hypertrophic cardiomyopathy; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; IV, intravenous; LOE, Level of Evidence; LV, left ventricular; N/A, not applicable; RVR, rapid ventricular response; and WPW, Wolff-Parkinson-White.

reoperations for bleeding, shorter hospital stays, and less prolonged ventilation. Minimally invasive stand-alone operations, bilateral pulmonary vein isolation, intraoperative confirmation of mapping, ablation of ganglionic plexi, and exclusion of the LAA procedures have been developed. Of 114 patients undergoing bilateral minithoracotomy surgical ablation of AF, 2 patients (1.8%) died within the perioperative period, and the overall complication rate was 10%.⁴⁵³ At the 6-month follow-up (ECG, Holter monitor, event monitor, or pacemaker interrogation), 52 of 60 patients (87%) with paroxysmal AF were in sinus rhythm and 43 of 60 patients (72%) were off antiarrhythmic drugs. In patients with persistent or long-standing persistent AF, the success rates of freedom from AF were lower, at 18 of 32 patients (56%) and 11 of 22 patients (50%), respectively.

The FAST (Atrial Fibrillation Catheter Ablation Versus Surgical Ablation Treatment) trial compared the outcomes of catheter ablation and surgical ablation in a randomized study design.⁴⁴⁰ Patients either had LA dilation and hypertension (42 patients, 33%) or failed prior catheter ablation (82 patients,

67%). Freedom from atrial arrhythmias was greater after surgical ablation compared with catheter ablation, but the complication rate after surgical ablation was higher. Decisions about the choice of catheter-based or surgical ablation must be made on the basis of patient preference and institutional experience and outcomes with each therapy.²⁸

7. Specific Patient Groups and AF

See Table 15 for a summary of recommendations for this section and [Online Data Supplement 17](#) for additional data on specific patient groups and AF.

7.1. Athletes

Paroxysmal or persistent AF is common in athletes and may be autonomically mediated or triggered by other supraventricular tachycardias.⁴⁵⁴ Contributing conditions such as hypertension and CAD should be considered, particularly for older athletes, and a transthoracic echocardiogram is helpful to evaluate for structural heart disease. Evaluation of the rate

of ventricular response during an episode of AF is warranted and may require ambulatory electrocardiographic monitoring and/or exercise testing to a level of exertion similar to that of the intended sport. Other therapies such as radiofrequency catheter ablation or a “pill-in-the-pocket” approach can be considered in athletes. Specifics of these therapies are considered in Section 6.1.3.⁴⁵⁵

7.2. Elderly

The prevalence of AF increases with age, and approximately 35% of patients with AF are ≥80 years of age.^{30,31} The elderly are a heterogeneous group with potential for multiple comorbidities (Table 3). It is critical to consider the implications of comorbidities to ensure that the patient’s overall goals of care are factored into management decisions. For the older patient with AF, symptoms may be minimal and somewhat atypical. The risk of stroke is increased in the elderly. For this reason, the CHA₂DS₂-VASc risk scoring system identifies 65 to 74 years of age as a minor risk factor for stroke and ≥75 years of age as a major risk factor for stroke (Section 4.1).

Because AF is often associated with minimal or no symptoms in this population and the clearance of antiarrhythmic medications is diminished, sensitivity to proarrhythmic effects, including bradyarrhythmias, is often increased. Therefore, a rate-control strategy is often preferred,³⁰ and direct-current cardioversion is less often warranted.⁴⁵⁶ Typically, rate control can be achieved with beta blockers or nondihydropyridine calcium channel antagonists. Care must be taken in these patients because they are often more susceptible to orthostatic hypotension or bradyarrhythmias and when AF is paroxysmal and sinus node dysfunction is more common. Comorbidities should also be considered. Digoxin can be useful for rate control in the relatively sedentary individual, but there are concerns about its risks (Section 5.1.3).

7.3. Hypertrophic Cardiomyopathy: Recommendations

Class I

1. Anticoagulation is indicated in patients with HCM with AF independent of the CHA₂DS₂-VASc score.^{50,457} (*Level of Evidence: B*)

Class IIa

1. Antiarrhythmic medications can be useful to prevent recurrent AF in patients with HCM. Amiodarone or disopyramide combined with a beta blocker or nondihydropyridine calcium channel antagonists are reasonable for therapy. (*Level of Evidence: C*)
2. AF catheter ablation can be beneficial in patients with HCM in whom a rhythm-control strategy is desired when antiarrhythmic drugs fail or are not tolerated.^{458–461} (*Level of Evidence: B*)

Class IIb

1. Sotalol, dofetilide, and dronedarone may be considered for a rhythm-control strategy in patients with HCM.¹¹ (*Level of Evidence: C*)

Patients with HCM are considered separately because their unique pathology distinguishes them from other patients with LV hypertrophy. HCM is defined on the basis of standard criteria such as the echocardiographic identification of a hypertrophied, nondilated LV in the absence of another cardiac or systemic disease capable of producing the magnitude of hypertrophy evident.⁴⁶² AF is relatively common in HCM, increases with age, and is often poorly tolerated symptomatically.⁵⁰ The incidence of AF is estimated at 2% per year in patients with HCM, and approximately two thirds of patients with both HCM and AF are paroxysmal.⁵⁰ AF is associated with increased mortality in patients with HCM (3% in patients with AF versus 1% in sinus rhythm per year)^{50,463} and is primarily due to HF. The HF risk associated with AF in patients with HCM is worse in patients with outflow obstruction and those who develop AF before 50 years of age.⁵⁰

There is an important risk of stroke and systemic embolism in patients with HCM and AF.^{50,464,465} In a study of 480 patients with HCM, the OR for stroke in those with AF was 17.7.⁵⁰ Although no randomized studies of anticoagulant therapy have been reported, the incidence of thromboembolism in patients with HCM and AF is high, and anticoagulation is indicated for these patients independent of their CHA₂DS₂-VASc (or CHADS₂) scores. Anticoagulation with direct thrombin or factor Xa inhibitors may represent another option to reduce the risk of thromboembolic events, but data for patients with HCM are not available.^{50,177,457}

Given the poor tolerance of AF in patients with HCM, a rhythm-control strategy is preferred. However, for those patients for whom a rate-control strategy is chosen, a nondihydropyridine calcium channel blocker, a beta blocker, or a combination of the 2 drugs is preferable. Digoxin, a positive inotrope, may increase the outflow gradient in HCM patients and should be avoided. There have been no systematic studies of the treatment of AF in patients with HCM, but various antiarrhythmic agents have been used, including disopyramide, propafenone, amiodarone, sotalol, dofetilide, and dronedarone. An implantable cardioverter-defibrillator may provide added safety with QT interval–prolonging drugs. Amiodarone or disopyramide in combination with ventricular rate-controlling agents is generally preferred.^{11,466}

Success and complication rates for AF catheter ablation appear to be similar for HCM and other forms of heart disease, but reported outcomes are likely influenced by selection bias.^{11,458,460} The surgical maze procedure for AF shows some success⁴⁶⁷; however, the role of a surgical maze procedure for patients undergoing other open chest surgical procedures (ie, septal myectomy) is unresolved.^{11,467}

7.4. AF Complicating ACS: Recommendations

Class I

1. Urgent direct-current cardioversion of new-onset AF in the setting of ACS is recommended for patients with hemodynamic compromise, ongoing ischemia, or inadequate rate control. (*Level of Evidence: C*)
2. Intravenous beta blockers are recommended to slow a rapid ventricular response to AF in patients with ACS who do not display HF, hemodynamic instability, or bronchospasm. (*Level of Evidence: C*)

3. For patients with ACS and AF with a CHA₂DS₂-VASc score of 2 or greater, anticoagulation with warfarin is recommended unless contraindicated. (*Level of Evidence: C*)

Class IIb

1. Administration of amiodarone or digoxin may be considered to slow a rapid ventricular response in patients with ACS and AF associated with severe LV dysfunction and HF or hemodynamic instability. (*Level of Evidence: C*)
2. Administration of nondihydropyridine calcium antagonists might be considered to slow a rapid ventricular response in patients with ACS and AF only in the absence of significant HF or hemodynamic instability. (*Level of Evidence: C*)

The incidence of AF in patients with ACS ranges from 10% to 21% and increases with patient age and severity of MI.^{138,468} In the Medicare population, AF is associated with increased in-hospital mortality (25.3% with AF versus 16.0% without AF), 30-day mortality (29.3% versus 19.1%), and 1-year mortality (48.3% versus 32.7%).¹³⁸ With multivariate adjustment, AF remains an independent predictor of mortality: in-hospital (OR: 1.21), 30-day (OR: 1.20), and 1-year (OR: 1.34).¹³⁸ Patients who develop AF during hospitalization have a worse prognosis than those with AF on admission.¹³⁸ Stroke rates are increased in patients with MI and AF compared with rates in those without AF (3.1% for those with AF versus 1.3% for those in normal sinus rhythm).⁴⁶⁸ Thus, AF is an independent predictor of poor long-term outcome in patients with ACS.^{469,470} Specific recommendations for management of patients with AF in the setting of ACS are based primarily on consensus because no adequate trials have tested alternative strategies.²⁰

Patients treated for ACS normally require dual antiplatelet therapy with aspirin plus other platelet inhibitors, such as clopidogrel, and may require the addition of warfarin or a novel oral anticoagulant ("triple therapy") as treatment of AF¹⁸⁶ (Section 4.3). In patients with long-standing AF or a moderate-to-high CHA₂DS₂-VASc score, efforts should be directed to minimize duration of triple therapy, and decisions about stent insertion should consider the potential requirement for long-term anticoagulant therapy. For patients who develop transient AF as a complication of ACS and who do not have a prior history of AF, the need for anticoagulation and the duration of oral anticoagulation should be based on the patient's CHA₂DS₂-VASc score. Use of dual antiplatelet therapy alone may be considered for patients with ACS who have AF and a low CHA₂DS₂-VASc score, with reconsideration of the indications for anticoagulation over time.^{199,323} An option is to consider the use of oral anticoagulation plus clopidogrel with or without aspirin.¹⁸⁶ The novel oral anticoagulants have not been evaluated in the context of AF and ACS, and thus no recommendation for their use can be made.

Urgent direct-current cardioversion is appropriate in patients with ACS presenting with new-onset AF and intrac-table ischemia, hemodynamic instability, or inadequate rate control. Intravenous administration of a beta blocker

is indicated for rate control in patients with ACS to reduce myocardial oxygen demands. Intravenous amiodarone is an appropriate alternative for rate control and may facilitate conversion to normal sinus rhythm. Digoxin may be considered in those with severe LV dysfunction and HF or hemodynamic instability. Systemic anticoagulation is indicated in those with large anterior infarcts and in survivors of ACS who develop persistent AF. Treatment with ACE inhibitors appears to reduce the incidence of AF in patients with LV dysfunction after ACS.^{471,472}

7.5. Hyperthyroidism: Recommendations

Class I

1. Beta blockers are recommended to control ventricular rate in patients with AF complicating thyrotoxicosis unless contraindicated. (*Level of Evidence: C*)
2. In circumstances in which a beta blocker cannot be used, a nondihydropyridine calcium channel antagonist is recommended to control the ventricular rate. (*Level of Evidence: C*)

AF is the most common arrhythmia in patients with hyperthyroidism (5% to 15% of patients) and is more frequent among those >60 years of age.^{152,473,474} Complications of AF in hyperthyroidism include HF and thromboembolism, although the correlation with thromboembolic disease is controversial.^{473–481} Treatment is directed primarily toward restoring a euthyroid state, which is usually associated with a spontaneous reversion of AF to sinus rhythm. Antiarrhythmic drugs and cardioversion often fail to achieve sustained sinus rhythm while thyrotoxicosis persists⁴⁸²; therefore, efforts to restore normal sinus rhythm may be deferred until the patient is euthyroid. Beta blockers are effective in controlling the ventricular rate in this situation, and treatment with beta blockers is particularly important in cases of thyroid storm; nondihydropyridine calcium channel antagonists are recommended for rate control.⁴⁸³ Although several studies reported thromboembolism in patients with thyrotoxicosis and AF, evidence suggests that embolic risk was not necessarily increased independent of other stroke risk factors.^{484,485} Anticoagulation for the patient with thyrotoxicosis and AF should be guided by CHA₂DS₂-VASc risk factors (Sections 4.1 and 4.1.1).

Hyperthyroidism and thyrotoxicosis can infrequently result from long-term amiodarone use. In the event of iatrogenic hyperthyroidism during treatment with amiodarone, the drug should be discontinued. The risks and benefits of treating patients with AF with a known history of thyroid disease with amiodarone should be carefully weighed before initiation of therapy, and patients should be monitored closely.⁴⁸⁶

7.6. Acute Noncardiac Illness

A number of acute noncardiac conditions are associated with AF (eg, hypertension, postoperative state, pulmonary embolism, viral infections). Management of the underlying condition and correction of contributing factors as first-line treatment is common to all of these scenarios,⁴⁸⁷ and for many of these patients AF will spontaneously terminate

with correction of the underlying condition. However, during acute illness, patients may require rate control with cardioversion, AV nodal blockers, and/or antiarrhythmic drugs if AF is poorly tolerated or rate control is not feasible. The specific rate- or rhythm-control agent(s) will depend on the underlying medical condition. Of note is that an elevated catecholamine state is common to many of these clinical circumstances, and unless contraindicated, a beta blocker is the preferred initial drug. The role of anticoagulation is less clear and likely disease specific and needs to be addressed on the basis of risk profile and duration of AF.

7.7. Pulmonary Disease: Recommendations

Class I

1. A nondihydropyridine calcium channel antagonist is recommended to control the ventricular rate in patients with AF and chronic obstructive pulmonary disease. (Level of Evidence: C)
2. Direct-current cardioversion should be attempted in patients with pulmonary disease who become hemodynamically unstable as a consequence of new-onset AF. (Level of Evidence: C)

Supraventricular arrhythmias, including AF, are common in patients with chronic obstructive pulmonary disease.^{488–490} AF should be distinguished from multifocal atrial tachycardia, which is unlikely to respond to electrical cardioversion but will often slow with treatment of the underlying disease and in response to nondihydropyridine calcium channel blockers.⁴⁹¹ Treatment of the underlying lung disease and correction of hypoxia and acid-base imbalance are of primary importance in this situation and represent first-line therapy. Antiarrhythmic drug therapy and cardioversion may be ineffective against AF until respiratory decompensation has been corrected. Theophylline and beta-adrenergic agonists can precipitate AF and make control of the ventricular response rate difficult. Non-beta-1-selective blockers, sotalol, propafenone, and adenosine are contraindicated in patients with bronchospasm. However, beta blockers, sotalol, or propafenone may be considered in patients with obstructive lung disease who develop AF and do not have bronchospasm. Rate control can usually be achieved safely with nondihydropyridine calcium channel antagonists or possibly amiodarone.²⁷⁵ Digoxin can be used with calcium channel blockers, particularly in those with preserved LVEF.⁴⁹² In patients refractory to drug therapy, AV nodal ablation and ventricular pacing may be necessary to control the ventricular rate. Anticoagulation, although not specifically studied in patients with AF due to pulmonary disease, is discussed in Section 4.2 for risk-based antithrombotic therapy.

7.8. WPW and Pre-Excitation Syndromes: Recommendations

Class I

1. Prompt direct-current cardioversion is recommended for patients with AF, WPW syndrome, and rapid ventricular response who are hemodynamically compromised.⁶⁴ (Level of Evidence: C)

2. Intravenous procainamide or ibutilide to restore sinus rhythm or slow the ventricular rate is recommended for patients with pre-excited AF and rapid ventricular response who are not hemodynamically compromised.⁶⁴ (Level of Evidence: C)
3. Catheter ablation of the accessory pathway is recommended in symptomatic patients with pre-excited AF, especially if the accessory pathway has a short refractory period that allows rapid antegrade conduction.⁶⁴ (Level of Evidence: C)

Class III: Harm

1. Administration of intravenous amiodarone, adenosine, digoxin (oral or intravenous), or nondihydropyridine calcium channel antagonists (oral or intravenous) in patients with WPW syndrome who have pre-excited AF is potentially harmful because these drugs accelerate the ventricular rate.^{493–495} (Level of Evidence: B)

AF is of specific concern in patients with WPW syndrome because of the potential for degeneration to ventricular fibrillation related to rapidly conducting anterograde accessory pathways. The risk of developing AF over 10 years in patients with WPW syndrome is estimated at 15%, although the mechanism of increased AF risk is poorly understood.^{496,497} Approximately 25% of patients with WPW syndrome have accessory pathways with short anterograde refractory periods (<250 msec), which are associated with a risk of rapid ventricular rates and ventricular fibrillation.^{498,499} Patients with multiple accessory pathways are also at greater risk of ventricular fibrillation.⁴⁹⁸ The safety and efficacy of catheter ablation of the accessory pathway is established⁶⁴; however, ablation of the accessory pathway does not always prevent AF, especially in older patients, and additional pharmacological or ablative therapy may be required. Once the accessory pathway has been eliminated, the process of selecting pharmacological therapy is the same as for patients without pre-excitation.

Specifics of antiarrhythmic therapies are described in Section 6. During AF, the ventricular rate is determined by competing conduction over the AV node and the accessory pathway(s). As with any unstable arrhythmia, cardioversion is recommended for hemodynamic instability.⁶⁴ Agents that slow AV nodal conduction without prolonging accessory pathway refractoriness can accelerate the ventricular rate and precipitate hemodynamic collapse and ventricular fibrillation in high-risk patients. Intravenous administration of ibutilide or procainamide may slow the rate of conduction over the accessory pathway, slow the ventricular rate, or may convert AF to sinus rhythm; it is recommended for hemodynamically stable patients in the setting of AF with conduction over an accessory pathway. Verapamil, diltiazem, adenosine, digoxin (oral or intravenous), and intravenous amiodarone can precipitate ventricular fibrillation and should not be used.^{493,495} Similarly, use of lidocaine in pre-excited AF is considered potentially harmful.⁵⁰⁰ Oral amiodarone can slow or block accessory pathway conduction during chronic oral therapy. Although beta blockers theoretically pose a similar potential risk, the data are sparse regarding administration of these agents in rapid AF in patients with WPW syndrome; nevertheless, they should be used with caution.^{494,501}

7.9. Heart Failure: Recommendations

Class I

1. Control of resting heart rate using either a beta blocker or nondihydropyridine calcium channel antagonist is recommended for patients with persistent or permanent AF and compensated HF with preserved ejection fraction (HFpEF).²⁶⁹ (Level of Evidence: B)
2. In the absence of pre-excitation, intravenous beta-blocker administration (or a nondihydropyridine calcium channel antagonist in patients with HFpEF) is recommended to slow the ventricular response to AF in the acute setting, with caution needed in patients with overt congestion, hypotension, or HF with reduced LVEF.^{502–505} (Level of Evidence: B)
3. In the absence of pre-excitation, intravenous digoxin or amiodarone is recommended to control heart rate acutely in patients with HF.^{277,503,506,507} (Level of Evidence: B)
4. Assessment of heart rate control during exercise and adjustment of pharmacological treatment to keep the rate in the physiological range is useful in symptomatic patients during activity. (Level of Evidence: C)
5. Digoxin is effective to control resting heart rate in patients with HF with reduced EF. (Level of Evidence: C)

Class IIa

1. A combination of digoxin and a beta blocker (or a nondihydropyridine calcium channel antagonist for patients with HFpEF) is reasonable to control resting and exercise heart rate in patients with AF.^{267,503} (Level of Evidence: B)
2. It is reasonable to perform AV node ablation with ventricular pacing to control heart rate when pharmacological therapy is insufficient or not tolerated.^{269,508,509} (Level of Evidence: B)
3. Intravenous amiodarone can be useful to control heart rate in patients with AF when other measures are unsuccessful or contraindicated. (Level of Evidence: C)
4. For patients with AF and rapid ventricular response causing or suspected of causing tachycardia-induced cardiomyopathy, it is reasonable to achieve rate control by either AV nodal blockade or a rhythm-control strategy.^{51,307,510} (Level of Evidence: B)
5. For patients with chronic HF who remain symptomatic from AF despite a rate-control strategy, it is reasonable to use a rhythm-control strategy. (Level of Evidence: C)

Class IIb

1. Oral amiodarone may be considered when resting and exercise heart rate cannot be adequately controlled using a beta blocker (or a nondihydropyridine calcium channel antagonist in patients with HFpEF) or digoxin, alone or in combination. (Level of Evidence: C)

2. AV node ablation may be considered when the rate cannot be controlled and tachycardia-mediated cardiomyopathy is suspected. (Level of Evidence: C)

Class III: Harm

1. AV node ablation should not be performed without a pharmacological trial to achieve ventricular rate control. (Level of Evidence: C)
2. For rate control, intravenous nondihydropyridine calcium channel antagonists, intravenous beta blockers, and dronedarone should not be administered to patients with decompensated HF. (Level of Evidence: C)

Patients with HF are more likely than the general population to develop AF,³⁸ and there is a direct relationship between the NYHA class and prevalence of AF in patients with HF, progressing from 4% in those who are NYHA class I to 40% in those who are NYHA class IV.⁵¹¹ AF is a strong independent risk factor for subsequent development of HF as well.^{38,512} In addition to those with HF and depressed EF, patients with HF due to diastolic dysfunction with HFpEF are also at greater risk for AF.⁵¹³ HF and AF can interact to perpetuate and exacerbate each other through mechanisms such as rate-dependent worsening of cardiac function, fibrosis, and activation of neurohumoral vasoconstrictors. AF can worsen symptoms in patients with HF and conversely, worsened HF can promote a rapid ventricular response in AF.

As with other patient populations, the main goals of therapy for those with AF and HF are prevention of thromboembolism and control of symptoms. Most patients with AF and HF can be expected to be candidates for systemic anticoagulation unless contraindicated (Section 4). General principles of management include correction of underlying causes of AF and HF as well as optimization of HF management. As in other patient populations, the issue of rate control versus rhythm control has been investigated. For patients who develop HF as a result of AF, a rhythm-control strategy should be pursued. AF with a rapid ventricular response is one of the few potentially reversible causes of HF. Therefore, a patient who presents with newly detected HF in the presence of AF with a rapid ventricular response should be presumed to have a rate-related cardiomyopathy until proved otherwise. In this situation, 2 strategies can be considered. One is rate control of the patient's AF to see if HF and EF improve. The other is to attempt to restore and maintain sinus rhythm. In this situation, it is common practice to initiate amiodarone and then arrange for cardioversion a month later. Amiodarone has the advantage of being both an effective rate-control medication and the most effective antiarrhythmic medication with a low risk of proarrhythmia.

In patients with HF who develop AF, a rhythm-control strategy is not superior to a rate-control strategy.⁵¹⁴ If rhythm control is chosen, AF catheter ablation in patients with HF may lead to an improvement in LV function and quality of life but is less likely to be effective than in patients with intact cardiac function.^{47,307}

Because of their favorable effect on morbidity and mortality in patients with systolic HF, beta blockers are the preferred agents for achieving rate control unless otherwise

contraindicated. Digoxin may be an effective adjunct to a beta blocker. Nondihydropyridine calcium antagonists, such as diltiazem, should be used with caution in those with depressed EF because of their negative inotropic effect. For those with HF and preserved EF, nondihydropyridine calcium antagonists can be effective for achieving rate control but may be more effective when used in combination with digoxin. For those patients for whom a rate-control strategy is chosen, AV node ablation and cardiac resynchronization therapy device placement can be useful when rate control cannot be achieved either because of drug inefficacy or intolerance.^{515–520}

7.10. Familial (Genetic) AF: Recommendation

Class IIb

1. For patients with AF and multigenerational family members with AF, referral to a tertiary care center for genetic counseling and testing may be considered. (Level of Evidence: C)

AF is heritable, and having an affected family member is associated with a 40% increased risk of the arrhythmia.^{155,521–524} Premature AF, defined as a first-degree relative with onset of AF before the age of 66, is associated with a doubling in risk of AF.¹⁵⁵ Thus, it is common, particularly among younger, healthier persons with AF, to observe families with AF. In the last 10 years, many mutations have been identified in individuals and families with AF.⁵²⁵ The implicated genes include a wide range of ion channels, signaling molecules, and related proteins; however, the role of these mutations in more common forms of AF appears limited. Population-based or genomewide association studies identified ≥ 9 distinct genetic loci for AF.^{156–159} Furthermore, combinations of AF-associated single nucleotide polymorphisms may identify individuals at high risk for arrhythmia.^{526,527} However, the role of these common genetic variants in risk stratification,^{155,528,529} assessment of disease progression, and determination of clinical outcomes^{157,530,531} is limited. Routine genetic testing related to AF is not indicated.⁵³²

7.11. Postoperative Cardiac and Thoracic Surgery: Recommendations

Class I

1. Treating patients who develop AF after cardiac surgery with a beta blocker is recommended unless contraindicated.^{533–536} (Level of Evidence: A)
2. A nondihydropyridine calcium channel blocker is recommended when a beta blocker is inadequate to achieve rate control in patients with postoperative AF.⁵³⁷ (Level of Evidence: B)

Class IIa

1. Preoperative administration of amiodarone reduces the incidence of AF in patients undergoing cardiac surgery and is reasonable as prophylactic therapy for patients at high risk for postoperative AF.^{538–540} (Level of Evidence: A)

2. It is reasonable to restore sinus rhythm pharmacologically with ibutilide or direct-current cardioversion in patients who develop postoperative AF, as advised for nonsurgical patients.⁵⁴¹ (Level of Evidence: B)
3. It is reasonable to administer antiarrhythmic medications in an attempt to maintain sinus rhythm in patients with recurrent or refractory postoperative AF, as advised for other patients who develop AF.⁵³⁷ (Level of Evidence: B)
4. It is reasonable to administer antithrombotic medication in patients who develop postoperative AF, as advised for nonsurgical patients.⁵⁴² (Level of Evidence: B)
5. It is reasonable to manage well-tolerated, new-onset postoperative AF with rate control and anticoagulation with cardioversion if AF does not revert spontaneously to sinus rhythm during follow-up. (Level of Evidence: C)

Class IIb

1. Prophylactic administration of sotalol may be considered for patients at risk of developing AF after cardiac surgery.^{536,543} (Level of Evidence: B)
2. Administration of colchicine may be considered for patients postoperatively to reduce AF after cardiac surgery.⁵⁴⁴ (Level of Evidence: B)

Postoperative AF occurs in 25% to 50% of patients after open heart surgery. Increased age is the most consistent risk factor.⁵⁴⁵ With the projected increase in the number of elderly patients undergoing cardiac operations, the incidence of postoperative AF is likely to increase. Postoperative AF is associated with stroke,⁵⁴⁶ increased cost,⁵⁴⁷ and mortality.⁵⁴⁸ Beta blockers, nondihydropyridine calcium channel blockers, and amiodarone are useful as treatments in patients with postoperative AF and may be initiated preoperatively in some patients.^{391,544}

In a meta-analysis of patients undergoing coronary revascularization, those who received preoperative statin therapy had less AF than those not treated with statins.³⁹¹ No published data exist for patients undergoing valvular or other heart surgery.

The COPPS (Colchicine for the Prevention of the Postpericardiotomy Syndrome) substudy examined the efficacy and safety of colchicine for AF prevention.⁵⁴⁴ In this multicenter trial, patients were randomized to colchicine with standard therapy or standard therapy alone. The primary endpoint was incidence of AF at 1 month postoperatively. Patients receiving colchicine had a reduced incidence of AF (12% versus 22% at 30 days postoperatively). The colchicine group also had a shorter length of hospital stay.

8. Evidence Gaps and Future Research Directions

The past decade has seen substantial progress in the understanding of mechanisms of AF, clinical implementation of ablation for maintaining sinus rhythm, and new drugs for stroke prevention. Further studies are needed to better inform clinicians about the risks and benefits of therapeutic options for an individual patient. Continued research is needed into the mechanisms that initiate and sustain AF. It is hoped that better understanding of these tissue and cellular mechanisms will lead to more defined approaches to treating and

abolishing AF. This includes new methodological approaches for AF ablation that would favorably impact survival, thromboembolism, and quality of life across different patient profiles. New pharmacological therapies are needed, including antiarrhythmic drugs that have atrial selectivity and drugs that target fibrosis, which will hopefully reach clinical evaluation. The successful introduction of new anticoagulants is encouraging, and further investigations will better inform clinical practices for optimizing beneficial applications and minimizing the risks of these agents, particularly in the elderly, in the presence of comorbidities and in the periprocedural period. Further investigations must be performed to better understand the links between the presence of AF, AF burden, and stroke risk, and to better define the relationship between AF and dementia. The roles of emerging surgical and procedural therapies to reduce stroke will be defined. Great promise lies in prevention. Future strategies for reversing the growing epidemic of AF will come from basic science and genetic, epidemiological, and clinical studies.

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References

- ACC/AHA Task Force on Practice Guidelines. Methodology manual and policies from the ACCF/AHA Task Force on Practice Guidelines. Cardiosource.com and My.americanheart.org Web sites. 2010. Available at: http://assets.cardiosource.com/Methodology_Manual_for_ACC_AHA_Writing_Committees.pdf and http://my.americanheart.org/idc/groups/ahamh-public/@wcm/@sop/documents/downloadable/ucm_319826.pdf. Accessed May 9, 2014.
- Committee on Standards for Systematic Reviews of Comparative Effectiveness Research, Institute of Medicine. Finding What Works in Health Care: Standards for Systematic Reviews. Washington, DC: The National Academies Press; 2011.
- Committee on Standards for Developing Trustworthy Clinical Practice Guidelines, Institute of Medicine. Clinical Practice Guidelines We Can Trust. Washington, DC: The National Academies Press; 2011.
- Fuster V, Ryden LE, Cannom DS, et al. 2011 ACCF/AHA/HRS focused updates incorporated into the ACC/AHA/ESC 2006 guidelines for the management of patients with atrial fibrillation: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines developed in partnership with the European Society of Cardiology and in collaboration with the European Heart Rhythm Association and the Heart Rhythm Society. *Circulation*. 2011;123:e269–367.
- Wann LS, Curtis AB, Ellenbogen KA, et al. 2011 ACCF/AHA/HRS focused update on the management of patients with atrial fibrillation (update on dabigatran): a report of the American College of Cardiology Foundation/American Heart Association Task Force on practice guidelines. *Circulation*. 2011;123:1144–50.
- Wann LS, Curtis AB, January CT, et al. 2011 ACCF/AHA/HRS focused update on the management of patients with atrial fibrillation (updating the 2006 guideline): a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2011;123:104–23.
- Al-Khatib SM, Allen Lapointe N, Chatterjee R, et al. Treatment of Atrial Fibrillation. Comparative Effectiveness Review 119. (Prepared by the Duke Evidence-based Practice Center under Contract No. 290-2007-10066-I.) AHRQ Publication No.13-EHC095-EF. Rockville, MD: Agency for Healthcare Research and Quality; June 2013. Available at: <http://www.effectivehealthcare.ahrq.gov/ehc/products/358/1559/atrial-fibrillation-report-130628.pdf>. Accessed August 14, 2014.
- Lopes RD, Crowley MJ, Shah BR, et al. Stroke Prevention in Atrial Fibrillation. Comparative Effectiveness Review No. 123. (Prepared by the Duke Evidence-based Practice Center under Contract No. 290-2007-10066-I.) AHRQ Publication No. 13-EHC113-EF. Rockville, MD: Agency for Healthcare Research and Quality; August 2013. Available at: <http://www.effectivehealthcare.ahrq.gov/ehc/products/352/1668/stroke-atrial-fibrillation-report-130821.pdf>. Accessed August 14, 2014.
- Chobanian AV, Bakris GL, Black HR, et al. Seventh report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. *Hypertension*. 2003;42:1206–52.
- Greenland P, Alpert JS, Beller GA, et al. 2010 ACCF/AHA guideline for assessment of cardiovascular risk in asymptomatic adults: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2010;122:e584–636.
- Hillis LD, Smith PK, Anderson JL, et al. 2011 ACCF/AHA guideline for coronary artery bypass graft surgery: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. Developed in collaboration with the American Association for Thoracic Surgery, Society of Cardiovascular Anesthesiologists, and Society of Thoracic Surgeons. *Circulation*. 2011;124:e652–735.
- Gersh BJ, Maron BJ, Bonow RO, et al. 2011 ACCF/AHA guideline for the diagnosis and treatment of hypertrophic cardiomyopathy: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2011;124:e783–831.
- Levine GN, Bates ER, Blankenship JC, et al. 2011 ACCF/AHA/SCAI guideline for percutaneous coronary intervention: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Society for Cardiovascular Angiography and Interventions. *Circulation*. 2011;124: e574–651.
- Smith SC Jr, Benjamin EJ, Bonow RO, et al. AHA/ACC secondary prevention and risk reduction therapy for patients with coronary and other atherosclerotic vascular disease: 2011 update: a guideline from the American Heart Association and American College of Cardiology Foundation. *Circulation*. 2011;124:2458–73.
- Skanes AC, Healey JS, Cairns JA, et al. Focused 2012 update of the Canadian Cardiovascular Society atrial fibrillation guidelines: recommendations for stroke prevention and rate/rhythm control. *Can J Cardiol*. 2012;28:125–36.
- Camm AJ, Lip GY, De Caterina R, et al. 2012 focused update of the ESC guidelines for the management of atrial fibrillation: an update of the 2010 ESC Guidelines for the management of atrial fibrillation*. Developed with the special contribution of the European Heart Rhythm Association. *Eur Heart J*. 2012;33:2719–47.
- Fihn SD, Gardin JM, Abrams J, et al. 2012 ACCF/AHA/ACP/AATS/PCNA/SCAI/STS guideline for the diagnosis and management of patients with stable ischemic heart disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on

- Practice Guidelines, and the American College of Physicians, American Association for Thoracic Surgery, Preventive Cardiovascular Nurses Association, Society for Cardiovascular Angiography and Interventions, and Society of Thoracic Surgeons. *Circulation*. 2012;126:e354-e471.
17. Eikelboom JW, Hirsh J, Spencer FA, et al. Antiplatelet drugs: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012;141:e89-119S.
 18. Tracy CM, Epstein AE, Darbar D, et al. 2012 ACCF/AHA/HRS focused update incorporated into the ACCF/AHA/HRS 2008 guidelines for device-based therapy of cardiac rhythm abnormalities: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society. *Circulation*. 2012;126:e354-471.
 19. Yancy CW, Jessup M, Bozkurt B, et al. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;128:e240-327.
 20. O'Gara PT, Kushner FG, Ascheim DD, et al. 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;127:e362-425.
 21. Anderson JL, Adams CD, Antman EM, et al. 2012 ACCF/AHA focused update incorporated into the ACCF/AHA 2007 guidelines for the management of patients with unstable angina/non-ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;127:e663-828.
 22. Nishimura RA, Otto CM, Bonow RO, et al. 2014 AHA/ACC guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;129:e521-643.
 23. Goff DC Jr, Lloyd-Jones DM, Bennett G, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;129(suppl 2):S49-73.
 24. Eckel RH, Jakicic JM, Ard JD, et al. 2013 AHA/ACC guideline on lifestyle management to reduce cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;129(suppl 2):S76-99.
 25. Jensen MD, Ryan DH, Apovian CM, et al. 2013 AHA/ACC/TOS guideline for the management of overweight and obesity in adults: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines and The Obesity Society. *Circulation*. 2014;129(suppl 2):S102-138.
 26. Stone NJ, Robinson J, Lichtenstein AH, et al. 2013 ACC/AHA guideline on the treatment of blood cholesterol to reduce atherosclerotic cardiovascular risk in adults: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;129(suppl 2):S1-45.
 27. Furie KL, Goldstein LB, Albers GW, et al. Oral antithrombotic agents for the prevention of stroke in nonvalvular atrial fibrillation: a science advisory for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2012;43:3442-53.
 28. Calkins H, Kuck KH, Cappato R, et al. 2012 HRS/EHRA/ECAS expert consensus statement on catheter and surgical ablation of atrial fibrillation: recommendations for patient selection, procedural techniques, patient management and follow-up, definitions, endpoints, and research trial design: a report of the Heart Rhythm Society (HRS) Task Force on Catheter and Surgical Ablation of Atrial Fibrillation. Developed in partnership with the European Heart Rhythm Association (EHRA), a registered branch of the European Society of Cardiology (ESC) and the European Cardiac Arrhythmia Society (ECAS); and in collaboration with the American College of Cardiology (ACC), American Heart Association (AHA), the Asia Pacific Heart Rhythm Society (APHRS), and the Society of Thoracic Surgeons (STS). *Heart Rhythm*. 2012;9:632-96.
 29. Wolf PA, Benjamin EJ, Belanger AJ, et al. Secular trends in the prevalence of atrial fibrillation: the Framingham Study. *Am Heart J*. 1996;131:790-5.
 30. Camm AJ, Kirchhof P, Lip GY, et al. Guidelines for the management of atrial fibrillation: the Task Force for the Management of Atrial Fibrillation of the European Society of Cardiology (ESC). *Eur Heart J*. 2010;31:2369-429.
 31. Go AS, Hylek EM, Phillips KA, et al. Prevalence of diagnosed atrial fibrillation in adults: national implications for rhythm management and stroke prevention: the AnTicoagulation and Risk Factors in Atrial Fibrillation (ATRIA) Study. *JAMA*. 2001;285:2370-5.
 32. McManus DD, Rienstra M, Benjamin EJ. An update on the prognosis of patients with atrial fibrillation. *Circulation*. 2012;126:e143-6.
 33. Lloyd-Jones DM, Wang TJ, Leip EP, et al. Lifetime risk for development of atrial fibrillation: the Framingham Heart Study. *Circulation*. 2004;110:1042-6.
 34. Alonso A, Agarwal SK, Soliman EZ, et al. Incidence of atrial fibrillation in whites and African-Americans: the Atherosclerosis Risk in Communities (ARIC) study. *Am Heart J*. 2009;158:111-7.
 35. Kannel WB, Wolf PA, Benjamin EJ, et al. Prevalence, incidence, prognosis, and predisposing conditions for atrial fibrillation: population-based estimates. *Am J Cardiol*. 1998;82:2N-9N.
 36. Wolf PA, Abbott RD, Kannel WB. Atrial fibrillation: a major contributor to stroke in the elderly. The Framingham Study. *Arch Intern Med*. 1987;147:1561-4.
 37. Miller PS, Andersson FL, Kalra L. Are cost benefits of anticoagulation for stroke prevention in atrial fibrillation underestimated? *Stroke*. 2005;36:360-6.
 38. Wang TJ, Larson MG, Levy D, et al. Temporal relations of atrial fibrillation and congestive heart failure and their joint influence on mortality: the Framingham Heart Study. *Circulation*. 2003;107:2920-5.
 39. Krahn AD, Manfreda J, Tate RB, et al. The natural history of atrial fibrillation: incidence, risk factors, and prognosis in the Manitoba Follow-Up Study. *Am J Med*. 1995;98:476-84.
 40. Stewart S, Hart CL, Hole DJ, et al. A population-based study of the long-term risks associated with atrial fibrillation: 20-year follow-up of the Renfrew/Paisley study. *Am J Med*. 2002;113:359-64.
 41. Ott A, Bretelet MM, de Bruyne MC, et al. Atrial fibrillation and dementia in a population-based study. The Rotterdam Study. *Stroke*. 1997;28:316-21.
 42. Go AS, Mozaffarian D, Roger VL, et al. Heart disease and stroke statistics—2014 update: a report from the American Heart Association. *Circulation*. 2014;129:e28-292.
 43. Kim MH, Johnston SS, Chu BC, et al. Estimation of total incremental health care costs in patients with atrial fibrillation in the United States. *Circ Cardiovasc Qual Outcomes*. 2011;4:313-20.
 44. Office of Information Products and Data Analytics CMMS. CMS administrative claims data, January 2011 - December 2011, From the Chronic Condition Warehouse. 2012. Available at: www.ccwdata.org. Accessed July 15, 2014.
 45. Ikeda T, Murai H, Kaneko S, et al. Augmented single-unit muscle sympathetic nerve activity in heart failure with chronic atrial fibrillation. *J Physiol*. 2012;590:509-18.
 46. Segerson NM, Sharma N, Smith ML, et al. The effects of rate and irregularity on sympathetic nerve activity in human subjects. *Heart Rhythm*. 2007;4:20-6.
 47. Hsu LF, Jais P, Sanders P, et al. Catheter ablation for atrial fibrillation in congestive heart failure. *N Engl J Med*. 2004;351:2373-83.
 48. Nabauer M, Gerth A, Limbourg T, et al. The Registry of the German Competence NETwork on Atrial Fibrillation: patient characteristics and initial management. *Europace*. 2009;11:423-34.
 49. Jabre P, Jouven X, Adnet F, et al. Atrial fibrillation and death after myocardial infarction: a community study. *Circulation*. 2011;123:2094-100.
 50. Olivetto I, Cecchi F, Casey SA, et al. Impact of atrial fibrillation on the clinical course of hypertrophic cardiomyopathy. *Circulation*. 2001;104:2517-24.
 51. Nerheim P, Birger-Botkin S, Piracha L, et al. Heart failure and sudden death in patients with tachycardia-induced cardiomyopathy and recurrent tachycardia. *Circulation*. 2004;110:247-52.
 52. Van Gelder IC, Crijns HJ, Blanksma PK, et al. Time course of hemodynamic changes and improvement of exercise tolerance after cardioversion of chronic atrial fibrillation unassociated with cardiac valve disease. *Am J Cardiol*. 1993;72:560-6.
 53. Gelzer AR, Moise NS, Vaidya D, et al. Temporal organization of atrial activity and irregular ventricular rhythm during spontaneous atrial fibrillation: an in vivo study in the horse. *J Cardiovasc Electrophysiol*. 2000;11:773-84.
 54. Kurian T, Ambrosi C, Hucker W, et al. Anatomy and electrophysiology of the human AV node. *Pacing Clin Electrophysiol*. 2010;33:754-62.
 55. Van Den Berg MP, Crijns HJ, Haaksma J, et al. Analysis of vagal effects on ventricular rhythm in patients with atrial fibrillation. *Clin Sci (Lond)*. 1994;86:531-5.
 56. Williams L, Frenneaux M. Syncope in hypertrophic cardiomyopathy: mechanisms and consequences for treatment. *Europace*. 2007;9:817-22.
 57. Ammass NM, Seward JB, Bailey KR, et al. Clinical profile and outcome of idiopathic restrictive cardiomyopathy. *Circulation*. 2000;101:2490-6.

58. Wang YC, Lin JL, Hwang JJ, et al. Left atrial dysfunction in patients with atrial fibrillation after successful rhythm control for >3 months. *Chest*. 2005;128:2551–6.
59. Brookes CI, White PA, Staples M, et al. Myocardial contractility is not constant during spontaneous atrial fibrillation in patients. *Circulation*. 1998;98:1762–8.
60. Clark DM, Plumb VJ, Epstein AE, et al. Hemodynamic effects of an irregular sequence of ventricular cycle lengths during atrial fibrillation. *J Am Coll Cardiol*. 1997;30:1039–45.
61. Healey JS, Connolly SJ, Gold MR, et al. Subclinical atrial fibrillation and the risk of stroke. *N Engl J Med*. 2012;366:120–9.
62. Santini M, Gasparini M, Landolina M, et al. Device-detected atrial tachyarrhythmias predict adverse outcome in real-world patients with implantable biventricular defibrillators. *J Am Coll Cardiol*. 2011;57:167–72.
63. Evans W, Swann P. Lone auricular fibrillation. *Br Heart J*. 1954;16:189–94.
64. Blomstrom-Lundqvist C, Scheinman MM, Aliot EM, et al. ACC/AHA/ESC guidelines for the management of patients with supraventricular arrhythmias—executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines and the European Society of Cardiology Committee for Practice Guidelines (Writing Committee to Develop Guidelines for the Management of Patients With Supraventricular Arrhythmias) developed in collaboration with NASPE-Heart Rhythm Society. *Circulation*. 2003;108:1871–909.
65. Saoudi N, Cosio F, Waldo A, et al. Classification of atrial flutter and regular atrial tachycardia according to electrophysiologic mechanism and anatomic bases: a statement from a joint expert group from the Working Group of Arrhythmias of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. *J Cardiovasc Electrophysiol*. 2001;12:852–66.
66. Hocini M, Shah AJ, Nault I, et al. Localized reentry within the left atrial appendage: arrhythmogenic role in patients undergoing ablation of persistent atrial fibrillation. *Heart Rhythm*. 2011;8:1853–61.
67. Miyazaki S, Shah AJ, Kobori A, et al. How to approach reentrant atrial tachycardia after atrial fibrillation ablation. *Circ Arrhythm Electrophysiol*. 2012;5:e1–7.
68. Wells JL Jr, MacLean WA, James TN, et al. Characterization of atrial flutter: studies in man after open heart surgery using fixed atrial electrodes. *Circulation*. 1979;60:665–73.
69. Waldo AL, Feld GK. Inter-relationships of atrial fibrillation and atrial flutter mechanisms and clinical implications. *J Am Coll Cardiol*. 2008;51:779–86.
70. Ishii Y, Gleva MJ, Gamache MC, et al. Atrial tachyarrhythmias after the maze procedure: incidence and prognosis. *Circulation*. 2004;110:II164–8.
71. Akar JG, Al-Chekakie MO, Hai A, et al. Surface electrocardiographic patterns and electrophysiologic characteristics of atrial flutter following modified radiofrequency MAZE procedures. *J Cardiovasc Electrophysiol*. 2007;18:349–55.
72. Chugh A, Latchamsetty R, Oral H, et al. Characteristics of cavotricuspid isthmus-dependent atrial flutter after left atrial ablation of atrial fibrillation. *Circulation*. 2006;113:609–15.
73. Perez FJ, Schubert CM, Parvez B, et al. Long-term outcomes after catheter ablation of cavo-tricuspid isthmus dependent atrial flutter: a meta-analysis. *Circ Arrhythm Electrophysiol*. 2009;2:393–401.
74. Ellis K, Wazni O, Marrouche N, et al. Incidence of atrial fibrillation post-cavotricuspid isthmus ablation in patients with typical atrial flutter: left-atrial size as an independent predictor of atrial fibrillation recurrence. *J Cardiovasc Electrophysiol*. 2007;18:799–802.
75. Gupta D, Earley MJ, Haywood GA, et al. Can atrial fibrillation with a coarse electrocardiographic appearance be treated with catheter ablation of the tricuspid valve-inferior vena cava isthmus? Results of a multicentre randomised controlled trial. *Heart*. 2007;93:688–93.
76. Knight BP, Michaud GF, Strickberger SA, et al. Electrocardiographic differentiation of atrial flutter from atrial fibrillation by physicians. *J Electrocardiol*. 1999;32:315–9.
77. Reithmann C, Hoffmann E, Spitzlberger G, et al. Catheter ablation of atrial flutter due to amiodarone therapy for paroxysmal atrial fibrillation. *Eur Heart J*. 2000;21:565–72.
78. Bertaglia E, Zoppo F, Bonso A, et al. Long term follow up of radiofrequency catheter ablation of atrial flutter: clinical course and predictors of atrial fibrillation occurrence. *Heart*. 2004;90:59–63.
79. McElderry HT, McGiffin DC, Plumb VJ, et al. Proarrhythmic aspects of atrial fibrillation surgery: mechanisms of postoperative macroreentrant tachycardias. *Circulation*. 2008;117:155–62.
80. Shah D. ECG manifestations of left atrial flutter. *Curr Opin Cardiol*. 2009;24:35–41.
81. Savelieva I, Kakouros N, Kourliouros A, et al. Upstream therapies for management of atrial fibrillation: review of clinical evidence and implications for European Society of Cardiology guidelines: part I: primary prevention. *Europace*. 2011;13:308–28.
82. Wakili R, Voigt N, Kaab S, et al. Recent advances in the molecular pathophysiology of atrial fibrillation. *J Clin Invest*. 2011;121:2955–68.
83. Kistler PM, Sanders P, Fynn SP, et al. Electrophysiologic and electro-anatomic changes in the human atrium associated with age. *J Am Coll Cardiol*. 2004;44:109–16.
84. Frustaci A, Chimenti C, Bellocci F, et al. Histological substrate of atrial biopsies in patients with lone atrial fibrillation. *Circulation*. 1997;96:1180–4.
85. Allesie M, Ausma J, Schotten U. Electrical, contractile and structural remodeling during atrial fibrillation. *Cardiovasc Res*. 2002;54:230–46.
86. Aviles RJ, Martin DO, Apperson-Hansen C, et al. Inflammation as a risk factor for atrial fibrillation. *Circulation*. 2003;108:3006–10.
87. Jesel L, Abbas M, Toti F, et al. Microparticles in atrial fibrillation: a link between cell activation or apoptosis, tissue remodelling and thrombogenicity. *Int J Cardiol*. 2013;168:660–9.
88. Burstein B, Nattel S. Atrial fibrosis: mechanisms and clinical relevance in atrial fibrillation. *J Am Coll Cardiol*. 2008;51:802–9.
89. Li D, Fareh S, Leung TK, et al. Promotion of atrial fibrillation by heart failure in dogs: atrial remodeling of a different sort. *Circulation*. 1999;100:87–95.
90. Akoum N, McGann C, Vergara G, et al. Atrial fibrosis quantified using late gadolinium enhancement MRI is associated with sinus node dysfunction requiring pacemaker implant. *J Cardiovasc Electrophysiol*. 2012;23:44–50.
91. Daccarett M, Badger TJ, Akoum N, et al. Association of left atrial fibrosis detected by delayed-enhancement magnetic resonance imaging and the risk of stroke in patients with atrial fibrillation. *J Am Coll Cardiol*. 2011;57:831–8.
92. Dickfeld T, Kato R, Zviman M, et al. Characterization of radiofrequency ablation lesions with gadolinium-enhanced cardiovascular magnetic resonance imaging. *J Am Coll Cardiol*. 2006;47:370–8.
93. McGann CJ, Kholmovski EG, Oakes RS, et al. New magnetic resonance imaging-based method for defining the extent of left atrial wall injury after the ablation of atrial fibrillation. *J Am Coll Cardiol*. 2008;52:1263–71.
94. Oakes RS, Badger TJ, Kholmovski EG, et al. Detection and quantification of left atrial structural remodeling with delayed-enhancement magnetic resonance imaging in patients with atrial fibrillation. *Circulation*. 2009;119:1758–67.
95. Peters DC, Wylie JV, Hauser TH, et al. Detection of pulmonary vein and left atrial scar after catheter ablation with three-dimensional navigated delayed enhancement MR imaging: initial experience. *Radiology*. 2007;243:690–5.
96. Haissaguerre M, Jais P, Shah DC, et al. Spontaneous initiation of atrial fibrillation by ectopic beats originating in the pulmonary veins. *N Engl J Med*. 1998;339:659–66.
97. Jais P, Haissaguerre M, Shah DC, et al. A focal source of atrial fibrillation treated by discrete radiofrequency ablation. *Circulation*. 1997;95:572–6.
98. Scherf D, Romano FJ, Terranova R. Experimental studies on auricular flutter and auricular fibrillation. *Am Heart J*. 1948;36:241–51.
99. Ehrlich JR, Cha TJ, Zhang L, et al. Cellular electrophysiology of canine pulmonary vein cardiomyocytes: action potential and ionic current properties. *J Physiol*. 2003;551:801–13.
100. Gherghiceanu M, Hinescu ME, Andrei F, et al. Interstitial Cajal-like cells (ICLC) in myocardial sleeves of human pulmonary veins. *J Cell Mol Med*. 2008;12:1777–81.
101. Levin MD, Lu MM, Petrenko NB, et al. Melanocyte-like cells in the heart and pulmonary veins contribute to atrial arrhythmia triggers. *J Clin Invest*. 2009;119:3420–36.
102. Nattel S, Dobrev D. The multidimensional role of calcium in atrial fibrillation pathophysiology: mechanistic insights and therapeutic opportunities. *Eur Heart J*. 2012;33:1870–7.
103. Hove-Madsen L, Llach A, Bayes-Genis A, et al. Atrial fibrillation is associated with increased spontaneous calcium release from the sarcoplasmic reticulum in human atrial myocytes. *Circulation*. 2004;110:1358–63.
104. Neef S, Dybkova N, Sossalla S, et al. CaMKII-dependent diastolic SR Ca²⁺ leak and elevated diastolic Ca²⁺ levels in right atrial myocardium of patients with atrial fibrillation. *Circ Res*. 2010;106:1134–44.
105. Voigt N, Li N, Wang Q, et al. Enhanced sarcoplasmic reticulum Ca²⁺ leak and increased Na⁺-Ca²⁺ exchanger function underlie

- delayed afterdepolarizations in patients with chronic atrial fibrillation. *Circulation*. 2012;125:2059–70.
106. Vest JA, Wehrens XH, Reiken SR, et al. Defective cardiac ryanodine receptor regulation during atrial fibrillation. *Circulation*. 2005;111:2025–32.
 107. Berenfeld O, Mandapati R, Dixit S, et al. Spatially distributed dominant excitation frequencies reveal hidden organization in atrial fibrillation in the Langendorff-perfused sheep heart. *J Cardiovasc Electrophysiol*. 2000;11:869–79.
 108. He B, Scherlag BJ, Nakagawa H, et al. The intrinsic autonomic nervous system in atrial fibrillation: a review. *ISRN Cardiol*. 2012;2012:490674.
 109. Hunter RJ, Diab I, Tayebjee M, et al. Characterization of fractionated atrial electrograms critical for maintenance of atrial fibrillation: a randomized, controlled trial of ablation strategies (the CFAE AF trial). *Circ Arrhythm Electrophysiol*. 2011;4:622–9.
 110. Mandapati R, Skanes A, Chen J, et al. Stable microreentrant sources as a mechanism of atrial fibrillation in the isolated sheep heart. *Circulation*. 2000;101:194–9.
 111. Narayan SM, Patel J, Mulpuru S, et al. Focal impulse and rotor modulation ablation of sustaining rotors abruptly terminates persistent atrial fibrillation to sinus rhythm with elimination on follow-up: a video case study. *Heart Rhythm*. 2012;9:1436–9.
 112. Narayan SM, Krummen DE, Shivkumar K, et al. Treatment of atrial fibrillation by the ablation of localized sources: CONFIRM (Conventional Ablation for Atrial Fibrillation With or Without Focal Impulse and Rotor Modulation) trial. *J Am Coll Cardiol*. 2012;60:628–36.
 113. Narayan SM, Krummen DE, Rappel WJ. Clinical mapping approach to diagnose electrical rotors and focal impulse sources for human atrial fibrillation. *J Cardiovasc Electrophysiol*. 2012;23:447–54.
 114. Cuculich PS, Wang Y, Lindsay BD, et al. Noninvasive characterization of epicardial activation in humans with diverse atrial fibrillation patterns. *Circulation*. 2010;122:1364–72.
 115. Moe GK, Abildskov JA. Atrial fibrillation as a self-sustaining arrhythmia independent of focal discharge. *Am Heart J*. 1959;58:59–70.
 116. Park HW, Shen MJ, Lin SF, et al. Neural mechanisms of atrial fibrillation. *Curr Opin Cardiol*. 2012;27:24–8.
 117. Choi EK, Shen MJ, Han S, et al. Intrinsic cardiac nerve activity and paroxysmal atrial tachyarrhythmia in ambulatory dogs. *Circulation*. 2010;121:2615–23.
 118. Shen MJ, Choi EK, Tan AY, et al. Neural mechanisms of atrial arrhythmias. *Nat Rev Cardiol*. 2012;9:30–9.
 119. Pappone C, Santinelli V, Manguso F, et al. Pulmonary vein denervation enhances long-term benefit after circumferential ablation for paroxysmal atrial fibrillation. *Circulation*. 2004;109:327–34.
 120. Scanavacca M, Pisani CF, Hachul D, et al. Selective atrial vagal denervation guided by evoked vagal reflex to treat patients with paroxysmal atrial fibrillation. *Circulation*. 2006;114:876–85.
 121. Pokushalov E, Romanov A, Katritsis DG, et al. Ganglionated plexus ablation vs linear ablation in patients undergoing pulmonary vein isolation for persistent/long-standing persistent atrial fibrillation: a randomized comparison. *Heart Rhythm*. 2013;10:1280–6.
 122. Coumel P, Attuel P, Lavalée J, et al. The atrial arrhythmia syndrome of vagal origin [in French]. *Arch Mal Coeur Vaiss*. 1978;71:645–56.
 123. Patton KK, Zacks ES, Chang JY, et al. Clinical subtypes of lone atrial fibrillation. *Pacing Clin Electrophysiol*. 2005;28:630–8.
 124. Frick M, Frykman V, Jensen-Ustad M, et al. Factors predicting success rate and recurrence of atrial fibrillation after first electrical cardioversion in patients with persistent atrial fibrillation. *Clin Cardiol*. 2001;24:238–44.
 125. Nattel S, Burstein B, Dobrev D. Atrial remodeling and atrial fibrillation: mechanisms and implications. *Circ Arrhythm Electrophysiol*. 2008;1:62–73.
 126. Wijffels MC, Kirchhof CJ, Dorland R, et al. Atrial fibrillation begets atrial fibrillation: a study in awake chronically instrumented goats. *Circulation*. 1995;92:1954–68.
 127. Goldstein RN, Ryu K, Khrestian C, et al. Prednisone prevents inducible atrial flutter in the canine sterile pericarditis model. *J Cardiovasc Electrophysiol*. 2008;19:74–81.
 128. Dudley SC Jr, Hoch NE, McCann LA, et al. Atrial fibrillation increases production of superoxide by the left atrium and left atrial appendage: role of the NADPH and xanthine oxidases. *Circulation*. 2005;112:1266–73.
 129. Carnes CA, Chung MK, Nakayama T, et al. Ascorbate attenuates atrial pacing-induced peroxynitrite formation and electrical remodeling and decreases the incidence of postoperative atrial fibrillation. *Circ Res*. 2001;89:E32–8.
 130. Healey JS, Baranchuk A, Crystal E, et al. Prevention of atrial fibrillation with angiotensin-converting enzyme inhibitors and angiotensin receptor blockers: a meta-analysis. *J Am Coll Cardiol*. 2005;45:1832–9.
 131. Ueng KC, Tsai TP, Yu WC, et al. Use of enalapril to facilitate sinus rhythm maintenance after external cardioversion of long-standing persistent atrial fibrillation: results of a prospective and controlled study. *Eur Heart J*. 2003;24:2090–8.
 132. Zaman AG, Kearney MT, Schechter C, et al. Angiotensin-converting enzyme inhibitors as adjunctive therapy in patients with persistent atrial fibrillation. *Am Heart J*. 2004;147:823–7.
 133. Komatsu T, Nakamura S, Suzuki O, et al. Long-term efficacy of combination therapy using antiarrhythmic agents and angiotensin converting enzyme inhibitor in patients with paroxysmal and persistent atrial fibrillation: importance of the timing of administration [in Japanese]. *J Cardiol*. 2003;41:73–80.
 134. Swedberg K, Zannad F, McMurray JJ, et al. Eplerenone and atrial fibrillation in mild systolic heart failure: results from the EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization And Survival Study in Heart Failure) study. *J Am Coll Cardiol*. 2012;59:1598–603.
 135. Huxley RR, Lopez FL, Folsom AR, et al. Absolute and attributable risks of atrial fibrillation in relation to optimal and borderline risk factors: the Atherosclerosis Risk in Communities (ARIC) study. *Circulation*. 2011;123:1501–8.
 136. Liakopoulos OJ, Choi YH, Kuhn EW, et al. Statins for prevention of atrial fibrillation after cardiac surgery: a systematic literature review. *J Thorac Cardiovasc Surg*. 2009;138:678–86.
 137. Mathew JP, Fontes ML, Tudor IC, et al. A multicenter risk index for atrial fibrillation after cardiac surgery. *JAMA*. 2004;291:1720–9.
 138. Rathore SS, Berger AK, Weinert KP, et al. Acute myocardial infarction complicated by atrial fibrillation in the elderly: prevalence and outcomes. *Circulation*. 2000;101:969–74.
 139. Benjamin EJ, Levy D, Vaziri SM, et al. Independent risk factors for atrial fibrillation in a population-based cohort: the Framingham Heart Study. *JAMA*. 1994;271:840–4.
 140. Frost L, Hune LJ, Vestergaard P. Overweight and obesity as risk factors for atrial fibrillation or flutter: the Danish Diet, Cancer, and Health Study. *Am J Med*. 2005;118:489–95.
 141. Wang TJ, Parise H, Levy D, et al. Obesity and the risk of new-onset atrial fibrillation. *JAMA*. 2004;292:2471–7.
 142. Gami AS, Hodge DO, Herges RM, et al. Obstructive sleep apnea, obesity, and the risk of incident atrial fibrillation. *J Am Coll Cardiol*. 2007;49:565–71.
 143. Heeringa J, Kors JA, Hofman A, et al. Cigarette smoking and risk of atrial fibrillation: the Rotterdam Study. *Am Heart J*. 2008;156:1163–9.
 144. Aizer A, Gaziano JM, Cook NR, et al. Relation of vigorous exercise to risk of atrial fibrillation. *Am J Cardiol*. 2009;103:1572–7.
 145. Mont L, Sambola A, Brugada J, et al. Long-lasting sport practice and lone atrial fibrillation. *Eur Heart J*. 2002;23:477–82.
 146. Frost L, Frost P, Vestergaard P. Work related physical activity and risk of a hospital discharge diagnosis of atrial fibrillation or flutter: the Danish Diet, Cancer, and Health Study. *Occup Environ Med*. 2005;62:49–53.
 147. Conen D, Tedrow UB, Cook NR, et al. Alcohol consumption and risk of incident atrial fibrillation in women. *JAMA*. 2008;300:2489–96.
 148. Frost L, Vestergaard P. Alcohol and risk of atrial fibrillation or flutter: a cohort study. *Arch Intern Med*. 2004;164:1993–8.
 149. Kodama S, Saito K, Tanaka S, et al. Alcohol consumption and risk of atrial fibrillation: a meta-analysis. *J Am Coll Cardiol*. 2011;57:427–36.
 150. Sawin CT, Geller A, Wolf PA, et al. Low serum thyrotropin concentrations as a risk factor for atrial fibrillation in older persons. *N Engl J Med*. 1994;331:1249–52.
 151. Cappola AR, Fried LP, Arnold AM, et al. Thyroid status, cardiovascular risk, and mortality in older adults. *JAMA*. 2006;295:1033–41.
 152. Frost L, Vestergaard P, Mosekilde L. Hyperthyroidism and risk of atrial fibrillation or flutter: a population-based study. *Arch Intern Med*. 2004;164:1675–8.
 153. Mitchell GF, Vasan RS, Keyes MJ, et al. Pulse pressure and risk of new-onset atrial fibrillation. *JAMA*. 2007;297:709–15.
 154. Marcus GM, Alonso A, Peralta CA, et al. European ancestry as a risk factor for atrial fibrillation in African Americans. *Circulation*. 2010;122:2009–15.
 155. Lubitz SA, Yin X, Fontes JD, et al. Association between familial atrial fibrillation and risk of new-onset atrial fibrillation. *JAMA*. 2010;304:2263–9.
 156. Ellinor PT, Lunetta KL, Albert CM, et al. Meta-analysis identifies six new susceptibility loci for atrial fibrillation. *Nat Genet*. 2012;44:670–5.
 157. Gudbjartsson DF, Holm H, Gretarsdottir S, et al. A sequence variant in ZFXH3 on 16q22 associates with atrial fibrillation and ischemic stroke. *Nat Genet*. 2009;41:876–8.

158. Gudbjartsson DF, Arnar DO, Helgadóttir A, et al. Variants conferring risk of atrial fibrillation on chromosome 4q25. *Nature*. 2007;448:353–7.
159. Benjamin EJ, Rice KM, Arking DE, et al. Variants in ZFHX3 are associated with atrial fibrillation in individuals of European ancestry. *Nat Genet*. 2009;41:879–81.
160. Pritchett AM, Jacobsen SJ, Mahoney DW, et al. Left atrial volume as an index of left atrial size: a population-based study. *J Am Coll Cardiol*. 2003;41:1036–43.
161. Cao JJ, Thach C, Manolio TA, et al. C-reactive protein, carotid intima-media thickness, and incidence of ischemic stroke in the elderly: the Cardiovascular Health Study. *Circulation*. 2003;108:166–70.
162. Patton KK, Ellinor PT, Heckbert SR, et al. N-terminal pro-B-type natriuretic peptide is a major predictor of the development of atrial fibrillation: the Cardiovascular Health Study. *Circulation*. 2009;120:1768–74.
163. Wang TJ, Larson MG, Levy D, et al. Plasma natriuretic peptide levels and the risk of cardiovascular events and death. *N Engl J Med*. 2004;350:655–63.
164. Klein AL, Grimm RA, Murray RD, et al. Use of transesophageal echocardiography to guide cardioversion in patients with atrial fibrillation. *N Engl J Med*. 2001;344:1411–20.
165. Manning WJ, Silverman DI, Keighley CS, et al. Transesophageal echocardiographically facilitated early cardioversion from atrial fibrillation using short-term anticoagulation: final results of a prospective 4.5-year study. *J Am Coll Cardiol*. 1995;25:1354–61.
166. Monahan K, Redline S. Role of obstructive sleep apnea in cardiovascular disease. *Curr Opin Cardiol*. 2011;26:541–7.
167. Ahmad Y, Lip GY, Apostolakis S. New oral anticoagulants for stroke prevention in atrial fibrillation: impact of gender, heart failure, diabetes mellitus and paroxysmal atrial fibrillation. *Expert Rev Cardiovasc Ther*. 2012;10:1471–80.
168. Chiang CE, Naditch-Brule L, Murin J, et al. Distribution and risk profile of paroxysmal, persistent, and permanent atrial fibrillation in routine clinical practice: insight from the real-life global survey evaluating patients with atrial fibrillation international registry. *Circ Arrhythm Electrophysiol*. 2012;5:632–9.
169. Flaker G, Ezekowitz M, Yusuf S, et al. Efficacy and safety of dabigatran compared to warfarin in patients with paroxysmal, persistent, and permanent atrial fibrillation: results from the RE-LY (Randomized Evaluation of Long-Term Anticoagulation Therapy) study. *J Am Coll Cardiol*. 2012;59:854–5.
170. Hohnloser SH, Duray GZ, Baber U, et al. Prevention of stroke in patients with atrial fibrillation: current strategies and future directions. *Eur Heart J*. 2007;10:H4–10.
171. Lip GY, Nieuwlaet R, Pisters R, et al. Refining clinical risk stratification for predicting stroke and thromboembolism in atrial fibrillation using a novel risk factor-based approach: the Euro Heart Survey on Atrial Fibrillation. *Chest*. 2010;137:263–72.
172. Olesen JB, Torp-Pedersen C, Hansen ML, et al. The value of the CHA2DS2-VASc score for refining stroke risk stratification in patients with atrial fibrillation with a CHADS2 score 0–1: a nationwide cohort study. *Thromb Haemost*. 2012;107:1172–9.
173. Mason PK, Lake DE, DiMarco JP, et al. Impact of the CHA2DS2-VASc score on anticoagulation recommendations for atrial fibrillation. *Am J Med*. 2012;125:603–6.
174. Cannegieter SC, Rosendaal FR, Wintzen AR, et al. Optimal oral anticoagulant therapy in patients with mechanical heart valves. *N Engl J Med*. 1995;333:11–7.
175. Acar J, Iung B, Boissel JP, et al. AREVA: multicenter randomized comparison of low-dose versus standard-dose anticoagulation in patients with mechanical prosthetic heart valves. *Circulation*. 1996;94:2107–12.
176. Hering D, Piper C, Bergemann R, et al. Thromboembolic and bleeding complications following St. Jude Medical valve replacement: results of the German Experience With Low-Intensity Anticoagulation Study. *Chest*. 2005;127:53–9.
177. Connolly SJ, Ezekowitz MD, Yusuf S, et al. Dabigatran versus warfarin in patients with atrial fibrillation. *N Engl J Med*. 2009;361:1139–51.
178. Patel MR, Mahaffey KW, Garg J, et al. Rivaroxaban versus warfarin in nonvalvular atrial fibrillation. *N Engl J Med*. 2011;365:883–91.
179. Granger CB, Alexander JH, McMurray JJ, et al. Apixaban versus warfarin in patients with atrial fibrillation. *N Engl J Med*. 2011;365:981–92.
180. Matchar DB, Jacobson A, Dolor R, et al. Effect of home testing of international normalized ratio on clinical events. *N Engl J Med*. 2010;363:1608–20.
181. Ezekowitz MD, James KE, Radford MJ, et al. Initiating and maintaining patients on warfarin anticoagulation: the importance of monitoring. *J Cardiovasc Pharmacol Ther*. 1999;4:3–8.
182. Hirsh J, Fuster V. Guide to anticoagulant therapy: part 2: oral anticoagulants. *Circulation*. 1994;89:1469–80.
183. Aguilar M, Hart R. Antiplatelet therapy for preventing stroke in patients with non-valvular atrial fibrillation and no previous history of stroke or transient ischemic attacks. *Cochrane Database Syst Rev*. 2005;(4):CD001925.
184. Hart RG, Pearce LA, Aguilar MI. Meta-analysis: antithrombotic therapy to prevent stroke in patients who have nonvalvular atrial fibrillation. *Ann Intern Med*. 2007;146:857–67.
185. Winkelmayer WC, Liu J, Setoguchi S, et al. Effectiveness and safety of warfarin initiation in older hemodialysis patients with incident atrial fibrillation. *Clin J Am Soc Nephrol*. 2011;6:2662–8.
186. Dewilde WJ, Oirbans T, Verheugt FW, et al. Use of clopidogrel with or without aspirin in patients taking oral anticoagulant therapy and undergoing percutaneous coronary intervention: an open-label, randomised, controlled trial. *Lancet*. 2013;381:1107–15.
187. Hariharan S, Madabushi R. Clinical pharmacology basis of deriving dosing recommendations for dabigatran in patients with severe renal impairment. *J Clin Pharmacol*. 2012;52:119S–25S.
188. Lehr T, Haertter S, Liesenfeld KH, et al. Dabigatran etexilate in atrial fibrillation patients with severe renal impairment: dose identification using pharmacokinetic modeling and simulation. *J Clin Pharmacol*. 2012;52:1373–8.
189. Connolly SJ, Eikelboom J, Joyner C, et al. Apixaban in patients with atrial fibrillation. *N Engl J Med*. 2011;364:806–17.
190. Van de Werf F, Brueckmann M, Connolly SJ, et al. A comparison of dabigatran etexilate with warfarin in patients with mechanical heart valves: the Randomized, phase II study to Evaluate the sAFety and pharmacokinetics of oral dabigatran etexilate in patients after heart valve replacement (RE-ALIGN). *Am Heart J*. 2012;163:931–7.
191. Ziegler PD, Glotzer TV, Daoud EG, et al. Incidence of newly detected atrial arrhythmias via implantable devices in patients with a history of thromboembolic events. *Stroke*. 2010;41:256–60.
192. Ziegler PD, Glotzer TV, Daoud EG, et al. Detection of previously undiagnosed atrial fibrillation in patients with stroke risk factors and usefulness of continuous monitoring in primary stroke prevention. *Am J Cardiol*. 2012;110:1309–14.
193. Glotzer TV, Hellkamp AS, Zimmerman J, et al. Atrial high rate episodes detected by pacemaker diagnostics predict death and stroke: report of the Atrial Diagnostics Ancillary Study of the M-Mode Selection Trial (MOST). *Circulation*. 2003;107:1614–9.
194. Glotzer TV, Daoud EG, Wyse DG, et al. The relationship between daily atrial tachyarrhythmia burden from implantable device diagnostics and stroke risk: the TRENDS study. *Circ Arrhythm Electrophysiol*. 2009;2:474–80.
195. Wolf PA, Abbott RD, Kannel WB. Atrial fibrillation as an independent risk factor for stroke: the Framingham Study. *Stroke*. 1991;22:983–8.
196. Lin HJ, Wolf PA, Kelly-Hayes M, et al. Stroke severity in atrial fibrillation: the Framingham Study. *Stroke*. 1996;27:1760–4.
197. Mant J, Hobbs FD, Fletcher K, et al. Warfarin versus aspirin for stroke prevention in an elderly community population with atrial fibrillation (the Birmingham Atrial Fibrillation Treatment of the Aged Study, BAFTA): a randomised controlled trial. *Lancet*. 2007;370:493–503.
198. Connolly S, Pogue J, Hart R, et al. Clopidogrel plus aspirin versus oral anticoagulation for atrial fibrillation in the Atrial Fibrillation Clopidogrel Trial with Irbesartan for prevention of Vascular Events (ACTIVE W): a randomised controlled trial. *Lancet*. 2006;367:1903–12.
199. Connolly SJ, Pogue J, Hart RG, et al. Effect of clopidogrel added to aspirin in patients with atrial fibrillation. *N Engl J Med*. 2009;360:2066–78.
200. Risk factors for stroke and efficacy of antithrombotic therapy in atrial fibrillation: analysis of pooled data from five randomized controlled trials. *Arch Intern Med*. 1994;154:1449–57.
201. Gage BF, Waterman AD, Shannon W, et al. Validation of clinical classification schemes for predicting stroke: results from the National Registry of Atrial Fibrillation. *JAMA*. 2001;285:2864–70.
202. Lip GY, Tse HF, Lane DA. Atrial fibrillation. *Lancet*. 2012;379:648–61.
203. Lane DA, Lip GY. Use of the CHA(2)DS(2)-VASc and HAS-BLED scores to aid decision making for thromboprophylaxis in nonvalvular atrial fibrillation. *Circulation*. 2012;126:860–5.
204. Gage BF, van Walraven C, Pearce L, et al. Selecting patients with atrial fibrillation for anticoagulation: stroke risk stratification in patients taking aspirin. *Circulation*. 2004;110:2287–92.
205. Friberg L, Benson L, Rosenqvist M, et al. Assessment of female sex as a risk factor in atrial fibrillation in Sweden: nationwide retrospective cohort study. *BMJ*. 2012;344:e3522.

206. Fang MC, Go AS, Chang Y, et al. A new risk scheme to predict warfarin-associated hemorrhage: the ATRIA (Anticoagulation and Risk Factors in Atrial Fibrillation) Study. *J Am Coll Cardiol*. 2011;58:395–401.
207. Gage BF, Yan Y, Milligan PE, et al. Clinical classification schemes for predicting hemorrhage: results from the National Registry of Atrial Fibrillation (NRAF). *Am Heart J*. 2006;151:713–9.
208. Ruiz-Gimenez N, Suarez C, Gonzalez R, et al. Predictive variables for major bleeding events in patients presenting with documented acute venous thromboembolism: findings from the RIETE Registry. *Thromb Haemost*. 2008;100:26–31.
209. Pisters R, Lane DA, Nieuwlaat R, et al. A novel user-friendly score (HAS-BLED) to assess 1-year risk of major bleeding in patients with atrial fibrillation: the Euro Heart Survey. *Chest*. 2010;138:1093–100.
210. Loewen P, Dahri K. Risk of bleeding with oral anticoagulants: an updated systematic review and performance analysis of clinical prediction rules. *Ann Hematol*. 2011;90:1191–200.
211. Stroke Prevention in Atrial Fibrillation Investigators. Stroke Prevention in Atrial Fibrillation Study: final results. *Circulation*. 1991;84:527–39.
212. Hart RG, Benavente O, McBride R, et al. Antithrombotic therapy to prevent stroke in patients with atrial fibrillation: a meta-analysis. *Ann Intern Med*. 1999;131:492–501.
213. Nutescu EA, Shapiro NL, Chevalier A, et al. A pharmacologic overview of current and emerging anticoagulants. *Cleve Clin J Med*. 2005;72 Suppl 1:S2–6.
214. Aguilar MI, Hart R. Oral anticoagulants for preventing stroke in patients with non-valvular atrial fibrillation and no previous history of stroke or transient ischemic attacks. *Cochrane Database Syst Rev*. 2005;(3):CD001927.
215. Aguilar MI, Hart R, Pearce LA. Oral anticoagulants versus antiplatelet therapy for preventing stroke in patients with non-valvular atrial fibrillation and no history of stroke or transient ischemic attacks. *Cochrane Database Syst Rev*. 2007;(3):CD006186.
216. Saxena R, Koudstaal PJ. Anticoagulants for preventing stroke in patients with nonrheumatic atrial fibrillation and a history of stroke or transient ischaemic attack. *Cochrane Database Syst Rev*. 2004;(2):CD000185.
217. Glazer NL, Dublin S, Smith NL, et al. Newly detected atrial fibrillation and compliance with antithrombotic guidelines. *Arch Intern Med*. 2007;167:246–52.
218. Lewis WR, Fonarow GC, LaBresh KA, et al. Differential use of warfarin for secondary stroke prevention in patients with various types of atrial fibrillation. *Am J Cardiol*. 2009;103:227–31.
219. Ogilvie IM, Newton N, Welner SA, et al. Underuse of oral anticoagulants in atrial fibrillation: a systematic review. *Am J Med*. 2010;123:638–45.
220. Waldo AL, Becker RC, Tapson VF, et al. Hospitalized patients with atrial fibrillation and a high risk of stroke are not being provided with adequate anticoagulation. *J Am Coll Cardiol*. 2005;46:1729–36.
221. Zimetbaum PJ, Thosani A, Yu HT, et al. Are atrial fibrillation patients receiving warfarin in accordance with stroke risk? *Am J Med*. 2010;123:446–53.
222. van Walraven C, Jennings A, Oake N, et al. Effect of study setting on anticoagulation control: a systematic review and metaregression. *Chest*. 2006;129:1155–66.
223. Baker WL, Cios DA, Sander SD, et al. Meta-analysis to assess the quality of warfarin control in atrial fibrillation patients in the United States. *J Manag Care Pharm*. 2009;15:244–52.
224. Agarwal S, Hachamovitch R, Menon V. Current trial-associated outcomes with warfarin in prevention of stroke in patients with nonvalvular atrial fibrillation: a meta-analysis. *Arch Intern Med*. 2012;172:623–31.
225. Diener HC, Connolly SJ, Ezekowitz MD, et al. Dabigatran compared with warfarin in patients with atrial fibrillation and previous transient ischaemic attack or stroke: a subgroup analysis of the RE-LY trial. *Lancet Neurol*. 2010;9:1157–63.
226. Limdi NA, Beasley TM, Baird MF, et al. Kidney function influences warfarin responsiveness and hemorrhagic complications. *J Am Soc Nephrol*. 2009;20:912–21.
227. Eikelboom JW, Wallentin L, Connolly SJ, et al. Risk of bleeding with 2 doses of dabigatran compared with warfarin in older and younger patients with atrial fibrillation: an analysis of the Randomized Evaluation of Long-Term Anticoagulant Therapy (RE-LY) trial. *Circulation*. 2011;123:2363–72.
228. Oldgren J, Alings M, Darius H, et al. Risks for stroke, bleeding, and death in patients with atrial fibrillation receiving dabigatran or warfarin in relation to the CHADS₂ score: a subgroup analysis of the RE-LY trial. *Ann Intern Med*. 2011;155:660–7, W204.
229. Southworth MR, Reichman ME, Unger EF. Dabigatran and postmarket reports of bleeding. *N Engl J Med*. 2013;368:1272–4.
230. Nagarakanti R, Ezekowitz MD, Oldgren J, et al. Dabigatran versus warfarin in patients with atrial fibrillation: an analysis of patients undergoing cardioversion. *Circulation*. 2011;123:131–6.
231. Uchino K, Hernandez AV. Dabigatran association with higher risk of acute coronary events: meta-analysis of noninferiority randomized controlled trials. *Arch Intern Med*. 2012;172:397–402.
232. Patel MR, Hellkamp AS, Lokhnygina Y, et al. Outcomes of discontinuing rivaroxaban compared with warfarin in patients with nonvalvular atrial fibrillation: analysis from the ROCKET AF Trial (Rivaroxaban Once-Daily, Oral, Direct Factor Xa Inhibition Compared With Vitamin K Antagonism for Prevention of Stroke and Embolism Trial in Atrial Fibrillation). *J Am Coll Cardiol*. 2013;61:651–8.
233. Wang X, Tirucherai G, Higen A, et al. Apixaban pharmacokinetics in subjects with end-stage renal disease on hemodialysis. *Clin Pharmacol Drug Dev*. 2014;1:187. Abstract.
234. Giugliano RP, Ruff CT, Braunwald E, et al. Edoxaban versus warfarin in patients with atrial fibrillation. *N Engl J Med*. 2013;369:2093–104.
235. Chan NC, Hirsh J, Ginsberg JS, et al. Betrixaban (PRT054021): pharmacology, dose selection and clinical studies. *Future Cardiol*. 2014;10:43–52.
236. US Food and Drug Administration. FDA Drug Safety Communication: Pradaxa (dabigatran etexilate mesylate) should not be used in patients with mechanical prosthetic heart valves. Available at: <http://www.fda.gov/Drugs/DrugSafety/ucm332912.htm>. 2013. Accessed May 23, 2014.
237. Eikelboom JW, Connolly SJ, Brueckmann M, et al. Dabigatran versus warfarin in patients with mechanical heart valves. *N Engl J Med*. 2013;369:1206–14.
238. Hart RG, Pearce LA, Asinger RW, et al. Warfarin in atrial fibrillation patients with moderate chronic kidney disease. *Clin J Am Soc Nephrol*. 2011;6:2599–604.
239. Agewall S, Cattaneo M, Collet JP, et al. Expert position paper on the use of proton pump inhibitors in patients with cardiovascular disease and antithrombotic therapy. *Eur Heart J*. 2013;34:1708–13, 1713a–1713b. Epub 2013 Feb 20.
240. Shanmugam N, Boerdlein A, Proff J, et al. Detection of atrial high-rate events by continuous home monitoring: clinical significance in the heart failure-cardiac resynchronization therapy population. *Europace*. 2012;14:230–7.
241. Birnie DH, Healey JS, Wells GA, et al. Pacemaker or defibrillator surgery without interruption of anticoagulation. *N Engl J Med*. 2013;368:2084–93.
242. Duke University. Effectiveness of Bridging Anticoagulation for Surgery (The BRIDGE Study) (NCT00786474). *Clinicaltrials.gov*. 2014. Available at: <http://clinicaltrials.gov/show/NCT00786474>. Accessed February 4, 2014.
243. Di Biase L, Burkhardt JD, Mohanty P, et al. Periprocedural stroke and management of major bleeding complications in patients undergoing catheter ablation of atrial fibrillation: the impact of periprocedural therapeutic international normalized ratio. *Circulation*. 2010;121:2550–6.
244. Lakkireddy D, Reddy YM, Di Biase L, et al. Feasibility and safety of dabigatran versus warfarin for periprocedural anticoagulation in patients undergoing radiofrequency ablation for atrial fibrillation: results from a multicenter prospective registry. *J Am Coll Cardiol*. 2012;59:1168–74.
245. Ahmed I, Gertner E, Nelson WB, et al. Continuing warfarin therapy is superior to interrupting warfarin with or without bridging anticoagulation therapy in patients undergoing pacemaker and defibrillator implantation. *Heart Rhythm*. 2010;7:745–9.
246. Cheng A, Nazarian S, Brinker JA, et al. Continuation of warfarin during pacemaker or implantable cardioverter-defibrillator implantation: a randomized clinical trial. *Heart Rhythm*. 2011;8:536–40.
247. Michaud GF, Pelosi F Jr, Noble MD, et al. A randomized trial comparing heparin initiation 6 h or 24 h after pacemaker or defibrillator implantation. *J Am Coll Cardiol*. 2000;35:1915–8.
248. Bernard ML, Shotwell M, Nietert PJ, et al. Meta-analysis of bleeding complications associated with cardiac rhythm device implantation. *Circ Arrhythm Electrophysiol*. 2012;5:468–74.
249. Hohnloser SH, Hijazi Z, Thomas L, et al. Efficacy of apixaban when compared with warfarin in relation to renal function in patients with atrial fibrillation: insights from the ARISTOTLE trial. *Eur Heart J*. 2012;33:2821–30.
250. Piccini JP, Stevens SR, Chang Y, et al. Renal dysfunction as a predictor of stroke and systemic embolism in patients with nonvalvular atrial fibrillation: validation of the R₂CHADS₂ index in the ROCKET AF (Rivaroxaban Once-daily, oral, direct factor Xa inhibition Compared with vitamin K antagonist for prevention of stroke and Embolism Trial in Atrial Fibrillation) and ATRIA (Anticoagulation and Risk factors In Atrial fibrillation) Study Cohorts. *Circulation*. 2013;127:224–32.

251. Hansen ML, Sorensen R, Clausen MT, et al. Risk of bleeding with single, dual, or triple therapy with warfarin, aspirin, and clopidogrel in patients with atrial fibrillation. *Arch Intern Med*. 2010;170:1433–41.
252. Karjalainen PP, Porela P, Ylitalo A, et al. Safety and efficacy of combined antiplatelet-warfarin therapy after coronary stenting. *Eur Heart J*. 2007;28:726–32.
253. Orford JL, Fasseas P, Melby S, et al. Safety and efficacy of aspirin, clopidogrel, and warfarin after coronary stent placement in patients with an indication for anticoagulation. *Am Heart J*. 2004;147:463–7.
254. Doyle BJ, Rihal CS, Gastineau DA, et al. Bleeding, blood transfusion, and increased mortality after percutaneous coronary intervention: implications for contemporary practice. *J Am Coll Cardiol*. 2009;53:2019–27.
255. Blackshear JL, Odell JA. Appendage obliteration to reduce stroke in cardiac surgical patients with atrial fibrillation. *Ann Thorac Surg*. 1996;61:755–9.
256. Reddy VY, Doshi SK, Siever H, et al. Percutaneous left atrial appendage closure for stroke prophylaxis in patients with atrial fibrillation: 2.3 year follow-up of the PROTECT AF Trial (Watchman left atrial appendage system for embolic protection in patients with atrial fibrillation) trial. *Circulation*. 2013;127:720–9.
257. Reddy VY, Mobius-Winkler S, Miller MA, et al. Left atrial appendage closure with the Watchman device in patients with a contraindication for oral anticoagulation: the ASAP study (ASA Plavix Feasibility Study With Watchman Left Atrial Appendage Closure Technology). *J Am Coll Cardiol*. 2013;61:2551–6.
258. Holmes DR. Final results of randomized trial of left atrial appendage closure versus warfarin for stroke/thromboembolic prevention in patients with non-valvular atrial fibrillation (PREVAIL). Released prior to: American College of Cardiology Scientific Session/12 Summit. 2013. Available at: http://my.americanheart.org/idc/groups/ahamapublic/@wcm/@sop/@scon/documents/downloadable/ucm_449994.pdf. Accessed May 23, 2014.
259. Park JW, Bethencourt A, Sievert H, et al. Left atrial appendage closure with Amplatzer cardiac plug in atrial fibrillation: initial European experience. *Catheter Cardiovasc Interv*. 2011;77:700–6.
260. Bartus K, Han FT, Bednarek J, et al. Percutaneous left atrial appendage suture ligation using the LARIAT device in patients with atrial fibrillation: initial clinical experience. *J Am Coll Cardiol*. 2013;62:108–18.
261. Heist EK, Refaat M, Danik SB, et al. Analysis of the left atrial appendage by magnetic resonance angiography in patients with atrial fibrillation. *Heart Rhythm*. 2006;3:1313–8.
262. Chatterjee S, Alexander JC, Pearson PJ, et al. Left atrial appendage occlusion: lessons learned from surgical and transcatheter experiences. *Ann Thorac Surg*. 2011;92:2283–92.
263. AtriClip LAA exclusion system with preloaded Gillinov-Cosgrove clip. US Food and Drug Administration. 2012. Available at: http://www.accessdata.fda.gov/cdrh_docs/pdf12/k122276.pdf. Accessed May 9, 2014.
264. Kanderian AS, Gillinov AM, Pettersson GB, et al. Success of surgical left atrial appendage closure: assessment by transesophageal echocardiography. *J Am Coll Cardiol*. 2008;52:924–9.
265. Healey JS, Crystal E, Lamy A, et al. Left Atrial Appendage Occlusion Study (LAAOS): results of a randomized controlled pilot study of left atrial appendage occlusion during coronary bypass surgery in patients at risk for stroke. *Am Heart J*. 2005;150:288–93.
266. Garcia-Fernandez MA, Perez-David E, Quiles J, et al. Role of left atrial appendage obliteration in stroke reduction in patients with mitral valve prosthesis: a transesophageal echocardiographic study. *J Am Coll Cardiol*. 2003;42:1253–8.
267. Farshi R, Kistner D, Sarma JS, et al. Ventricular rate control in chronic atrial fibrillation during daily activity and programmed exercise: a crossover open-label study of five drug regimens. *J Am Coll Cardiol*. 1999;33:304–10.
268. Steinberg JS, Katz RJ, Bren GB, et al. Efficacy of oral diltiazem to control ventricular response in chronic atrial fibrillation at rest and during exercise. *J Am Coll Cardiol*. 1987;9:405–11.
269. Olshansky B, Rosenfeld LE, Warner AL, et al. The Atrial Fibrillation Follow-up Investigation of Rhythm Management (AFFIRM) study: approaches to control rate in atrial fibrillation. *J Am Coll Cardiol*. 2004;43:1201–8.
270. Abrams J, Allen J, Allin D, et al. Efficacy and safety of esmolol vs propranolol in the treatment of supraventricular tachyarrhythmias: a multicenter double-blind clinical trial. *Am Heart J*. 1985;110:913–22.
271. Ellenbogen KA, Dias VC, Plumb VJ, et al. A placebo-controlled trial of continuous intravenous diltiazem infusion for 24-hour heart rate control during atrial fibrillation and atrial flutter: a multicenter study. *J Am Coll Cardiol*. 1991;18:891–7.
272. Siu CW, Lau CP, Lee WL, et al. Intravenous diltiazem is superior to intravenous amiodarone or digoxin for achieving ventricular rate control in patients with acute uncomplicated atrial fibrillation. *Crit Care Med*. 2009;37:2174–9.
273. Platia EV, Michelson EL, Porterfield JK, et al. Esmolol versus verapamil in the acute treatment of atrial fibrillation or atrial flutter. *Am J Cardiol*. 1989;63:925–9.
274. Van Gelder IC, Groeneweld HF, Crijns HJ, et al. Lenient versus strict rate control in patients with atrial fibrillation. *N Engl J Med*. 2010;362:1363–73.
275. Delle KG, Geppert A, Neunteufl T, et al. Amiodarone versus diltiazem for rate control in critically ill patients with atrial tachyarrhythmias. *Crit Care Med*. 2001;29:1149–53.
276. Hou ZY, Chang MS, Chen CY, et al. Acute treatment of recent-onset atrial fibrillation and flutter with a tailored dosing regimen of intravenous amiodarone: a randomized, digoxin-controlled study. *Eur Heart J*. 1995;16:521–8.
277. Clemon HF, Wood MA, Gilligan DM, et al. Intravenous amiodarone for acute heart rate control in the critically ill patient with atrial tachyarrhythmias. *Am J Cardiol*. 1998;81:594–8.
278. Ozcan C, Jahangir A, Friedman PA, et al. Long-term survival after ablation of the atrioventricular node and implantation of a permanent pacemaker in patients with atrial fibrillation. *N Engl J Med*. 2001;344:1043–51.
279. Weerasooriya R, Davis M, Powell A, et al. The Australian Intervention Randomized Control of Rate in Atrial Fibrillation Trial (AIRCRAFT). *J Am Coll Cardiol*. 2003;41:1697–702.
280. Wood MA, Brown-Mahoney C, Kay GN, et al. Clinical outcomes after ablation and pacing therapy for atrial fibrillation: a meta-analysis. *Circulation*. 2000;101:1138–44.
281. Gulamhusein S, Ko P, Carruthers SG, et al. Acceleration of the ventricular response during atrial fibrillation in the Wolff-Parkinson-White syndrome after verapamil. *Circulation*. 1982;65:348–54.
282. Connolly SJ, Camm AJ, Halperin JL, et al. Dronedronarone in high-risk permanent atrial fibrillation. *N Engl J Med*. 2011;365:2268–76.
283. Kober L, Torp-Pedersen C, McMurray JJ, et al. Increased mortality after dronedronarone therapy for severe heart failure. *N Engl J Med*. 2008;358:2678–87.
284. Demircan C, Cikrikler HI, Engindeniz Z, et al. Comparison of the effectiveness of intravenous diltiazem and metoprolol in the management of rapid ventricular rate in atrial fibrillation. *Emerg Med J*. 2005;22:411–4.
285. Khand AU, Rankin AC, Martin W, et al. Carvedilol alone or in combination with digoxin for the management of atrial fibrillation in patients with heart failure? *J Am Coll Cardiol*. 2003;42:1944–51.
286. Phillips BG, Gandhi AJ, Sanoski CA, et al. Comparison of intravenous diltiazem and verapamil for the acute treatment of atrial fibrillation and atrial flutter. *Pharmacotherapy*. 1997;17:1238–45.
287. Schamroth L. Immediate effects of intravenous verapamil on atrial fibrillation. *Cardiovasc Res*. 1971;5:419–24.
288. Lundstrom T, Ryden L. Ventricular rate control and exercise performance in chronic atrial fibrillation: effects of diltiazem and verapamil. *J Am Coll Cardiol*. 1990;16:86–90.
289. Jacob AS, Nielsen DH, Gianelly RE. Fatal ventricular fibrillation following verapamil in Wolff-Parkinson-White syndrome with atrial fibrillation. *Ann Emerg Med*. 1985;14:159–60.
290. Jordaens L, Trouerbach J, Calle P, et al. Conversion of atrial fibrillation to sinus rhythm and rate control by digoxin in comparison to placebo. *Eur Heart J*. 1997;18:643–8.
291. Kochiadakis GE, Kanoupakis EM, Kalebubas MD, et al. Sotalol vs metoprolol for ventricular rate control in patients with chronic atrial fibrillation who have undergone digitalization: a single-blinded crossover study. *Europace*. 2001;3:73–9.
292. Koh KK, Kwon KS, Park HB, et al. Efficacy and safety of digoxin alone and in combination with low-dose diltiazem or betaxolol to control ventricular rate in chronic atrial fibrillation. *Am J Cardiol*. 1995;75:88–90.
293. Corley SD, Epstein AE, DiMarco JP, et al. Relationships between sinus rhythm, treatment, and survival in the Atrial Fibrillation Follow-Up Investigation of Rhythm Management (AFFIRM) Study. *Circulation*. 2004;109:1509–13.
294. Rathore SS, Curtis JP, Wang Y, et al. Association of serum digoxin concentration and outcomes in patients with heart failure. *JAMA*. 2003;289:871–8.

295. Whitbeck MG, Charnigo RJ, Khaled P, et al. Increased mortality among patients taking digoxin-analysis from the AFFIRM study. *Eur Heart J*. 2013;34:1481–8.
296. The Digitalis Investigation Group. The effect of digoxin on mortality and morbidity in patients with heart failure. *N Engl J Med*. 1997;336:525–33.
297. Gheorghiadu M, Fonarow GC, Van Veldhuisen DJ, et al. Lack of evidence of increased mortality among patients with atrial fibrillation taking digoxin: findings from post hoc propensity-matched analysis of the AFFIRM trial. *Eur Heart J*. 2013;34:1489–97.
298. Hofmann R, Steinwender C, Kammler J, et al. Effects of a high dose intravenous bolus amiodarone in patients with atrial fibrillation and a rapid ventricular rate. *Int J Cardiol*. 2006;110:27–32.
299. Tse HF, Lam YM, Lau CP, et al. Comparison of digoxin versus low-dose amiodarone for ventricular rate control in patients with chronic atrial fibrillation. *Clin Exp Pharmacol Physiol*. 2001;28:446–50.
300. Wellens HJ, Brugada P, Abdollah H, et al. A comparison of the electrophysiologic effects of intravenous and oral amiodarone in the same patient. *Circulation*. 1984;69:120–4.
301. Nebojsa M, Dragan S, Nebojsa A, et al. Lethal outcome after intravenous administration of amiodarone in patient with atrial fibrillation and ventricular preexcitation. *J Cardiovasc Electrophysiol*. 2011;22:1077–8.
302. Badshah A, Mirza B, Janjua M, et al. Amiodarone-induced torsade de pointes in a patient with Wolff-Parkinson-White syndrome. *Hellenic J Cardiol*. 2009;50:224–6.
303. Davy JM, Herold M, Hoglund C, et al. Dronedronarone for the control of ventricular rate in permanent atrial fibrillation: the Efficacy and safety of dRonedArone for the cOntrol of ventricular rate during atrial fibrillation (ERATO) study. *Am Heart J*. 2008;156:527–9.
304. Brignole M, Menozzi C, Gianfranchi L, et al. Assessment of atrioventricular junction ablation and VVIR pacemaker versus pharmacological treatment in patients with heart failure and chronic atrial fibrillation: a randomized, controlled study. *Circulation*. 1998;98:953–60.
305. Kay GN, Ellenbogen KA, Giudici M, et al. The Ablate and Pace Trial: a prospective study of catheter ablation of the AV conduction system and permanent pacemaker implantation for treatment of atrial fibrillation. APT Investigators. *J Interv Card Electrophysiol*. 1998;2:121–35.
306. Brignole M, Gianfranchi L, Menozzi C, et al. Assessment of atrioventricular junction ablation and DDDR mode-switching pacemaker versus pharmacological treatment in patients with severely symptomatic paroxysmal atrial fibrillation: a randomized controlled study. *Circulation*. 1997;96:2617–24.
307. Khan MN, Jais P, Cummings J, et al. Pulmonary-vein isolation for atrial fibrillation in patients with heart failure. *N Engl J Med*. 2008;359:1778–85.
308. Evans GT Jr, Scheinman MM, Bardy G, et al. Predictors of in-hospital mortality after DC catheter ablation of atrioventricular junction: results of a prospective, international, multicenter study. *Circulation*. 1991;84:1924–37.
309. Geelen P, Brugada J, Andries E, et al. Ventricular fibrillation and sudden death after radiofrequency catheter ablation of the atrioventricular junction. *Pacing Clin Electrophysiol*. 1997;20:343–8.
310. Wang RX, Lee HC, Hodge DO, et al. Effect of pacing method on risk of sudden death after atrioventricular node ablation and pacemaker implantation in patients with atrial fibrillation. *Heart Rhythm*. 2013;10:696–701.
311. Curtis AB, Worley SJ, Adamson PB, et al. Biventricular pacing for atrioventricular block and systolic dysfunction. *N Engl J Med*. 2013;368:1585–93.
312. Leon AR, Greenberg JM, Kanuru N, et al. Cardiac resynchronization in patients with congestive heart failure and chronic atrial fibrillation: effect of upgrading to biventricular pacing after chronic right ventricular pacing. *J Am Coll Cardiol*. 2002;39:1258–63.
313. Van Gelder IC, Hagens VE, Bosker HA, et al. A comparison of rate control and rhythm control in patients with recurrent persistent atrial fibrillation. *N Engl J Med*. 2002;347:1834–40.
314. Singh BN, Singh SN, Reda DJ, et al. Amiodarone versus sotalol for atrial fibrillation. *N Engl J Med*. 2005;352:1861–72.
315. Hagens VE, Ranchor AV, Van Sonderen E, et al. Effect of rate or rhythm control on quality of life in persistent atrial fibrillation: results from the Rate Control Versus Electrical Cardioversion (RACE) Study. *J Am Coll Cardiol*. 2004;43:241–7.
316. de Vos CB, Pisters R, Nieuwlaar R, et al. Progression from paroxysmal to persistent atrial fibrillation clinical correlates and prognosis. *J Am Coll Cardiol*. 2010;55:725–31.
317. Cosio FG, Aliot E, Botto GL, et al. Delayed rhythm control of atrial fibrillation may be a cause of failure to prevent recurrences: reasons for change to active antiarrhythmic treatment at the time of the first detected episode. *Europace*. 2008;10:21–7.
318. Kirchhof P, Bax J, Blomstrom-Lundquist C, et al. Early and comprehensive management of atrial fibrillation: executive summary of the proceedings from the 2nd AFNET-EHRA consensus conference “research perspectives in AF.” *Eur Heart J*. 2009;30:2969–77c.
319. Van Gelder IC, Haegeli LM, Brandes A, et al. Rationale and current perspective for early rhythm control therapy in atrial fibrillation. *Europace*. 2011;13:1517–25.
320. Moreyra E, Finkelhor RS, Cebul RD. Limitations of transesophageal echocardiography in the risk assessment of patients before nonanticoagulated cardioversion from atrial fibrillation and flutter: an analysis of pooled trials. *Am Heart J*. 1995;129:71–5.
321. Gallagher MM, Hennessy BJ, Edvardsson N, et al. Embolic complications of direct current cardioversion of atrial arrhythmias: association with low intensity of anticoagulation at the time of cardioversion. *J Am Coll Cardiol*. 2002;40:926–33.
322. Jaber WA, Prior DL, Thamilarasan M, et al. Efficacy of anticoagulation in resolving left atrial and left atrial appendage thrombi: a transesophageal echocardiographic study. *Am Heart J*. 2000;140:150–6.
323. You JJ, Singer DE, Howard PA, et al. Antithrombotic therapy for atrial fibrillation: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141:e531–75S.
324. Piccini JP, Stevens SR, Lokhnygina Y, et al. Outcomes following cardioversion and atrial fibrillation ablation in patients treated with rivaroxaban and warfarin in the ROCKET AF Trial. *J Am Coll Cardiol*. 2013;61:1998–2006.
325. Flaker G, Lopes RD, Al-Khatib SM, et al. Efficacy and safety of apixaban in patients following cardioversion for atrial fibrillation: insights from the ARISTOTLE trial (Apixaban for Reduction in Stroke and Other Thromboembolic Events in Atrial Fibrillation). *J Am Coll Cardiol*. 2014;63:1082–7.
326. von Besser K, Mills AM. Is discharge to home after emergency department cardioversion safe for the treatment of recent-onset atrial fibrillation? *Ann Emerg Med*. 2011;58:517–20.
327. Oral H, Souza JJ, Michaud GF, et al. Facilitating transthoracic cardioversion of atrial fibrillation with ibutilide pretreatment. *N Engl J Med*. 1999;340:1849–54.
328. Alboni P, Botto GL, Baldi N, et al. Outpatient treatment of recent-onset atrial fibrillation with the “pill-in-the-pocket” approach. *N Engl J Med*. 2004;351:2384–91.
329. Ellenbogen KA, Clemon HF, Stambler BS, et al. Efficacy of ibutilide for termination of atrial fibrillation and flutter. *Am J Cardiol*. 1996;78:42–5.
330. Khan IA. Single oral loading dose of propafenone for pharmacological cardioversion of recent-onset atrial fibrillation. *J Am Coll Cardiol*. 2001;37:542–7.
331. Patsilinos S, Christou A, Kafkas N, et al. Effect of high doses of magnesium on converting ibutilide to a safe and more effective agent. *Am J Cardiol*. 2010;106:673–6.
332. Singh S, Zoble RG, Yellen L, et al. Efficacy and safety of oral dofetilide in converting to and maintaining sinus rhythm in patients with chronic atrial fibrillation or atrial flutter: the Symptomatic Atrial Fibrillation Investigative Research on Dofetilide (SAFIRE-D) study. *Circulation*. 2000;102:2385–90.
333. Stambler BS, Wood MA, Ellenbogen KA, et al. Efficacy and safety of repeated intravenous doses of ibutilide for rapid conversion of atrial flutter or fibrillation. Ibutilide Repeat Dose Study Investigators. *Circulation*. 1996;94:1613–21.
334. Khan IA, Mehta NJ, Gowda RM. Amiodarone for pharmacological cardioversion of recent-onset atrial fibrillation. *Int J Cardiol*. 2003;89:239–48.
335. Letelier LM, Udol K, Ena J, et al. Effectiveness of amiodarone for conversion of atrial fibrillation to sinus rhythm: a meta-analysis. *Arch Intern Med*. 2003;163:777–85.
336. Pedersen OD, Bagger H, Keller N, et al. Efficacy of dofetilide in the treatment of atrial fibrillation-flutter in patients with reduced left ventricular function: a Danish Investigations of Arrhythmia and Mortality on Dofetilide (DIAMOND) Substudy. *Circulation*. 2001;104:292–6.
337. Bianconi L, Mennuni M, Lukic V, et al. Effects of oral propafenone administration before electrical cardioversion of chronic atrial fibrillation: a placebo-controlled study. *J Am Coll Cardiol*. 1996;28:700–6.
338. Mittal S, Ayati S, Stein KM, et al. Transthoracic cardioversion of atrial fibrillation: comparison of rectilinear biphasic versus damped sine wave monophasic shocks. *Circulation*. 2000;101:1282–7.

339. Kirchhof P, Andresen D, Bosch R, et al. Short-term versus long-term antiarrhythmic drug treatment after cardioversion of atrial fibrillation (Flec-SL): a prospective, randomised, open-label, blinded endpoint assessment trial. *Lancet*. 2012;380:238–46.
340. Gallagher MM, Guo XH, Poloniecki JD, et al. Initial energy setting, outcome and efficiency in direct current cardioversion of atrial fibrillation and flutter. *J Am Coll Cardiol*. 2001;38:1498–504.
341. Berger M, Schweitzer P. Timing of thromboembolic events after electrical cardioversion of atrial fibrillation or flutter: a retrospective analysis. *Am J Cardiol*. 1998;82:1545–7, A8.
342. Airaksinen K, Grönberg T, Nuotio I, et al. Thromboembolic complications after cardioversion of acute atrial fibrillation: the FinCV Study. *J Am Coll Cardiol*. 2013;62:1187–92.
343. Weigner MJ, Thomas LR, Patel U, et al. Early cardioversion of atrial fibrillation facilitated by transesophageal echocardiography: short-term safety and impact on maintenance of sinus rhythm at 1 year. *Am J Med*. 2001;110:694–702.
344. Wu LA, Chandrasekaran K, Friedman PA, et al. Safety of expedited anticoagulation in patients undergoing transesophageal echocardiographic-guided cardioversion. *Am J Med*. 2006;119:142–6.
345. Weigner MJ, Caulfield TA, Danias PG, et al. Risk for clinical thromboembolism associated with conversion to sinus rhythm in patients with atrial fibrillation lasting less than 48 hours. *Ann Intern Med*. 1997;126:615–20.
346. Murray KT. Cardiovascular drugs: ibutilide. *Circulation*. 1998;97:493–7.
347. Lafuente-Lafuente C, Mouly S, Longas-Tejero MA, et al. Antiarrhythmics for maintaining sinus rhythm after cardioversion of atrial fibrillation. *Cochrane Database Syst Rev*. 2007;(4):CD005049.
348. Channer KS, Birchall A, Steeds RP, et al. A randomized placebo-controlled trial of pre-treatment and short- or long-term maintenance therapy with amiodarone supporting DC cardioversion for persistent atrial fibrillation. *Eur Heart J*. 2004;25:144–50.
349. Galperin J, Elizari MV, Chiale PA, et al. Efficacy of amiodarone for the termination of chronic atrial fibrillation and maintenance of normal sinus rhythm: a prospective, multicenter, randomized, controlled, double blind trial. *J Cardiovasc Pharmacol Ther*. 2001;6:341–50.
350. Hohnloser SH, Crijns HJ, van Eickels M, et al. Effect of dronedarone on cardiovascular events in atrial fibrillation. *N Engl J Med*. 2009;360:668–78.
351. Singh BN, Connolly SJ, Crijns HJ, et al. Dronedrone for maintenance of sinus rhythm in atrial fibrillation or flutter. *N Engl J Med*. 2007;357:987–99.
352. Touboul P, Brugada J, Capucci A, et al. Dronedrone for prevention of atrial fibrillation: a dose-ranging study. *Eur Heart J*. 2003;24:1481–7.
353. Van Gelder IC, Crijns HJ, Van Gilst WH, et al. Efficacy and safety of flecainide acetate in the maintenance of sinus rhythm after electrical cardioversion of chronic atrial fibrillation or atrial flutter. *Am J Cardiol*. 1989;64:1317–21.
354. Roy D, Talajic M, Dorian P, et al. Amiodarone to prevent recurrence of atrial fibrillation. Canadian Trial of Atrial Fibrillation Investigators. *N Engl J Med*. 2000;342:913–20.
355. Bellandi F, Simonetti I, Leoncini M, et al. Long-term efficacy and safety of propafenone and sotalol for the maintenance of sinus rhythm after conversion of recurrent symptomatic atrial fibrillation. *Am J Cardiol*. 2001;88:640–5.
356. Dogan A, Ergene O, Nazli C, et al. Efficacy of propafenone for maintaining sinus rhythm in patients with recent onset or persistent atrial fibrillation after conversion: a randomized, placebo-controlled study. *Acta Cardiol*. 2004;59:255–61.
357. Pritchett EL, Page RL, Carlson M, et al. Efficacy and safety of sustained-release propafenone (propafenone SR) for patients with atrial fibrillation. *Am J Cardiol*. 2003;92:941–6.
358. Benditt DG, Williams JH, Jin J, et al. Maintenance of sinus rhythm with oral d,l-sotalol therapy in patients with symptomatic atrial fibrillation and/or atrial flutter. d,l-Sotalol Atrial Fibrillation/Flutter Study Group. *Am J Cardiol*. 1999;84:270–7.
359. Freemantle N, Lafuente-Lafuente C, Mitchell S, et al. Mixed treatment comparison of dronedrone, amiodarone, sotalol, flecainide, and propafenone, for the management of atrial fibrillation. *Europace*. 2011;13:329–45.
360. Piccini JP, Hasselblad V, Peterson ED, et al. Comparative efficacy of dronedrone and amiodarone for the maintenance of sinus rhythm in patients with atrial fibrillation. *J Am Coll Cardiol*. 2009;54:1089–95.
361. Le Heuzey JY, De Ferrari GM, Radzik D, et al. A short-term, randomized, double-blind, parallel-group study to evaluate the efficacy and safety of dronedrone versus amiodarone in patients with persistent atrial fibrillation: the DIONYSOS study. *J Cardiovasc Electrophysiol*. 2010;21:597–605.
362. Maintenance of sinus rhythm in patients with atrial fibrillation: an AFFIRM substudy of the first antiarrhythmic drug. *J Am Coll Cardiol*. 2003;42:20–9.
363. Calkins H, Reynolds MR, Spector P, et al. Treatment of atrial fibrillation with antiarrhythmic drugs or radiofrequency ablation: two systematic literature reviews and meta-analyses. *Circ Arrhythm Electrophysiol*. 2009;2:349–61.
364. Zimetbaum P. Antiarrhythmic drug therapy for atrial fibrillation. *Circulation*. 2012;125:381–9.
365. Lafuente-Lafuente C, Mouly S, Longas-Tejero MA, et al. Antiarrhythmic drugs for maintaining sinus rhythm after cardioversion of atrial fibrillation: a systematic review of randomized controlled trials. *Arch Intern Med*. 2006;166:719–28.
366. Kirchhof P, Franz MR, Bardai A, et al. Giant T-U waves precede torsades de pointes in long QT syndrome: a systematic electrocardiographic analysis in patients with acquired and congenital QT prolongation. *J Am Coll Cardiol*. 2009;54:143–9.
367. Kao DP, Hiatt WR, Krantz MJ. Proarrhythmic potential of dronedrone: emerging evidence from spontaneous adverse event reporting. *Pharmacotherapy*. 2012;32:767–71.
368. Echt DS, Liebson PR, Mitchell LB, et al. Mortality and morbidity in patients receiving encainide, flecainide, or placebo: the Cardiac Arrhythmia Suppression Trial. *N Engl J Med*. 1991;324:781–8.
369. Roden D. Antiarrhythmic drugs. In: Brunton L, Lazo J, Parker KL, eds. Goodman and Gilman's The Pharmacological Basis of Therapeutics. New York, NY: McGraw-Hill; 2005:899–932.
370. Freeman BL, Hardy MA. Multiplanar phalangeal and metatarsal osteotomies for hallux rigidus. *Clin Podiatr Med Surg*. 2011;28:329–44.
371. Bardy GH, Lee KL, Mark DB, et al. Amiodarone or an implantable cardioverter-defibrillator for congestive heart failure. *N Engl J Med*. 2005;352:225–37.
372. Aliot E, Capucci A, Crijns HJ, et al. Twenty-five years in the making: flecainide is safe and effective for the management of atrial fibrillation. *Europace*. 2011;13:161–73.
373. Torp-Pedersen C, Møller M, Bloch-Thomsen PE, et al. Dofetilide in patients with congestive heart failure and left ventricular dysfunction. Danish Investigations of Arrhythmia and Mortality on Dofetilide Study Group. *N Engl J Med*. 1999;341:857–65.
374. Karlson BW, Torstensson I, Abjorn C, et al. Disopyramide in the maintenance of sinus rhythm after electroconversion of atrial fibrillation: a placebo-controlled one-year follow-up study. *Eur Heart J*. 1988;9:284–90.
375. Sherrid MV, Barac I, McKenna WJ, et al. Multicenter study of the efficacy and safety of disopyramide in obstructive hypertrophic cardiomyopathy. *J Am Coll Cardiol*. 2005;45:1251–8.
376. Nergårdh AK, Rosenqvist M, Nordlander R, et al. Maintenance of sinus rhythm with metoprolol CR initiated before cardioversion and repeated cardioversion of atrial fibrillation: a randomized double-blind placebo-controlled study. *Eur Heart J*. 2007;28:1351–7.
377. Fabritz L, Kirchhof P. Predictable and less predictable unwanted cardiac drugs effects: individual pre-disposition and transient precipitating factors. *Basic Clin Pharmacol Toxicol*. 2010;106:263–8.
378. Hauser TH, Pinto DS, Josephson ME, et al. Safety and feasibility of a clinical pathway for the outpatient initiation of antiarrhythmic medications in patients with atrial fibrillation or atrial flutter. *Am J Cardiol*. 2003;91:1437–41.
379. Schneider MP, Hua TA, Bohm M, et al. Prevention of atrial fibrillation by renin-angiotensin system inhibition: a meta-analysis. *J Am Coll Cardiol*. 2010;55:2299–307.
380. Ducharme A, Swedberg K, Pfeffer MA, et al. Prevention of atrial fibrillation in patients with symptomatic chronic heart failure by candesartan in the Candesartan in Heart failure: Assessment of Reduction in Mortality and morbidity (CHARM) program. *Am Heart J*. 2006;151:985–91.
381. Jibrini MB, Molnar J, Arora RR. Prevention of atrial fibrillation by way of abrogation of the renin-angiotensin system: a systematic review and meta-analysis. *Am J Ther*. 2008;15:36–43.
382. Patti G, Chello M, Candura D, et al. Randomized trial of atorvastatin for reduction of postoperative atrial fibrillation in patients undergoing cardiac surgery: results of the ARMYDA-3 (Atorvastatin for Reduction of MYocardial Dysrhythmia After cardiac surgery) study. *Circulation*. 2006;114:1455–61.

383. Goette A, Schon N, Kirchhof P, et al. Angiotensin II-antagonist in paroxysmal atrial fibrillation (ANTIPAF) trial. *Circ Arrhythm Electrophysiol*. 2012;5:43–51.
384. Savelieva I, Kourliouros A, Camm J. Primary and secondary prevention of atrial fibrillation with statins and polyunsaturated fatty acids: review of evidence and clinical relevance. *Naunyn Schmiedeberg Arch Pharmacol*. 2010;381:1–13.
385. Mozaffarian D, Marchioli R, Macchia A, et al. Fish oil and postoperative atrial fibrillation: the Omega-3 Fatty Acids for Prevention of Post-operative Atrial Fibrillation (OPERA) randomized trial. *JAMA*. 2012;308:2001–11.
386. Savelieva I, Kakouros N, Kourliouros A, et al. Upstream therapies for management of atrial fibrillation: review of clinical evidence and implications for European Society of Cardiology guidelines: part II: secondary prevention. *Europace*. 2011;13:610–25.
387. Yamashita T, Inoue H, Okumura K, et al. Randomized trial of angiotensin II-receptor blocker vs. dihydropyridine calcium channel blocker in the treatment of paroxysmal atrial fibrillation with hypertension (J-RHYTHM II study). *Europace*. 2011;13:473–9.
388. Macchia A, Grancelli H, Varini S, et al. Omega-3 fatty acids for the prevention of recurrent symptomatic atrial fibrillation: results of the FORWARD (Randomized Trial to Assess Efficacy of PUFA for the Maintenance of Sinus Rhythm in Persistent Atrial Fibrillation) trial. *J Am Coll Cardiol*. 2013;61:463–8.
389. Fauchier L, Pierre B, de Labriolle A, et al. Antiarrhythmic effect of statin therapy and atrial fibrillation: a meta-analysis of randomized controlled trials. *J Am Coll Cardiol*. 2008;51:828–35.
390. Liu T, Li L, Korantzopoulos P, et al. Statin use and development of atrial fibrillation: a systematic review and meta-analysis of randomized clinical trials and observational studies. *Int J Cardiol*. 2008;126:160–70.
391. Liakopoulos OJ, Kuhn EW, Slottosch I, et al. Preoperative statin therapy for patients undergoing cardiac surgery. *Cochrane Database Syst Rev*. 2012;(4):CD008493.
392. Bonanno C, Paccanaro M, La Vecchia L, et al. Efficacy and safety of catheter ablation versus antiarrhythmic drugs for atrial fibrillation: a meta-analysis of randomized trials. *J Cardiovasc Med (Hagerstown)*. 2010;11:408–18.
393. Nair GM, Nery PB, Diwakaramenon S, et al. A systematic review of randomized trials comparing radiofrequency ablation with antiarrhythmic medications in patients with atrial fibrillation. *J Cardiovasc Electrophysiol*. 2009;20:138–44.
394. Parkash R, Tang AS, Sapp JL, et al. Approach to the catheter ablation technique of paroxysmal and persistent atrial fibrillation: a meta-analysis of the randomized controlled trials. *J Cardiovasc Electrophysiol*. 2011;22:729–38.
395. Piccini JP, Lopes RD, Kong MH, et al. Pulmonary vein isolation for the maintenance of sinus rhythm in patients with atrial fibrillation: a meta-analysis of randomized, controlled trials. *Circ Arrhythm Electrophysiol*. 2009;2:626–33.
396. Jais P, Cauchemez B, Macle L, et al. Catheter ablation versus antiarrhythmic drugs for atrial fibrillation: the A4 study. *Circulation*. 2008;118:2498–505.
397. Wilber DJ, Pappone C, Neuzil P, et al. Comparison of antiarrhythmic drug therapy and radiofrequency catheter ablation in patients with paroxysmal atrial fibrillation: a randomized controlled trial. *JAMA*. 2010;303:333–40.
398. Stabile G, Bertaglia E, Senatore G, et al. Catheter ablation treatment in patients with drug-refractory atrial fibrillation: a prospective, multicentre, randomized, controlled study (Catheter Ablation For The Cure Of Atrial Fibrillation Study). *Eur Heart J*. 2006;27:216–21.
399. Oral H, Pappone C, Chugh A, et al. Circumferential pulmonary-vein ablation for chronic atrial fibrillation. *N Engl J Med*. 2006;354:934–41.
400. Mont L, Bisbal F, Hernandez-Madrid A, et al. Catheter ablation vs. antiarrhythmic drug treatment of persistent atrial fibrillation: a multicentre, randomized, controlled trial (SARA study). *Eur Heart J*. 2014;35:501–7.
401. Wazni OM, Marrouche NF, Martin DO, et al. Radiofrequency ablation vs antiarrhythmic drugs as first-line treatment of symptomatic atrial fibrillation: a randomized trial. *JAMA*. 2005;293:2634–40.
402. Morillo C, Verma A, Kuck K, et al. Radiofrequency ablation vs antiarrhythmic drugs as first-line treatment of paroxysmal atrial fibrillation (RAAFT 2): a randomized trial. *JAMA*. 2014;311:692–700.
403. Cosedis NJ, Johannessen A, Raatikainen P, et al. Radiofrequency ablation as initial therapy in paroxysmal atrial fibrillation. *N Engl J Med*. 2012;367:1587–95.
404. Haissaguerre M, Hocini M, Sanders P, et al. Catheter ablation of long-lasting persistent atrial fibrillation: clinical outcome and mechanisms of subsequent arrhythmias. *J Cardiovasc Electrophysiol*. 2005;16:1138–47.
405. Terasawa T, Balk EM, Chung M, et al. Systematic review: comparative effectiveness of radiofrequency catheter ablation for atrial fibrillation. *Ann Intern Med*. 2009;151:191–202.
406. Ip S, Terasawa T, Balk EM, et al. Comparative effectiveness of radiofrequency catheter ablation for atrial fibrillation. Comparative Effectiveness Review No.15. 2009. Available at: <http://effectivehealthcare.ahrq.gov/index.cfm/search-for-guides-reviews-and-reports/?pageaction=displayproduct&productid=323>. Accessed May 9, 2014.
407. Packer DL, Kowal RC, Whealan KR, et al. Cryoballoon ablation of pulmonary veins for paroxysmal atrial fibrillation: first results of the North American Arctic Front (STOP AF) Pivotal Trial. *J Am Coll Cardiol*. 2013;61:1713–23.
408. Leong-Sit P, Zado E, Callans DJ, et al. Efficacy and risk of atrial fibrillation ablation before 45 years of age. *Circ Arrhythm Electrophysiol*. 2010;3:452–7.
409. Wokhlu A, Monahan KH, Hodge DO, et al. Long-term quality of life after ablation of atrial fibrillation: the impact of recurrence, symptom relief, and placebo effect. *J Am Coll Cardiol*. 2010;55:2308–16.
410. Jons C, Hansen PS, Johannessen A, et al. The Medical Antiarrhythmic Treatment or Radiofrequency Ablation in Paroxysmal Atrial Fibrillation (MANTRA-PAF) trial: clinical rationale, study design, and implementation. *Europace*. 2009;11:917–23.
411. Gaztanaga L, Frankel DS, Kohari M, et al. Time to recurrence of atrial fibrillation influences outcome following catheter ablation. *Heart Rhythm*. 2013;10:2–9.
412. Medi C, Sparks PB, Morton JB, et al. Pulmonary vein antral isolation for paroxysmal atrial fibrillation: results from long-term follow-up. *J Cardiovasc Electrophysiol*. 2011;22:137–41.
413. Ouyang F, Tilz R, Chun J, et al. Long-term results of catheter ablation in paroxysmal atrial fibrillation: lessons from a 5-year follow-up. *Circulation*. 2010;122:2368–77.
414. Sawhney N, Anousheh R, Chen WC, et al. Five-year outcomes after segmental pulmonary vein isolation for paroxysmal atrial fibrillation. *Am J Cardiol*. 2009;104:366–72.
415. Wokhlu A, Hodge DO, Monahan KH, et al. Long-term outcome of atrial fibrillation ablation: impact and predictors of very late recurrence. *J Cardiovasc Electrophysiol*. 2010;21:1071–8.
416. Hakalahti A, Uusimaa P, Ylitalo K, et al. Catheter ablation of atrial fibrillation in patients with therapeutic oral anticoagulation treatment. *Europace*. 2011;13:640–5.
417. Kwak JJ, Pak HN, Jang JK, et al. Safety and convenience of continuous warfarin strategy during the periprocedural period in patients who underwent catheter ablation of atrial fibrillation. *J Cardiovasc Electrophysiol*. 2010;21:620–5.
418. Wazni OM, Beheiry S, Fahmy T, et al. Atrial fibrillation ablation in patients with therapeutic international normalized ratio: comparison of strategies of anticoagulation management in the periprocedural period. *Circulation*. 2007;116:2531–4.
419. Winkle RA, Mead RH, Engel G, et al. The use of dabigatran immediately after atrial fibrillation ablation. *J Cardiovasc Electrophysiol*. 2012;23:264–8.
420. Kim JS, She F, Jongnarangsin K, et al. Dabigatran vs warfarin for radiofrequency catheter ablation of atrial fibrillation. *Heart Rhythm*. 2013;10:483–9.
421. Bassiouny M, Saliba W, Rickard J, et al. Use of dabigatran for periprocedural anticoagulation in patients undergoing catheter ablation for atrial fibrillation. *Circ Arrhythm Electrophysiol*. 2013;6:460–6.
422. Eitel C, Koch J, Sommer P, et al. Novel oral anticoagulants in a real-world cohort of patients undergoing catheter ablation of atrial fibrillation. *Europace*. 2013;15:1587–93.
423. Verma A, Champagne J, Sapp J, et al. Discerning the incidence of symptomatic and asymptomatic episodes of atrial fibrillation before and after catheter ablation (DISCERN AF): a prospective, multicenter study. *JAMA Intern Med*. 2013;173:149–56.
424. Hindricks G, Piorkowski C, Tanner H, et al. Perception of atrial fibrillation before and after radiofrequency catheter ablation: relevance of asymptomatic arrhythmia recurrence. *Circulation*. 2005;112:307–13.
425. Themistoclakis S, Corrado A, Marchlinski FE, et al. The risk of thromboembolism and need for oral anticoagulation after successful atrial fibrillation ablation. *J Am Coll Cardiol*. 2010;55:735–43.

426. Oral H, Chugh A, Ozaydin M, et al. Risk of thromboembolic events after percutaneous left atrial radiofrequency ablation of atrial fibrillation. *Circulation*. 2006;114:759–65.
427. Saad EB, D'Avila A, Costa IP, et al. Very low risk of thromboembolic events in patients undergoing successful catheter ablation of atrial fibrillation with a CHADS2 score ≤ 3 : a long-term outcome study. *Circ Arrhythm Electrophysiol*. 2011;4:615–21.
428. Bunch TJ, May HT, Bair TL, et al. Atrial fibrillation ablation patients have long-term stroke rates similar to patients without atrial fibrillation regardless of CHADS2 score. *Heart Rhythm*. 2013;10:1272–7.
429. Jones DG, Haldar SK, Hussain W, et al. A randomized trial to assess catheter ablation versus rate control in the management of persistent atrial fibrillation in heart failure. *J Am Coll Cardiol*. 2013;61:1894–903.
430. Dagres N, Varounis C, Gaspar T, et al. Catheter ablation for atrial fibrillation in patients with left ventricular systolic dysfunction: a systematic review and meta-analysis. *J Card Fail*. 2011;17:964–70.
431. Cha YM, Wokhlu A, Asirvatham SJ, et al. Success of ablation for atrial fibrillation in isolated left ventricular diastolic dysfunction: a comparison to systolic dysfunction and normal ventricular function. *Circ Arrhythm Electrophysiol*. 2011;4:724–32.
432. Wilton SB, Fundytus A, Ghali WA, et al. Meta-analysis of the effectiveness and safety of catheter ablation of atrial fibrillation in patients with versus without left ventricular systolic dysfunction. *Am J Cardiol*. 2010;106:1284–91.
433. Cappato R, Calkins H, Chen SA, et al. Updated worldwide survey on the methods, efficacy, and safety of catheter ablation for human atrial fibrillation. *Circ Arrhythm Electrophysiol*. 2010;3:32–8.
434. Arbelo E, Brugada J, Hindricks G, et al. ESC-EURObservational Research Programme: the Atrial Fibrillation Ablation Pilot Study, conducted by the European Heart Rhythm Association. *Europace*. 2012;14:1094–103.
435. Shah RU, Freeman JV, Shilane D, et al. Procedural complications, re-hospitalizations, and repeat procedures after catheter ablation for atrial fibrillation. *J Am Coll Cardiol*. 2012;59:143–9.
436. Hoyt H, Bhonsale A, Chilukuri K, et al. Complications arising from catheter ablation of atrial fibrillation: temporal trends and predictors. *Heart Rhythm*. 2011;8:1869–74.
437. Piccini JP, Sinner MF, Greiner MA, et al. Outcomes of medicare beneficiaries undergoing catheter ablation for atrial fibrillation. *Circulation*. 2012;126:2200–7.
438. Gaita F, Caponi D, Pianelli M, et al. Radiofrequency catheter ablation of atrial fibrillation: a cause of silent thromboembolism? Magnetic resonance imaging assessment of cerebral thromboembolism in patients undergoing ablation of atrial fibrillation. *Circulation*. 2010;122:1667–73.
439. Herrera SC, Deneke T, Hocini M, et al. Incidence of asymptomatic intracranial embolic events after pulmonary vein isolation: comparison of different atrial fibrillation ablation technologies in a multicenter study. *J Am Coll Cardiol*. 2011;58:681–8.
440. Boersma LV, Castella M, van Boven W, et al. Atrial fibrillation catheter ablation versus surgical ablation treatment (FAST): a 2-center randomized clinical trial. *Circulation*. 2012;125:23–30.
441. Prasad SM, Maniar HS, Camillo CJ, et al. The Cox maze III procedure for atrial fibrillation: long-term efficacy in patients undergoing lone versus concomitant procedures. *J Thorac Cardiovasc Surg*. 2003;126:1822–8.
442. Damiano RJ Jr, Schwartz FH, Bailey MS, et al. The Cox maze IV procedure: predictors of late recurrence. *J Thorac Cardiovasc Surg*. 2011;141:113–21.
443. Khargi K, Deneke T, Haardt H, et al. Saline-irrigated, cooled-tip radiofrequency ablation is an effective technique to perform the maze procedure. *Ann Thorac Surg*. 2001;72:S1090–5.
444. Jessurun ER, Van Hemel NM, Defauw JJ, et al. A randomized study of combining maze surgery for atrial fibrillation with mitral valve surgery. *J Cardiovasc Surg (Torino)*. 2003;44:9–18.
445. Akpinar B, Guden M, Sagbas E, et al. Combined radiofrequency modified maze and mitral valve procedure through a port access approach: early and mid-term results. *Eur J Cardiothorac Surg*. 2003;24:223–30.
446. Schuetz A, Schulze CJ, Sarvanakis KK, et al. Surgical treatment of permanent atrial fibrillation using microwave energy ablation: a prospective randomized clinical trial. *Eur J Cardiothorac Surg*. 2003;24:475–80.
447. Vasconcelos JT, Scanavacca MI, Sampaio RO, et al. Surgical treatment of atrial fibrillation through isolation of the left atrial posterior wall in patients with chronic rheumatic mitral valve disease: a randomized study with control group. *Arq Bras Cardiol*. 2004;83:211–8.
448. de Lima GG, Kalil RA, Leiria TL, et al. Randomized study of surgery for patients with permanent atrial fibrillation as a result of mitral valve disease. *Ann Thorac Surg*. 2004;77:2089–94.
449. Doukas G, Samani NJ, Alexiou C, et al. Left atrial radiofrequency ablation during mitral valve surgery for continuous atrial fibrillation: a randomized controlled trial. *JAMA*. 2005;294:2323–9.
450. Abreu Filho CA, Lisboa LA, Dallan LA, et al. Effectiveness of the maze procedure using cooled-tip radiofrequency ablation in patients with permanent atrial fibrillation and rheumatic mitral valve disease. *Circulation*. 2005;112:120–5.
451. Blomstrom-Lundqvist C, Johansson B, Berglin E, et al. A randomized double-blind study of epicardial left atrial cryoablation for permanent atrial fibrillation in patients undergoing mitral valve surgery: the SWEDish Multicentre Atrial Fibrillation study (SWEDMAF). *Eur Heart J*. 2007;28:2902–8.
452. Ad N, Suri RM, Gammie JS, et al. Surgical ablation of atrial fibrillation trends and outcomes in North America. *J Thorac Cardiovasc Surg*. 2012;144:1051–60.
453. Edgerton JR, McClelland JH, Duke D, et al. Minimally invasive surgical ablation of atrial fibrillation: six-month results. *J Thorac Cardiovasc Surg*. 2009;138:109–13.
454. Hoogsteen J, Schep G, Van Hemel NM, et al. Paroxysmal atrial fibrillation in male endurance athletes: a 9-year follow up. *Europace*. 2004;6:222–8.
455. Calvo N, Mont L, Tamborero D, et al. Efficacy of circumferential pulmonary vein ablation of atrial fibrillation in endurance athletes. *Europace*. 2010;12:30–6.
456. Wyse DG. Pharmacotherapy for rhythm management in elderly patients with atrial fibrillation. *J Interv Card Electrophysiol*. 2009;25:25–9.
457. Maron BJ, Olivetto I, Bellone P, et al. Clinical profile of stroke in 900 patients with hypertrophic cardiomyopathy. *J Am Coll Cardiol*. 2002;39:301–7.
458. Bunch TJ, Munger TM, Friedman PA, et al. Substrate and procedural predictors of outcomes after catheter ablation for atrial fibrillation in patients with hypertrophic cardiomyopathy. *J Cardiovasc Electrophysiol*. 2008;19:1009–14.
459. Di Donna P, Olivetto I, Delcre SD, et al. Efficacy of catheter ablation for atrial fibrillation in hypertrophic cardiomyopathy: impact of age, atrial remodelling, and disease progression. *Europace*. 2010;12:347–55.
460. Gaita F, Di Donna P, Olivetto I, et al. Usefulness and safety of transcatheter ablation of atrial fibrillation in patients with hypertrophic cardiomyopathy. *Am J Cardiol*. 2007;99:1575–81.
461. Kilicaslan F, Verma A, Saad E, et al. Efficacy of catheter ablation of atrial fibrillation in patients with hypertrophic obstructive cardiomyopathy. *Heart Rhythm*. 2006;3:275–80.
462. Maron BJ. Hypertrophic cardiomyopathy. *Lancet*. 1997;350:127–33.
463. Cecchi F, Olivetto I, Monterege A, et al. Hypertrophic cardiomyopathy in Tuscany: clinical course and outcome in an unselected regional population. *J Am Coll Cardiol*. 1995;26:1529–36.
464. Higashikawa M, Nakamura Y, Yoshida M, et al. Incidence of ischemic strokes in hypertrophic cardiomyopathy is markedly increased if complicated by atrial fibrillation. *Jpn Circ J*. 1997;61:673–81.
465. Savage DD, Seides SF, Maron BJ, et al. Prevalence of arrhythmias during 24-hour electrocardiographic monitoring and exercise testing in patients with obstructive and nonobstructive hypertrophic cardiomyopathy. *Circulation*. 1979;59:866–75.
466. Tendra M, Wycisk A, Schneeweiss A, et al. Effect of sotalol on arrhythmias and exercise tolerance in patients with hypertrophic cardiomyopathy. *Cardiology*. 1993;82:335–42.
467. Chen MS, McCarthy PM, Lever HM, et al. Effectiveness of atrial fibrillation surgery in patients with hypertrophic cardiomyopathy. *Am J Cardiol*. 2004;93:373–5.
468. Crenshaw BS, Ward SR, Granger CB, et al. Atrial fibrillation in the setting of acute myocardial infarction: the GUSTO-I experience. Global Utilization of Streptokinase and TPA for Occluded Coronary Arteries. *J Am Coll Cardiol*. 1997;30:406–13.
469. Goldberg RJ, Seeley D, Becker RC, et al. Impact of atrial fibrillation on the in-hospital and long-term survival of patients with acute myocardial infarction: a community-wide perspective. *Am Heart J*. 1990;119:996–1001.
470. Behar S, Zahavi Z, Goldbourt U, et al. Long-term prognosis of patients with paroxysmal atrial fibrillation complicating acute myocardial infarction. SPRINT Study Group. *Eur Heart J*. 1992;13:45–50.
471. Pedersen OD, Bagger H, Kober L, et al. The occurrence and prognostic significance of atrial fibrillation/flutter following acute myocardial

- infarction. TRACE Study Group. TRAndolapril Cardiac Evaluation. *Eur Heart J*. 1999;20:748–54.
472. McMurray J, Kober L, Robertson M, et al. Antiarrhythmic effect of carvedilol after acute myocardial infarction: results of the Carvedilol Post-Infarct Survival Control in Left Ventricular Dysfunction (CAPRICORN) trial. *J Am Coll Cardiol*. 2005;45:525–30.
473. Klein I, Danzi S. Thyroid disease and the heart. *Circulation*. 2007;116:1725–35.
474. Petersen P. Thromboembolic complications in atrial fibrillation. *Stroke*. 1990;21:4–13.
475. Klein I, Ojamaa K. Thyroid hormone and the cardiovascular system. *N Engl J Med*. 2001;344:501–9.
476. Siu CW, Pong V, Zhang X, et al. Risk of ischemic stroke after new-onset atrial fibrillation in patients with hyperthyroidism. *Heart Rhythm*. 2009;6:169–73.
477. Taylor FC, Cohen H, Ebrahim S. Systematic review of long term anticoagulation or antiplatelet treatment in patients with non-rheumatic atrial fibrillation. *BMJ*. 2001;322:321–6.
478. Bar-Sela S, Ehrenfeld M, Eliakim M. Arterial embolism in thyrotoxicosis with atrial fibrillation. *Arch Intern Med*. 1981;141:1191–2.
479. Hurley DM, Hunter AN, Hewett MJ, et al. Atrial fibrillation and arterial embolism in hyperthyroidism. *Aust N Z J Med*. 1981;11:391–3.
480. Staffurth JS, Gibberd MC, Fui SN. Arterial embolism in thyrotoxicosis with atrial fibrillation. *Br Med J*. 1977;2:688–90.
481. Yuen RW, Gutteridge DH, Thompson PL, et al. Embolism in thyrotoxic atrial fibrillation. *Med J Aust*. 1979;1:630–1.
482. Agner T, Almdal T, Thorsteinsson B, et al. A reevaluation of atrial fibrillation in thyrotoxicosis. *Dan Med Bull*. 1984;31:157–9.
483. Clozel JP, Danchin N, Genton P, et al. Effects of propranolol and of verapamil on heart rate and blood pressure in hyperthyroidism. *Clin Pharmacol Ther*. 1984;36:64–9.
484. Traube E, Coplan NL. Embolic risk in atrial fibrillation that arises from hyperthyroidism: review of the medical literature. *Tex Heart Inst J*. 2011;38:225–8.
485. Petersen P, Hansen JM. Stroke in thyrotoxicosis with atrial fibrillation. *Stroke*. 1988;19:15–8.
486. Bahn RS, Burch HB, Cooper DS, et al. Hyperthyroidism and other causes of thyrotoxicosis: management guidelines of the American Thyroid Association and American Association of Clinical Endocrinologists. *Endocr Pract*. 2011;17:456–520.
487. Dunning J, Treasure T, Versteegh M, et al. Guidelines on the prevention and management of de novo atrial fibrillation after cardiac and thoracic surgery. *Eur J Cardiothorac Surg*. 2006;30:852–72.
488. Hudson LD, Kurt TL, Petty TL, et al. Arrhythmias associated with acute respiratory failure in patients with chronic airway obstruction. *Chest*. 1973;63:661–5.
489. Fuso L, Incalzi RA, Pistelli R, et al. Predicting mortality of patients hospitalized for acutely exacerbated chronic obstructive pulmonary disease. *Am J Med*. 1995;98:272–7.
490. Shih HT, Webb CR, Conway WA, et al. Frequency and significance of cardiac arrhythmias in chronic obstructive lung disease. *Chest*. 1988;94:44–8.
491. McCord J, Borzak S. Multifocal atrial tachycardia. *Chest*. 1998;113:203–9.
492. Goldman S, Probst P, Selzer A, et al. Inefficacy of “therapeutic” serum levels of digoxin in controlling the ventricular rate in atrial fibrillation. *Am J Cardiol*. 1975;35:651–5.
493. Boriani G, Biffi M, Frabetti L, et al. Ventricular fibrillation after intravenous amiodarone in Wolff-Parkinson-White syndrome with atrial fibrillation. *Am Heart J*. 1996;131:1214–6.
494. Kim RJ, Gerling BR, Kono AT, et al. Precipitation of ventricular fibrillation by intravenous diltiazem and metoprolol in a young patient with occult Wolff-Parkinson-White syndrome. *Pacing Clin Electrophysiol*. 2008;31:776–9.
495. Simonian SM, Lotfipour S, Wall C, et al. Challenging the superiority of amiodarone for rate control in Wolff-Parkinson-White and atrial fibrillation. *Intern Emerg Med*. 2010;5:421–6.
496. Fukatani M, Tanigawa M, Mori M, et al. Prediction of a fatal atrial fibrillation in patients with asymptomatic Wolff-Parkinson-White pattern. *Jpn Circ J*. 1990;54:1331–9.
497. Pietersen AH, Andersen ED, Sandoe E. Atrial fibrillation in the Wolff-Parkinson-White syndrome. *Am J Cardiol*. 1992;70:38–43A.
498. Klein GJ, Bashore TM, Sellers TD, et al. Ventricular fibrillation in the Wolff-Parkinson-White syndrome. *N Engl J Med*. 1979;301:1080–5.
499. Zardini M, Yee R, Thakur RK, et al. Risk of sudden arrhythmic death in the Wolff-Parkinson-White syndrome: current perspectives. *Pacing Clin Electrophysiol*. 1994;17:966–75.
500. Akhtar M, Gilbert CJ, Shenasa M. Effect of lidocaine on atrioventricular response via the accessory pathway in patients with Wolff-Parkinson-White syndrome. *Circulation*. 1981;63:435–41.
501. Dolla E, Levy S, Cointe R, et al. Oral propranolol in Wolff-Parkinson-White syndrome: electrophysiological data [in French]. *Arch Mal Coeur Vaiss*. 1991;84:917–21.
502. Balser JR, Martinez EA, Winters BD, et al. Beta-adrenergic blockade accelerates conversion of postoperative supraventricular tachyarrhythmias. *Anesthesiology*. 1998;89:1052–9.
503. Tamariz LJ, Bass EB. Pharmacological rate control of atrial fibrillation. *Cardiol Clin*. 2004;22:35–45.
504. Lewis RV, McMurray J, McDewitt DG. Effects of atenolol, verapamil, and xamoterol on heart rate and exercise tolerance in digitalised patients with chronic atrial fibrillation. *J Cardiovasc Pharmacol*. 1989;13:1–6.
505. Hunt SA, Abraham WT, Chin MH, et al. 2009 Focused update incorporated into the ACC/AHA 2005 guidelines for the diagnosis and management of heart failure in adults: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2009;119:e391–479.
506. Roberts SA, Diaz C, Nolan PE, et al. Effectiveness and costs of digoxin treatment for atrial fibrillation and flutter. *Am J Cardiol*. 1993;72:567–73.
507. Segal JB, McNamara RL, Miller MR, et al. The evidence regarding the drugs used for ventricular rate control. *J Fam Pract*. 2000;49:47–59.
508. Feld GK, Fleck RP, Fujimura O, et al. Control of rapid ventricular response by radiofrequency catheter modification of the atrioventricular node in patients with medically refractory atrial fibrillation. *Circulation*. 1994;90:2299–307.
509. Williamson BD, Man KC, Daoud E, et al. Radiofrequency catheter modification of atrioventricular conduction to control the ventricular rate during atrial fibrillation. *N Engl J Med*. 1994;331:910–7.
510. Gentles PJ, Sauer WH, Gerstenfeld EP, et al. Reversal of left ventricular dysfunction following ablation of atrial fibrillation. *J Cardiovasc Electrophysiol*. 2007;18:9–14.
511. Maisel WH, Stevenson LW. Atrial fibrillation in heart failure: epidemiology, pathophysiology, and rationale for therapy. *Am J Cardiol*. 2003;91:2–8D.
512. Dickstein K, Cohen-Solal A, Filippatos G, et al. ESC guidelines for the diagnosis and treatment of acute and chronic heart failure 2008: the Task Force for the Diagnosis and Treatment of Acute and Chronic Heart Failure 2008 of the European Society of Cardiology. Developed in collaboration with the Heart Failure Association of the ESC (HFA). *Eur Heart J*. 2008;29:2388–442.
513. Tsang TS, Gersh BJ, Appleton CP, et al. Left ventricular diastolic dysfunction as a predictor of the first diagnosed nonvalvular atrial fibrillation in 840 elderly men and women. *J Am Coll Cardiol*. 2002;40:1636–44.
514. Roy D, Talajic M, Nattel S, et al. Rhythm control versus rate control for atrial fibrillation and heart failure. *N Engl J Med*. 2008;358:2667–77.
515. Upadhyay GA, Choudhry NK, Auricchio A, et al. Cardiac resynchronization in patients with atrial fibrillation: a meta-analysis of prospective cohort studies. *J Am Coll Cardiol*. 2008;52:1239–46.
516. Wilton SB, Leung AA, Ghali WA, et al. Outcomes of cardiac resynchronization therapy in patients with versus those without atrial fibrillation: a systematic review and meta-analysis. *Heart Rhythm*. 2011;8:1088–94.
517. Gasparini M, Auricchio A, Regoli F, et al. Four-year efficacy of cardiac resynchronization therapy on exercise tolerance and disease progression: the importance of performing atrioventricular junction ablation in patients with atrial fibrillation. *J Am Coll Cardiol*. 2006;48:734–43.
518. Doshi RN, Daoud EG, Fellows C, et al. Left ventricular-based cardiac stimulation post AV nodal ablation evaluation (the PAVE study). *J Cardiovasc Electrophysiol*. 2005;16:1160–5.
519. Brignole M, Botto G, Mont L, et al. Cardiac resynchronization therapy in patients undergoing atrioventricular junction ablation for permanent atrial fibrillation: a randomized trial. *Eur Heart J*. 2011;32:2420–9.
520. Brignole M, Gammage M, Puggioni E, et al. Comparative assessment of right, left, and biventricular pacing in patients with permanent atrial fibrillation. *Eur Heart J*. 2005;26:712–22.
521. Fox CS, Parise H, D’Agostino RB Sr, et al. Parental atrial fibrillation as a risk factor for atrial fibrillation in offspring. *JAMA*. 2004;291:2851–5.
522. Darbar D, Herron KJ, Ballew JD, et al. Familial atrial fibrillation is a genetically heterogeneous disorder. *J Am Coll Cardiol*. 2003;41:2185–92.

523. Christophersen IE, Ravn LS, Budtz-Joergensen E, et al. Familial aggregation of atrial fibrillation: a study in Danish twins. *Circ Arrhythm Electrophysiol*. 2009;2:378–83.
524. Ellinor PT, Yoerger DM, Ruskin JN, et al. Familial aggregation in lone atrial fibrillation. *Hum Genet*. 2005;118:179–84.
525. Lubitz SA, Ozcan C, Magnani JW, et al. Genetics of atrial fibrillation: implications for future research directions and personalized medicine. *Circ Arrhythm Electrophysiol*. 2010;3:291–9.
526. Lubitz SA, Sinner MF, Lunetta KL, et al. Independent susceptibility markers for atrial fibrillation on chromosome 4q25. *Circulation*. 2010;122:976–84.
527. Lubitz SA, Lunetta KL, Lin H, et al. Novel genetic markers associate with atrial fibrillation risk in Europeans and Japanese. *J Am Coll Cardiol*. 2014;63:1200–10.
528. Smith JG, Newton-Cheh C, Almgren P, et al. Genetic polymorphisms for estimating risk of atrial fibrillation in the general population: a prospective study. *Arch Intern Med*. 2012;172:742–4.
529. Everett BM, Cook NR, Conen D, et al. Novel genetic markers improve measures of atrial fibrillation risk prediction. *Eur Heart J*. 2013;34:2243–51.
530. Gretarsdottir S, Thorleifsson G, Manolescu A, et al. Risk variants for atrial fibrillation on chromosome 4q25 associate with ischemic stroke. *Ann Neurol*. 2008;64:402–9.
531. Lemmens R, Buysschaert I, Geelen V, et al. The association of the 4q25 susceptibility variant for atrial fibrillation with stroke is limited to stroke of cardioembolic etiology. *Stroke*. 2010;41:1850–7.
532. Ackerman MJ, Priori SG, Willems S, et al. HRS/EHRA expert consensus statement on the state of genetic testing for the channelopathies and cardiomyopathies. *Heart Rhythm*. 2011;8:1308–39.
533. Crystal E, Garfinkle MS, Connolly SS, et al. Interventions for preventing post-operative atrial fibrillation in patients undergoing heart surgery. *Cochrane Database Syst Rev*. 2004;(4):CD003611.
534. Yoshioka I, Sakurai M, Namai A, et al. Postoperative treatment of carvedilol following low dose landiolol has preventive effect for atrial fibrillation after coronary artery bypass grafting. *Thorac Cardiovasc Surg*. 2009;57:464–7.
535. Davis EM, Packard KA, Hilleman DE. Pharmacologic prophylaxis of postoperative atrial fibrillation in patients undergoing cardiac surgery: beyond beta-blockers. *Pharmacotherapy*. 2010;30:274–318e.
536. Koniari I, Apostolakis E, Rogkakou C, et al. Pharmacologic prophylaxis for atrial fibrillation following cardiac surgery: a systematic review. *J Cardiothorac Surg*. 2010;5:121.
537. Hilleman DE, Hunter CB, Mohiuddin SM, et al. Pharmacological management of atrial fibrillation following cardiac surgery. *Am J Cardiovasc Drugs*. 2005;5:361–9.
538. Daoud EG, Strickberger SA, Man KC, et al. Preoperative amiodarone as prophylaxis against atrial fibrillation after heart surgery. *N Engl J Med*. 1997;337:1785–91.
539. Guarnieri T, Nolan S, Gottlieb SO, et al. Intravenous amiodarone for the prevention of atrial fibrillation after open heart surgery: the Amiodarone Reduction in Coronary Heart (ARCH) trial. *J Am Coll Cardiol*. 1999;34:343–7.
540. Mitchell LB, Exner DV, Wyse DG, et al. Prophylactic Oral Amiodarone for the Prevention of Arrhythmias that Begin Early After Revascularization, Valve Replacement, or Repair: PAPABEAR: a randomized controlled trial. *JAMA*. 2005;294:3093–100.
541. Vanderlugt JT, Mattioni T, Denker S, et al. Efficacy and safety of ibutilide fumarate for the conversion of atrial arrhythmias after cardiac surgery. *Circulation*. 1999;100:369–75.
542. Al-Khatib SM, Hafley G, Harrington RA, et al. Patterns of management of atrial fibrillation complicating coronary artery bypass grafting: results from the Project of Ex-vivo Vein graft ENGINEering via Transfection IV (PREVENT-IV) Trial. *Am Heart J*. 2009;158:792–8.
543. Shepherd J, Jones J, Frampton GK, et al. Intravenous magnesium sulphate and sotalol for prevention of atrial fibrillation after coronary artery bypass surgery: a systematic review and economic evaluation. *Health Technol Assess*. 2008;12:iii–iv, ix–95.
544. Imazio M, Brucato A, Ferrazzi P, et al. Colchicine reduces postoperative atrial fibrillation: results of the Colchicine for the Prevention of the Postpericardiotomy Syndrome (COPPS) atrial fibrillation substudy. *Circulation*. 2011;124:2290–5.
545. Echahidi N, Pibarot P, O'Hara G, et al. Mechanisms, prevention, and treatment of atrial fibrillation after cardiac surgery. *J Am Coll Cardiol*. 2008;51:793–801.
546. Mathew JP, Parks R, Savino JS, et al. Atrial fibrillation following coronary artery bypass graft surgery: predictors, outcomes, and resource utilization. MultiCenter Study of Perioperative Ischemia Research Group. *JAMA*. 1996;276:300–6.
547. Aranki SF, Shaw DP, Adams DH, et al. Predictors of atrial fibrillation after coronary artery surgery: current trends and impact on hospital resources. *Circulation*. 1996;94:390–7.
548. Almassi GH, Schowalter T, Nicolosi AC, et al. Atrial fibrillation after cardiac surgery: a major morbid event? *Ann Surg*. 1997;226:501–11.

KEY WORDS: AHA Scientific Statements ■ atrial fibrillation ■ cardio-renal physiology/pathophysiology ■ cardiovascular surgery: transplantation, ventricular assistance, cardiomyopathy ■ epidemiology ■ full revision ■ health policy and outcome research ■ other atrial fibrillation

Appendix 1. Author Relationships With Industry and Other Entities (Relevant)—2014 AHA/ACC/HRS Guideline for the Management of Patients With Atrial Fibrillation

Committee Member	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Voting Recusals by Section*
Craig T. January (Chair)	University of Wisconsin-Madison—Professor of Medicine, Cardiovascular Medicine Division	None	None	None	None	None	None	None
L. Samuel Wann (Vice Chair)	Columbia St. Mary's Cardiovascular Physicians—Clinical Cardiologist	• United Healthcare	None	None	None	None	None	4.1 5.0 6.3 7.3 7.10
Joseph S. Alpert	University of Arizona Health Sciences Center—Professor of Medicine	• Bayer Pharmaceuticals (DSMB)† • Boehringer Ingelheim • Daiichi-Sankyo • Johnson & Johnson • Roche Diagnostics • Sanofi-aventis • Servier Pharmaceuticals	None	None	None	None	None	4.1 5.0
Hugh Calkins	Johns Hopkins Hospital—Professor of Medicine, Director of Electrophysiology	• AtriCure • Biosense Webster • CareCore • iRhythm • Medtronic‡ • Sanofi-aventis	None	None	None	None	None	5.0 6.3 7.8
Joaquin E. Cigarroa	Oregon Health and Science University—Clinical Professor; Clinical Chief of Cardiology	None	None	None	None	None	None	None
Joseph C. Cleveland, Jr	University of Colorado—Professor of Surgery; Denver Veterans Affairs Hospital—Chief, Cardiac Surgery	None	None	None	None	None	None	None
Jamie B. Conti	University of Florida—Professor of Medicine; Division of Cardiovascular Medicine—Chief	None	None	None	• Boston Scientific‡ • Medtronic‡ • St. Jude Medical‡	• Boston Scientific‡ • Medtronic‡ • St. Jude Medical‡	None	5.0 6.3 7.8
Patrick T. Ellinor	Massachusetts General Hospital Heart Center, Cardiac Arrhythmia Service—Director	None	None	None	None	None	None	None
Michael D. Ezekowitz	Jefferson Medical College—Professor	• ARYx Therapeutics‡ • AstraZeneca • Boehringer Ingelheim‡ • Bristol-Myers Squibb‡ • Daiichi-Sankyo‡ • Eisai • Johnson & Johnson‡ • Medtronic‡ • Pfizer‡ • Portola‡ • Sanofi-aventis‡	None	None	• ARYx Therapeutics‡ • Boehringer Ingelheim‡ • Daiichi-Sankyo† • Portola‡	None	None	4.1 5.0 6.3 7.8

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Appendix 1. Continued

Committee Member	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Voting Recusals by Section*
Michael E. Field	University of Wisconsin School of Medicine and Public Health—Assistant Professor of Medicine, Director of Cardiac Arrhythmia Service	None	None	None	None	None	None	None
Katherine T. Murray	Vanderbilt University School of Medicine, Divisions of Clinical Pharmacology and Cardiology—Professor of Medicine	None	None	None	• GlaxoSmithKline†	None	None	None
Ralph L. Sacco	University of Miami, Miller School of Medicine, Department of Neurology—Chairman	• Boehringer Ingelheim†§	None	None	None	None	None	None
William G. Stevenson	Brigham and Women's Hospital, Cardiac Arrhythmia Program—Director; Harvard Medical School—Professor of Medicine	None	None	• Biosense Webster—Needle Ablation Patent†	• Biosense Webster‡	None	None	5.0 6.3 7.8
Patrick J. Tchou	Cleveland Clinic Foundation—Section of Cardiac Electrophysiology and Pacing, Department of Cardiovascular Medicine Heart and Vascular Institute	None	None	None	None	None	None	None
Cynthia M. Tracy	George Washington University Medical Center—Associate Director and Professor of Medicine	None	None	None	None	None	None	None
Clyde W. Yancy	Northwestern University, Feinberg School of Medicine—Magerstadt Professor of Medicine; Division of Cardiology—Chief	None	None	None	None	None	None	None

This table represents the relationships of committee members with industry and other entities that were determined to be relevant to this document. These relationships were reviewed and updated in conjunction with all meetings and/or conference calls of the writing committee during the document development process. The table does not necessarily reflect relationships with industry at the time of publication. A person is deemed to have a significant interest in a business if the interest represents ownership of $\geq 5\%$ of the voting stock or share of the business entity, or ownership of $\geq \$10\,000$ of the fair market value of the business entity; or if funds received by the person from the business entity exceed 5% of the person's gross income for the previous year. Relationships that exist with no financial benefit are also included for the purpose of transparency. Relationships in this table are modest unless otherwise noted.

According to the ACC/AHA, a person has a *relevant* relationship IF: a) the *relationship or interest* relates to the same or similar subject matter, intellectual property or asset, topic, or issue addressed in the *document*; or b) the *company/entity* (with whom the relationship exists) makes a drug, drug class, or device addressed in the *document*, or makes a competing drug or device addressed in the *document*; or c) the *person, or a member of the person's household*, has a reasonable potential for financial, professional, or other personal gain or loss as a result of the issues/content addressed in the *document*.

*Writing committee members are required to recuse themselves from voting on sections to which their specific relationships with industry and other entities may apply.

†No financial benefit.

‡Indicates significant relationship.

§Dr. Sacco's relationship with Boehringer Ingelheim was added just after final balloting of the recommendations and before organizational review, so it was not relevant during the writing or voting stages of the guideline's development.

ACC indicates American College of Cardiology; AHA, American Heart Association; DSMB, data safety monitoring board; and HRS, Heart Rhythm Society.

Appendix 2. Reviewer Relationships With Industry and Other Entities (Relevant)—2014 AHA/ACC/HRS Guideline for the Management of Patients With Atrial Fibrillation

Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
A. John Camm	Official Reviewer—HRS	St. George's, University of London—Professor of Clinical Cardiology	• Bayer • Biotronik • Boehringer Ingelheim • Boston Scientific • Bristol-Myers Squibb • ChanRx • Daiichi-Sankyo • Forest Laboratories • Johnson & Johnson • Medtronic • Novartis* • Sanofi-aventis • Servier • St. Jude Medical • Takeda • Xention	• Pfizer	None	• Biotronik† • Servier (DSMB) • St. Jude Medical (DSMB)	None	None
John Fisher	Official Reviewer—AHA	Albert Einstein College of Medicine—Professor of Medicine	• Medtronic*	None	None	None	• Biotronik* • Boston Scientific* • Medtronic* • St. Jude Medical*	None
Jonathan L. Halperin	Official Reviewer—ACC/AHA Task Force on Practice Guidelines	Mt. Sinai Medical Center—Professor of Medicine	• AstraZeneca • Bayer • Biotronik* • Boehringer Ingelheim* • Boston Scientific • Bristol-Myers Squibb • Daiichi-Sankyo • Janssen Pharmaceuticals • Johnson & Johnson • Medtronic • Pfizer • Sanofi-aventis	None	None	None	None	None
Jose Joglar	Official Reviewer—AHA	UT Southwestern Medical Center—Associate Professor of Internal Medicine	None	None	None	None	• Medtronic* • St. Jude Medical*	None

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Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Peter Kowey	Official Reviewer—HRS	Lankenau Medical Office Building—Chief of Cardiology	• Astellas† • AstraZeneca* • Boehringer Ingelheim* • Bristol-Myers Squibb • Daiichi-Sankyo* • Forest Laboratories • GlaxoSmithKline* • Johnson & Johnson* • Medtronic • Merck* • Pfizer* • Portola • Sanofi-aventis*	None	• CardioNet*	None	None	None
John Strobel	Official Reviewer—ACC Board of Governors	Premier Healthcare, LLC—Clinical Cardiac EP Indiana University—Assistant Clinical Professor of Medicine	None	• Boehringer Ingelheim • Bristol-Myers Squibb • Pfizer • Sanofi-aventis	None	None	None	• Plaintiff, ICD, 2012
Stuart Winston	Official Reviewer—ACC Board of Trustees	Michigan Heart, P. C. Michigan Heart and Vascular Institute—Cardiologist	None	None	None	None	• Biotronik† • Medtronic†	None
James R. Edgerton	Organizational Reviewer—STS	The Heart Hospital Baylor Plano—Cardiologist; University of Texas at Arlington—Adjunct Assistant Clinical Professor	None	• AtriCure*	None	None	None	None
Jeffrey L. Anderson	Content Reviewer—ACC/AHA Task Force on Practice Guidelines	Intermountain Medical Center—Associate Chief of Cardiology	• The Medicines Company • Sanofi-aventis	None	None	None	None	None
Nancy Berg	Content Reviewer—ACC EP Section Leadership Council	Park Nicollet Health Services—Registered Nurse	• Medtronic	None	None	• Mayo Clinic	• Medtronic†	None
Emmanouil Brilakis	Content Reviewer—ACC Interventional Section Leadership Council	UT Southwestern Medical School—Director, Cardiac Catheterization Laboratory, VA North Texas Healthcare System	• Boston Scientific* • Bridgepoint Medical* • Janssen Pharmaceuticals • Sanofi-aventis • St. Jude Medical	None	None	None	• Abbott Vascular† • AstraZeneca† • Cordis* • Daiichi-Sankyo* • Medtronic* • The Medicines Company*	None

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Appendix 2. Continued

Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Yong-Mei Cha	Content Reviewer—AHA	Mayo Clinic, Division of Cardiovascular Diseases— Professor of Medicine	None	None	None	None	None	None
Jafna Cox	Content Reviewer— ACC Board of Governors	Queen Elizabeth II Health Sciences Center—Professor, Departments of Medicine, Community Health, and Epidemiology	• AstraZeneca • Bayer • Boehringer Ingelheim	None	None	• Bayer* • Pfizer*	None	None
Anne Curtis	Content Reviewer	University of Buffalo— Charles and Mary Bauer Professor of Medicine	• Biosense Webster • Bristol-Myers Squibb • Medtronic* • Pfizer • Sanofi-aventis • St. Jude Medical	None	None	None	None	None
Lesley H. Curtis	Content Reviewer— ACC/AHA Task Force on Practice Guidelines	Duke University School of Medicine— Associate Professor of Medicine	None	None	None	None	• Medtronic* • GE Healthcare* • GlaxoSmithKline* • Johnson & Johnson*	None
Kenneth Ellenbogen	Content Reviewer	VCU Medical Center—Director, Clinical EP Laboratory	• Biosense Webster • Biotronik* • Boston Scientific* • Cameron Health • Janssen Pharmaceuticals • Medtronic* • Sanofi-aventis • St. Jude Medical	None	None	• Biosense Webster* • Boston Scientific* • Medtronic* • Sanofi-aventis*	• Biosense Webster* • Boston Scientific* • CardioNet • Medtronic* • Sanofi-aventis* • St. Jude Medical*	• Represented hospital, ICD, 2012
N.A. Mark Estes III	Content Reviewer	Tufts University School of Medicine— Professor of Medicine	• Boston Scientific* • Medtronic	None	None	• Boston Scientific	• Boston Scientific* • Medtronic* • St. Jude Medical*	None
Gregg Fonarow	Content Reviewer	Ahmanson—UCLA Cardiomyopathy Center, Division of Cardiology	• Boston Scientific • Johnson & Johnson • The Medicines Company • Medtronic	None	None	• Novartis*	• Medtronic†	None
Valentin Fuster	Content Reviewer	Mount Sinai School of Medicine— Director, Zena and Michael A. Wiener Cardiovascular Institute	None	None	None	None	None	None

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Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Richard Goodman	Content Reviewer—HHS	HHS Office of the Assistant Secretary for Health and National Center for Chronic Disease Prevention and Health Promotion Centers for Disease Control and Prevention—Senior Medical Advisor	None	None	None	None	None	None
Judith S. Hochman	Content Reviewer—ACC/AHA Task Force on Practice Guidelines	New York University School of Medicine—Clinical Chief of Cardiology	• GlaxoSmithKline • Janssen Pharmaceuticals	None	None	None	None	None
Warren Jackman	Content Reviewer	University of Oklahoma Health Sciences Center for Cardiac Arrhythmia Research Institute—Professor of Medicine	• Biosense Webster* • Endosense* • VytronUS*	• Biotronik* • Boston Scientific*	• Rhythmia Medical*	• Boston Scientific* • Rhythmia Medical*	None	None
Samuel Jones	Content Reviewer—ACC Board of Governors	USUHS—Associate Professor of Medicine	None	None	None	None	• Medtronic† • St. Jude Medical†	None
Paulus Kirchhof	Content Reviewer—HRS	University of Birmingham, School of Clinical and Experimental Medicine—Chair in Cardiovascular Medicine	None	None	None	• Sanofi-aventis (DSMB)	None	None
Bradley Knight	Content Reviewer	Northwestern Medical Center Division of Cardiology—Director of Clinical Cardiac EP	• Boston Scientific • Cameron Health†	• Biosense Webster • Biotronik • Boston Scientific • Medtronic	None	• Catheter Robotics	None	• Plaintiff, pacemaker surgery, 2012
Austin Kutscher	Content Reviewer	Hunterdon Cardiovascular Associates—Cardiologist	• Pfizer	• Bristol-Myers Squibb • Forest Laboratories	None	• Boehringer Ingelheim • Bristol-Myers Squibb	None	None
Gregory Michaud	Content Reviewer	Harvard Medical School, Brigham and Women's Hospital—Assistant Professor	• Boston Scientific • Medtronic	None	None	• Boston Scientific* • St. Jude Medical*	None	None
William Miles	Content Reviewer	University of Florida, Department of Medicine—Cardiologist	None	None	None	• Medtronic—STOP-AF (PI) • Zoll Medical	None	None
Simone Musco	Content Reviewer—ACC Board of Governors	Saint Patrick Hospital—Cardiologist	None	• Bristol-Myers Squibb • Sanofi-aventis	None	None	None	None

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Appendix 2. Continued

Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Brian Olshansky	Content Reviewer—ACC EP Section Leadership Council	University of Iowa Hospital— Professor of Medicine	• Boehringer Ingelheim • Boston Scientific • Guidant • Medtronic* • Sanofi-aventis	None	None	• Boston Scientific (DSMB) • Sanofi-aventis (DSMB)	None	None
Huseyin Murat Ozdemir	Content Reviewer—AIG	Gazi University School of Medicine— Professor of Cardiology	• Bayer • Boehringer Ingelheim • Bristol-Myers Squibb • Novartis • Pfizer • Servier	None	None	None	None	None
Douglas Packer	Content Reviewer	Mayo Foundation St. Mary's Hospital Complex— Professor of Medicine	• Abiomed† • Biosense Webster† • Boston Scientific† • InfoBionic† • Johnson & Johnson† • Medtronic† • Janssen Pharmaceuticals† • Sanofi-aventis† • Siemens† • St. Jude Medical†	None	None	• Biosense Webster* • Boston Scientific* • CardioFocus • Endosense* • Hansen Medical • Medtronic* • Siemens • St. Jude Medical* • Thermedical*	• St. Jude Medical*	None
Richard Page	Content Reviewer	University of Wisconsin Hospital and Clinics—Chair, Department of Medicine	None	None	None	None	None	None
Robert Page	Content Reviewer—AHA PharmD	University of Colorado School of Pharmacy— Associate Professor	None	None	None	None	None	None
Gurusher Panjra	Content Reviewer—ACC HF and Transplant Section Leadership Council	George Washington University— Assistant Professor of Medicine	None	None	None	None	None	None
Eric Prystowsky	Content Reviewer—HRS	St. Vincent Hospital and Health Center— Director, Clinical EP Laboratory	• Bard* • Medtronic*	None	• CardioNet* • Topera* • Stereotaxis*	None	• CardioNet* • Stereotaxis*	None
Pasala Ravichandran	Content Reviewer—ACC Surgeons' Council	Oregon Health and Science University— Associate Professor	None	None	None	None	None	None
Anitra Romfh	Content Reviewer—ACC Adult Congenital and Pediatric Cardiology Section Leadership Council	Children's Hospital Boston— Cardiologist	None	None	None	None	None	None

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Appendix 2. Continued

Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Elizabeth Saarel	Content Reviewer—ACC Adult Congenital and Pediatric Cardiology Section Leadership Council	University of Utah School of Medicine and Primary Children's Medical Center—Associate Professor	None	None	None	None	None	None
Marcel Salive	Content Reviewer—HHS	National Institute on Aging, Division of Geriatrics and Clinical Gerontology	None	None	• Express Scripts*	None	None	None
John Sapp	Content Reviewer—HRS	Dalhousie University—Director of EP	• Biosense Webster	None	None	• Biosense Webster* • St. Jude Medical*	None	None
Frank W. Sellke	Content Reviewer—ACC/AHA Task Force on Practice Guidelines	Cardiovascular Institute, Rhode Island Hospital and Lifespan—Chief of Cardiothoracic Surgery	None	None	None	None	• The Medicines Company	None
Win-Kuang Shen	Content Reviewer—ACC/AHA Task Force on Practice Guidelines	Mayo Clinic Arizona—Professor of Medicine, Consultant	None	None	None	None	None	None
David J. Slotwiner	Content Reviewer	Long Island Jewish Medical Center—Associate Director, EP Laboratory	None	None	None	None	• Boston Scientific	None
Jonathan Steinberg	Content Reviewer	Valley Health System Arrhythmia Institute—Director; Columbia University College of Physicians and Surgeons—Professor of Medicine	• Ambucor • Biosense Webster • Boston Scientific • Medtronic	• Bristol-Myers Squibb* • Sanofi-aventis	None	• Biosense Webster* • Janssen Pharmaceuticals • Medtronic*	None	None
Vinod Thourani	Content Reviewer—ACC Surgeons' Council	Emory University School of Medicine—Associate Professor of Cardiothoracic Surgery	• Edwards Lifesciences • Sorin • St. Jude Medical	None	• Apica Cardiovascular†	• Maquet	None	None
Mellanie True Hills	Content Reviewer—Patient Advocate	StopAfib.org—Speaker and Chief Executive Officer	• AtriCure	None	None	None	• Bayer* • Boehringer Ingelheim* • Janssen Pharmaceuticals* • Johnson & Johnson* • Medtronic • Sanofi-aventis*	None

(Continued)

Appendix 2. Continued

Reviewer	Representation	Employment	Consultant	Speaker's Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Albert Waldo	Content Reviewer—HRS	Case Western Reserve University—The Walter H. Pritchard Professor of Cardiology, Professor of Medicine, and Professor of Biomedical Engineering	• Abbott Vascular • AtriCure • Biosense Webster • Biotronik • Daiichi-Sankyo • Gilead • Janssen Pharmaceuticals* • Merck • Pfizer • Sanofi-aventis	• Janssen Pharmaceuticals* • Sanofi-aventis*	None	• Biotronik • Daiichi-Sankyo • Gilead* • St. Jude Medical*	None	None

This table represents the relationships of reviewers with industry and other entities that were disclosed at the time of peer review and determined to be relevant to this document. It does not necessarily reflect relationships with industry at the time of publication. A person is deemed to have a significant interest in a business if the interest represents ownership of $\geq 5\%$ of the voting stock or share of the business entity, or ownership of $\geq \$10\,000$ of the fair market value of the business entity; or if funds received by the person from the business entity exceed 5% of the person's gross income for the previous year. A relationship is considered to be modest if it is less than significant under the preceding definition. Relationships that exist with no financial benefit are also included for the purpose of transparency. Relationships in this table are modest unless otherwise noted. Names are listed in alphabetical order within each category of review.

According to the ACC/AHA, a person has a *relevant* relationship IF: a) the *relationship or interest* relates to the same or similar subject matter, intellectual property or asset, topic, or issue addressed in the *document*; or b) the *company/entity* (with whom the relationship exists) makes a drug, drug class, or device addressed in the *document*, or makes a competing drug or device addressed in the *document*; or c) the *person*, or a member of the person's household, has a reasonable potential for financial, professional, or other personal gain or loss as a result of the issues/content addressed in the *document*.

*Significant relationship.

†No financial benefit.

ACC indicates American College of Cardiology; AHA, American Heart Association; AIG, Association of International Governors; DSMB, data safety monitoring board; EP, electrophysiology; HF, heart failure; HHS, Health and Human Services; HRS, Heart Rhythm Society; ICD, implantable cardioverter-defibrillator; PharmD, Doctor of Pharmacy; PI, principal investigator; STOP-AF, Sustained Treatment Of Paroxysmal Atrial Fibrillation; STS, Society of Thoracic Surgeons; UCLA, University of California, Los Angeles; USUHS, Uniformed Services University of the Health Sciences; UT, University of Texas; VA, Veterans Affairs; and VCU, Virginia Commonwealth University.

Appendix 3. Abbreviations

ACE = angiotensin-converting enzyme
 ACS = acute coronary syndrome
 AF = atrial fibrillation
 ARB = angiotensin-receptor blocker
 AV = atrioventricular
 bpm = beats per minute
 CAD = coronary artery disease
 CKD = chronic kidney disease
 CrCl = creatinine clearance
 ECG = electrocardiogram/electrocardiographic
 EF = ejection fraction
 HCM = hypertrophic cardiomyopathy
 HF = heart failure
 HFpEF = heart failure with preserved ejection fraction
 INR = international normalized ratio
 LA = left atrial/atrium
 LAA = left atrial appendage
 LMWH = low-molecular-weight heparin
 LV = left ventricular
 LVEF = left ventricular ejection fraction
 RCT = randomized controlled trial
 RV = right ventricular
 TEE = transesophageal echocardiography
 TIA = transient ischemic attack
 TTR = time in therapeutic range
 UFH = unfractionated heparin
 WPW = Wolff-Parkinson-White

Appendix 4. Initial Clinical Evaluation in Patients With AF

Minimum Evaluation

- | | |
|--|--|
| 1. History and physical examination, to define | <ul style="list-style-type: none"> • Presence and nature of symptoms associated with AF • Clinical type of AF (paroxysmal, persistent, or permanent) • Onset of first symptomatic attack or date of discovery of AF • Frequency, duration, precipitating factors, and modes of initiation or termination of AF • Response to any pharmacological agents that have been administered • Presence of any underlying heart disease or reversible conditions (eg, hyperthyroidism or alcohol consumption) |
| 2. ECG, to identify | <ul style="list-style-type: none"> • Rhythm (verify AF) • LVH • P-wave duration and morphology or fibrillatory waves • Pre-excitation • Bundle-branch block • Prior MI • Other atrial arrhythmias • To measure and follow R-R, QRS, and QT intervals in conjunction with antiarrhythmic drug therapy |
| 3. TTE, to identify | <ul style="list-style-type: none"> • VHD • LA and RA size • LV and RV size and function • Peak RV pressure (pulmonary hypertension) • LV hypertrophy • LA thrombus (low sensitivity) • Pericardial disease |
| 4. Blood tests of thyroid, renal, and hepatic function | <ul style="list-style-type: none"> • For a first episode of AF • When ventricular rate is difficult to control |

Additional Testing (1 or several tests may be necessary)

- | | |
|----------------------------------|---|
| 1. 6-min walk test | <ul style="list-style-type: none"> • If adequacy of rate control is in question |
| 2. Exercise testing | <ul style="list-style-type: none"> • If adequacy of rate control is in question • To reproduce exercise-induced AF • To exclude ischemia before treatment of selected patients with a type IC* antiarrhythmic drug |
| 3. Holter or event monitoring | <ul style="list-style-type: none"> • If diagnosis of type of arrhythmia is in question • As a means of evaluating rate control |
| 4. TEE | <ul style="list-style-type: none"> • To identify LA thrombus (in LAA) • To guide cardioversion |
| 5. Electrophysiological study | <ul style="list-style-type: none"> • To clarify the mechanism of wide-QRS-complex tachycardia • To identify a predisposing arrhythmia such as atrial flutter or paroxysmal supraventricular tachycardia • To seek sites for curative AF ablation or AV conduction block/modification |
| 6. Chest radiograph, to evaluate | <ul style="list-style-type: none"> • Lung parenchyma, when clinical findings suggest an abnormality • Pulmonary vasculature, when clinical findings suggest an abnormality |

*Type IC refers to the Vaughan Williams classification of antiarrhythmic drugs.

AF indicates atrial fibrillation; AV, atrioventricular; ECG, electrocardiogram; LA, left atrial; LAA, left atrial appendage; LV, left ventricular; LVH, left ventricular hypertrophy; MI, myocardial infarction; RA, right atrial; RV, right ventricular; TEE, transesophageal echocardiography; TTE, transthoracic echocardiogram; and VHD, valvular heart disease.

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